Digital Design & Computer Arch.

Lecture 1: Introduction: Fundamentals, Transistors, Gates

Prof. Onur Mutlu

ETH Zürich

Spring 2026

19 February 2026

Brief Self Introduction

- Onur Mutlu
  - Full Professor @ ETH Zurich ITET (INFK), since Sept 2015
  - Strecker Professor @ Carnegie Mellon University ECE (CS), 2009-2016, 2016-…
  - Started the Comp Arch Research Group @ Microsoft Research, 2006-2009
  - Worked @ Google, VMware, Microsoft Research, Intel, AMD, Stanford
  - PhD in Computer Engineering from University of Texas at Austin in 2006
  - BS in Computer Engineering & Psychology from University of Michigan in 2000
  - https://people.inf.ethz.ch/omutlu/ omutlu@gmail.com

- Research and Teaching in:
  - **Computer architecture, systems, hardware security, bioinformatics**
  - Memory and storage systems
  - Robust & dependable hardware systems: security, safety, predictability, reliability
  - Hardware/software cooperation
  - New computing paradigms; architectures with emerging technologies/devices
  - Architectures for bioinformatics, genomics, health, medicine, AI/ML
  - …

My Co-Instructor

- Dr. Mohammad Sadrosadati
  - Senior Researcher and Lecturer @ SAFARI Research Group, ETHZ
  - PhD in Computer Engineering from Sharif University of Technology in 2019
  - MS in Computer Engineering from Sharif University of Technology in 2014
  - BS in Computer Engineering from Sharif University of Technology in 2012
  - mohammad.sadrosadati@safari.ethz.ch

- Research & Teaching Areas
  - Computer Architecture
  - Memory/Storage Systems
  - Near-Data Processing
  - Heterogeneous System Architecture
  - Bioinformatics
  - Interconnection Network

Our Head Teaching Assistant (I)

- Dr. Konstantina Koliogeorgi
  - Head Teaching Assistant
  - Senior Lecturer & Researcher @ SAFARI
  - PhD from National Technical University of Athens in 2023
  - kkoliogeorgi@safari.ethz.ch

- Research & Teaching Areas
  - Hardware/Software Co-Design
  - Heterogeneous System Architecture
  - Reconfigurable Computing and Architectures
  - Hardware Acceleration
  - Optimized Architectures for Genome analysis
  - High Level Synthesis Tools
  - Design Space Exploration

Our Head Teaching Assistant (II)

- Dr. Zhiheng Yue
  - Head Teaching Assistant
  - Senior Lecturer & Researcher @ SAFARI
  - PhD from Tsinghua University in 2024
  - zhiyue@ethz.ch

- Research & Teaching Areas
  - Processing-in-memory
  - SRAM/DRAM
  - 3D Integration
  - Design technology co-optimization
  - AI Accelerator
  - Physically unclonable function

Our Head Teaching Assistant for Labs (I)

- Ataberk Olgun
  - Head Teaching Assistant (Labs)
  - Senior PhD Student @ SAFARI
  - https://ataberkolgun.com/
  - aolgun@ethz.ch

- Research and Teaching Areas
  - Computer Architecture
  - Memory Systems
  - DRAM
  - Hardware Robustness (Security, Safety, Reliability, Availability)
  - FPGAs
  - Hardware/Software Cooperation

Our Head Teaching Assistant for Labs (II)

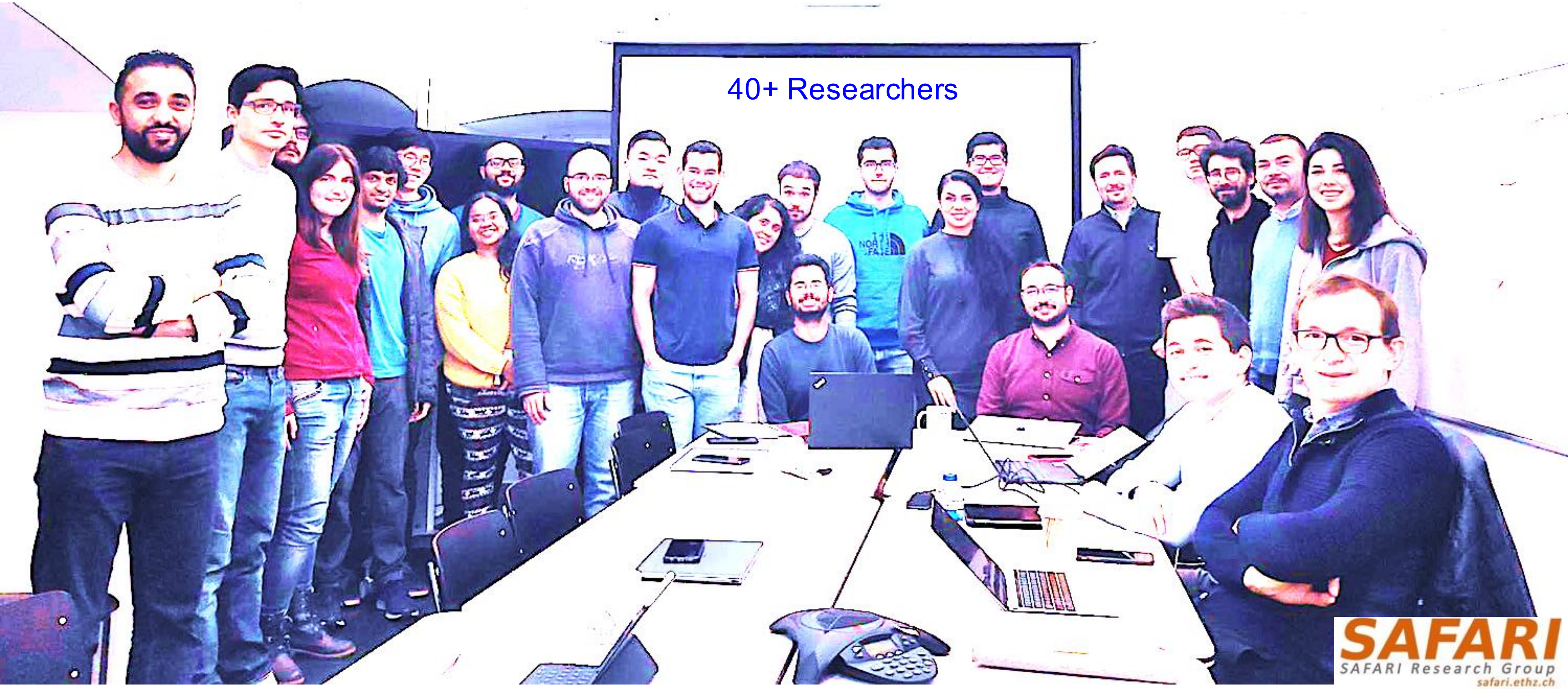
- Mayank Kabra
  - Head Teaching Assistant (Labs)
  - PhD Student @ SAFARI
  - kabram@ethz.ch

- Research and Teaching Areas
  - Privacy Enhancing Technologies
  - Memory and Storage-Centric Computing
  - Cryptographic Algorithm-Hardware Codesign
  - Memory Technologies (DRAM, SSD)
  - Computer Architecture

SAFARI Research Group

Computer architecture, HW/SW, systems, bioinformatics, security, memory

https://safari.ethz.ch/safari-newsletter-april-2020/



Think BIG, Aim HIGH!

https://safari.ethz.ch

SAFARI Newsletter January 2021 Edition

- https://safari.ethz.ch/safari-newsletter-january-2021/

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SAFARI Research Group

Newsletter
January 2021

***Think Big, Aim High, and
Have a Wonderful 2021!***

Dear SAFARI friends,

Happy New Year! We are excited to share our group highlights with you in this second edition of the SAFARI newsletter (You can find the first edition from April 2020 **here**). 2020 has

SAFARI Newsletter December 2021 Edition

- https://safari.ethz.ch/safari-newsletter-december-2021/

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Think Big, Aim High

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December 2021

SAFARI Newsletter June 2023 Edition

- https://safari.ethz.ch/safari-newsletter-june-2023/

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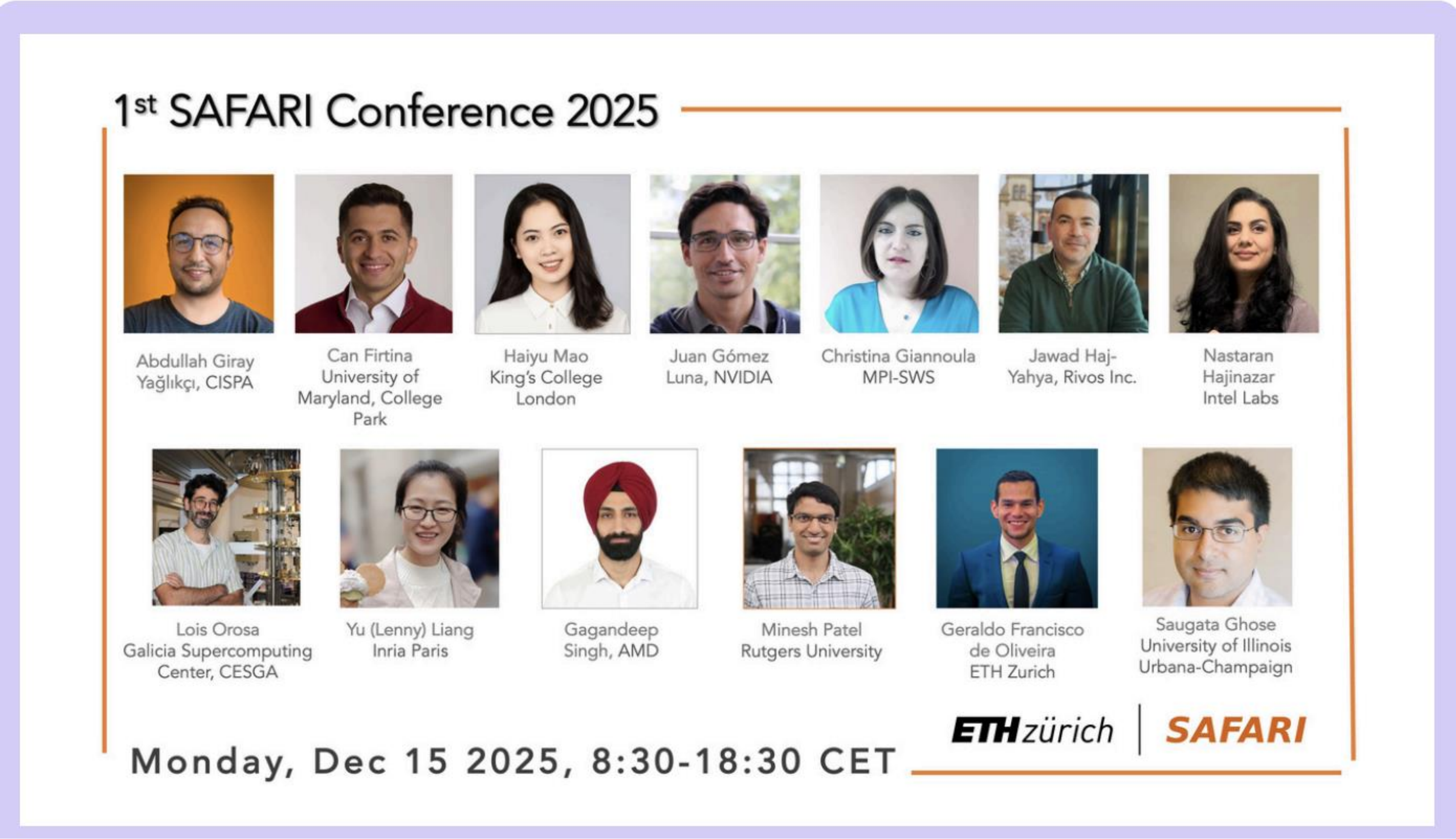
June 2023

SAFARI Newsletter July 2024 Edition

- https://safari.ethz.ch/safari-newsletter-july-2024/

SAFARI Newsletter December 2025 Edition

- https://people.inf.ethz.ch/omutlu/pub/safari-newsletter-december-2025.pdf



SAFARI Introduction & Research

Computer architecture, HW/SW, systems, bioinformatics, security, memory

SAFARI Research Group
safari.ethz.ch

Think BIG, Aim HIGH!

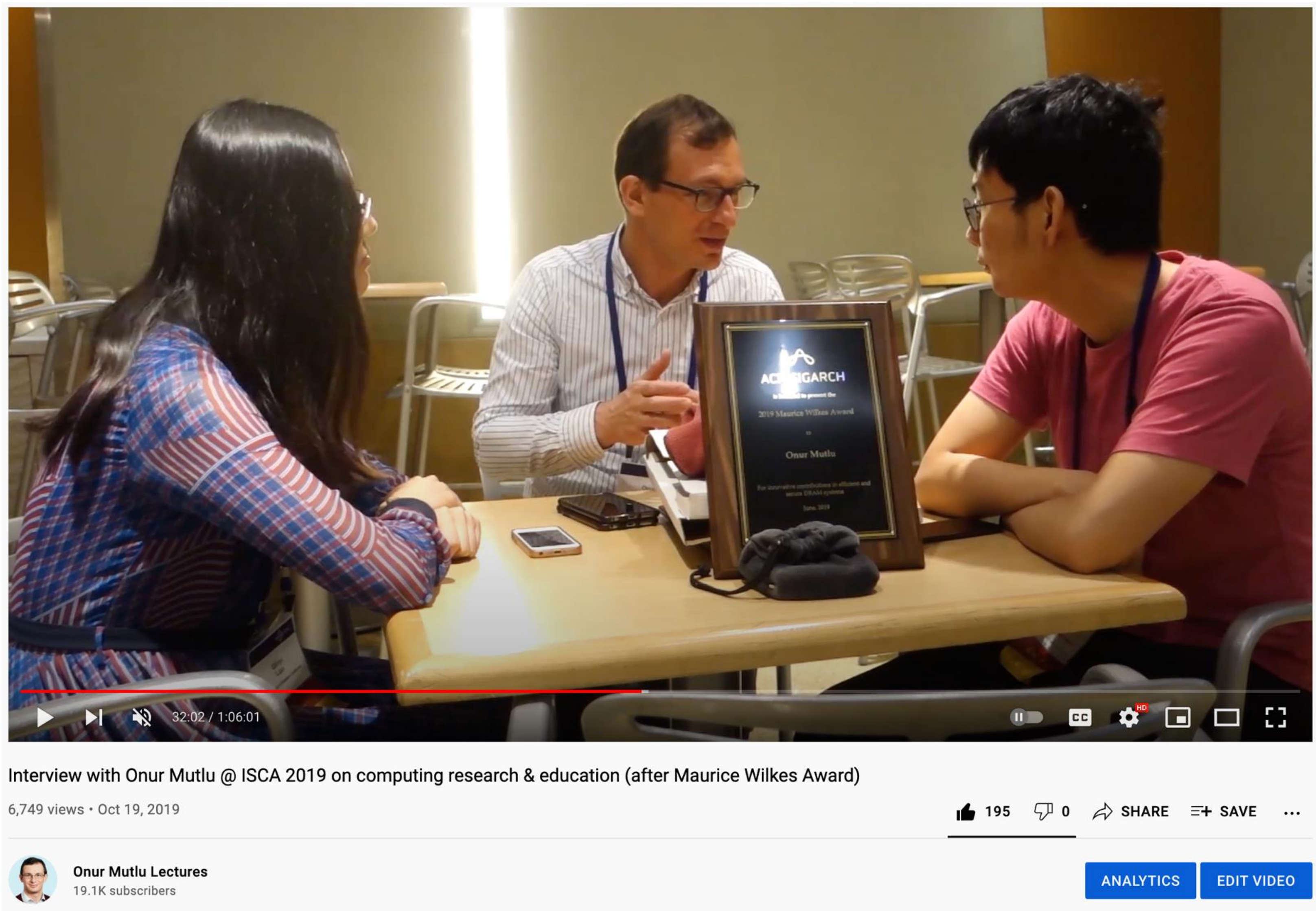
https://www.youtube.com/watch?v=mV2OuB2djEs

SAFARI PhD and Post-Doc Alumni

- Rahul Bera (ETH Zurich), MICRO 2022 Best Paper Award, ISCA 2024 Best Paper Award, HPCA 2026 Distinguished Artifact Award
- Geraldo Francisco de Oliveira Jr. (Huawei Research Zurich, Senior Researcher)
- Can Firtina (Univ. of Maryland, Assistant Professor), ETH Doctoral Medal 2025
- Lukas Breitwieser (CERN), 2025 Impact Award of the Research Software Engineering Society, Best artifact award at PPoPP 2023
- A. Giray Yaglikci (CISPA), 2025 SIGMICRO Dissertation Award, DSN Carter Award Best Thesis 2025; IEEE TCCA/ACM SIGARCH Dissertation Award HM 2025, PACT 2023 SRC Winner, Intel HWSec Award Finalist 2021, HOST PhD Competition Finalist 2024, VTS PhD Competition Finalist 2025, HPCAHoF 2025, ISCA HoF 2025, Best Artifact award at HPCA 2025/ISCA 2023…
- Hasan Hassan (Rivos), EDAA Outstanding Dissertation Award 2023; S&P 2020 Best Paper Award, 2020 Pwnie Award, IEEE Micro TP HM 2020
- Christina Giannoula (MPI-SWS), NTUA Best Dissertation Award 2023
- Minesh Patel (Rutgers, Asst. Prof.), DSN Carter Award Best Thesis 2022; ETH Doctoral Medal 2023; MICRO'20 & DSN'19 Best Paper Awards; ISCA HoF 2021
- Damla Senol Cali (Bionano Genomics), SRC TECHCON 2019 Best Student Presentation Award; RECOMB-Seq 2018 Best Poster Award
- Nastaran Hajinazar (Intel)
- Gagandeep Singh (AMD/Xilinx), FPL 2020 Best Paper Award Finalist
- Amirali Boroumand (Stanford Univ → Google), SRC TECHCON 2018 Best Presentation Award
- Jeremie Kim (Apple), EDAA Outstanding Dissertation Award 2020; IEEE Micro Top Picks 2019; ISCA/MICRO HoF 2021
- Nandita Vijaykumar (Univ. of Toronto, Assistant Professor), ISCA Hall of Fame 2021
- Kevin Hsieh (Microsoft Research, Principal Researcher)
- Justin Meza (Facebook), HiPEAC 2015 Best Student Presentation Award; ICCD 2012 Best Paper Award
- Mohammed Alser (Georgia State Univ., Assistant Professor), IEEE Turkey Best PhD Thesis Award 2018
- Yixin Luo (Google → OpenAI), HPCA 2015 Best Paper Session
- Kevin Chang (Facebook), SRC TECHCON 2016 Best Student Presentation Award
- Rachata Ausavarungnirun (KMUNTB, Assistant Professor → MangoBoost), NOCS 2015 and NOCS 2012 Best Paper Award Finalist
- Gennady Pekhimenko (Univ. of Toronto, Associate Professor), ISCA Hall of Fame 2021; ASPLOS 2015 SRC Winner
- Vivek Seshadri (Microsoft Research, Principal Researcher)
- Donghyuk Lee (NVIDIA Research, Senior Researcher), HPCA Hall of Fame 2018
- Yoongu Kim (Software Robotics → Google), IFIP JCL Award'24, TCAD'19 Top Pick Award; IEEE Micro Top Picks'10; HPCA'10 Best Paper Session
- Lavanya Subramanian (Intel Labs → Facebook → OpenAI)

- Samira Khan (Univ. of Virginia, Assistant Professor → Google), HPCA 2014 Best Paper Session, DSN 2025 Test of Time Award
- Saugata Ghose (Univ. of Illinois, Assistant Professor), DFRWS-EU 2017 Best Paper Award, HPCA HoF 2024, MICRO HoF 2024
- Jawad Haj-Yahya (Huawei Research Zurich, Principal Researcher → Rivos)
- Lois Orosa (Galicia Supercomputing Center, Director)
- Jisung Park (POSTECH, Assistant Professor)
- Gagandeep Singh (AMD/Xilinx, Researcher)
- Juan Gomez-Luna (NVIDIA, Researcher), ISPASS 2023 Best Paper Session, MICRO HoF 2022
- Mohammed Alser (Georgia State Univ., Assistant Professor), IEEE Turkey Best PhD Thesis Award 2018
- Haiyu Mao (King's College, Assistant Professor)
- Yu (Lenny) Liang (INRIA Paris, Researcher)

An Interview on Computing Futures



https://www.youtube.com/watch?v=8ffSEKZhmvo

An Interview at ISCA 2023

TCuARCH meets Prof. Onur Mutlu - 2023

Onur Mutlu Lectures
46.5K subscribers

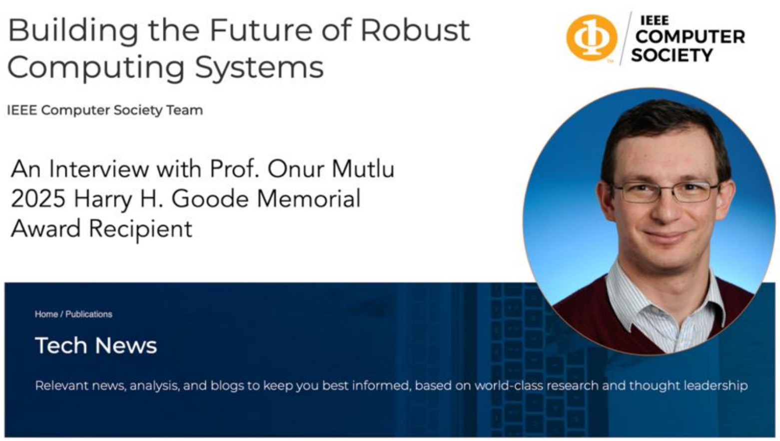
Analytics | Edit video | 27 | Share | Promote | Download | Clip

454 views Premiered May 6, 2024

Prof. Onur Mutlu (https://people.inf.ethz.ch/omutlu/) shares his mindset seeing his life and career as a researcher, mentor, and advisor. His last remark should resonate with many young researchers. "Do what you are really excited about. Everything will follow."

 https://www.youtube.com/watch?v=6Hpn4SAX0dI

Some More Recent Interviews (I)



Building the Future of Robust Computing Systems: Interview with Onur Mutlu

Posted on **September 24, 2025** by **ewent**

Read the latest interview with Onur Mutlu in the IEEE Computer Society INSIDER Tech news on receiving the 2025 Harry H. Goode Memorial Award for his seminal contributions to computer architecture research and practice, especially in memory systems. He discusses his work on RowHammer and research that bridges the gap between academia and industry, while taking a look towards building the future of robust computing systems.

Interview: Building the Future of Robust Computing Systems, IEEE Computer Society Team, INSIDER news, 09/23/2025

SAFARI News: Onur Mutlu receives the 2025 Harry H. Goode Memorial Award

Home / Publications / Tech News / INSIDER

Building the Future of Robust Computing Systems

IEEE Computer Society Team

Published 09/23/2025

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An Interview with Dr. Onur Mutlu – 2025 Harry H. Goode Memorial Award Recipient

Dr. Onur Mutlu, Professor of Computer Science at ETH Zurich, has done research in computer architecture, computing systems, hardware security, memory and storage systems, and bioinformatics and is an innovator whose work has improved microprocessors and memory and storage systems used by billions of people.

Your identification of the RowHammer vulnerability has had a significant impact on hardware security and memory systems. What led you to this discovery, and how has it influenced subsequent research in memory systems?

RowHammer is a fascinating phenomenon that affects all modern main memory (i.e., DRAM) chips used in almost all computers today (comprising a more than 120 billion USD market size). It is the fact that repeatedly accessing one DRAM row causes bitflips in physically nearby DRAM rows that should not change at all, leading to data corruption. We have identified the phenomenon and analyzed it in detail for the first time in 2012. We have continued to study it since then, discovering new properties as well as new read disturbance mechanisms along the way, such as RowPress (in 2020-2023) and Variable Read Disturbance (in 2024-2025). We have come a long way, yet the phenomenon still fascinates me and there is a lot more to do in fundamentally understanding and very efficiently solving it, especially as memory technology scales to much denser capacities using increasingly smaller cells to store each bit of data. And, not only us, the phenomenon continues to fascinate researchers across hardware security, design automation, computer architecture, dependable systems, device analysis communities. Many works are published each year on understanding, analyzing, modeling the phenomenon as well as solving it using new and creative methods across the system stack (spanning both hardware and software). For example, top security and computer architecture conferences have been almost regularly having dedicated RowHammer sessions (sometimes multiple of them) since 2020.

https://safari.ethz.ch/building-the-future-of-robust-computing-systems-interview-with-onur-mutlu/

https://www.computer.org/publications/tech-news/insider-membership-news/build-robust-computing-onur-mutlu

Beyond the Bits: Cultivating Minds and Memory Systems

IEEE Computer Society
71,833 followers

October 18, 2025

In this article, Onur Mutlu shares his innovative work in memory and storage systems, highlighting how his research pushes the boundaries of what modern computing hardware can achieve.

Looking ahead, what emerging trends in memory and storage systems excite you the most, and why?

Memory (including storage) is the major energy and performance bottleneck today and its importance will only increase. There are many exciting topics to work on, some of which I outlined earlier. Changing the computing paradigm to be memory-centric is one of the most exciting developments and trends that we discussed earlier. It can fundamentally enable ultra-efficient systems by getting rid of the huge energy and performance waste we have in computing systems today. Emerging integration and packaging technologies (e.g., 3D heterogeneous integration, monolithic 3D, hybrid bonding, etc.) can greatly help memory-centric computing by enabling many types of processing and memory components to be tightly integrated/packaged together. I am quite excited about revisiting how architectures should be designed in the presence of such heterogeneous integration/packaging technologies along with very efficient interconnect technologies (e.g., photonic interconnects). Such new hardware paradigms (and their thoughtful novel combinations) can enable fundamentally more efficient AI algorithms & models or biomedical/health applications (and other general-purpose workloads) that were not possible or envisioned before.

A major research topic today is to design systems that are driven by AI. In particular, I am quite excited about rethinking how system architectures can be designed using AI/ML techniques, especially for online control tasks that are all over hardware systems we design today. Many architectural controllers today use human-designed heuristics and policies. These policies tend to be very simplistic and they cannot improve over time with data. One goal in my research has been to design data-driven architectures. A data-driven architecture enables the machine itself to learn the best policies for managing itself and executing programs. Controllers in such an architecture, when needed, are data-driven autonomous agents that automatically learn far-sighted control policies. A prime example of such a controller is the reinforcement learning based self-optimizing memory controllers (which we published at ISCA 2008), reinforcement learning based hardware prefetchers (MICRO 2021) and SSD data placement mechanisms (ISCA 2022). Such ML-based controllers not only improve performance and efficiency under a wide variety of conditions and workloads but also reduce the hardware and system designer's burden in designing sophisticated controllers. We believe an intelligent architecture will consist of a collection of such intelligent AI/ML agents that perform automatic data-driven online policy learning, including learning how to best coordinate with each other to make decisions that benefit the overall system. Such machines

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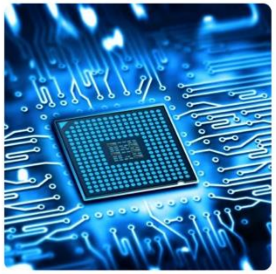
The Next Computing Revolution: Bringing Processing Inside Memory

By Onur Mutlu on September 30, 2025



Dr. Onur Mutlu is a renowned computer scientist and Professor at ETH Zurich whose pioneering research in computer architecture, memory systems, and hardware security has shaped industry standards and influenced technologies used by billions worldwide.

In this article, he shares his insights on advancing memory-centric computing, exploring the challenges, breakthroughs, and future possibilities of next-generation computer memory systems.

With advancements in memory-centric (i.e., processing-in-memory) architectures, what challenges do you foresee in integrating these systems into mainstream computing?

Memory-centric computing (or processing-in-memory) is another fascinating research area where we can change the paradigm of how we do computing. Moving data between memory and processor consumes orders of magnitude more energy than computation. Many results from real systems show that most (e.g., >90%) of the energy spent executing major AI models comes from data movement and memory access, not computation performed on the data. Unfortunately, our existing processor-centric design paradigm is the major fundamental cause of significant data movement across the entire system. All data needs to be processed in computation units, which today are very far away from memory, storage, and sensing units. If we want to make computing truly efficient and high performance (and also robust and sustainable), we should minimize data movement. We can do so by placing computation capability near and inside memory structures (especially high-density memories like DRAM and flash), and thus greatly (i.e., by orders of magnitude) enhance energy efficiency, performance, robustness, security, and sustainability of almost all computing systems we build.

We have been doing research in this area for at least 15 years. We have written many papers and designed many new promising processing-in-memory techniques, advocating for a paradigm shift and demonstrating great energy and performance benefits with processing-in-memory. Processing data near, and in, where data resides or is produced simply makes sense, from a very fundamental first-principles standpoint. Three important works that overview the area, written 12 years apart, are the following:

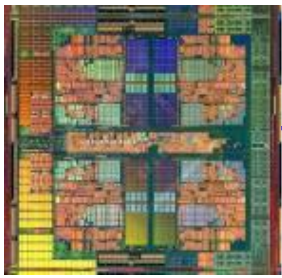
Onur Mutlu, Ataberk Olgun, and İsmail Emir Yüksel, *"Memory-Centric Computing: Solving Computing's Memory Problem"* Invited Paper in Proceedings of the 17th IEEE International Memory Workshop (IMW), Monterey, CA, USA, May 2025.

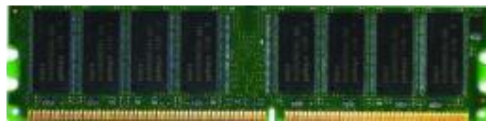
Suggestions on Education, Research, Growth

https://www.youtube.com/watch?v=83tlorht7Mc&list=PL5Q2soXY2Zi8D_5MGV6EnXEJHnV2YFBJl&index=54

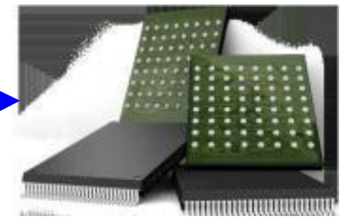
SAFARI Research Group: Current Mission

Computer architecture, HW/SW, systems, bioinformatics, security



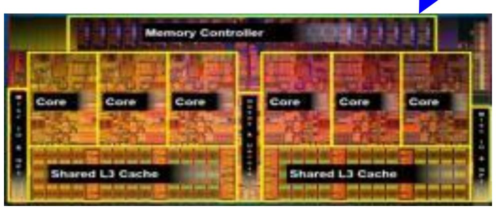


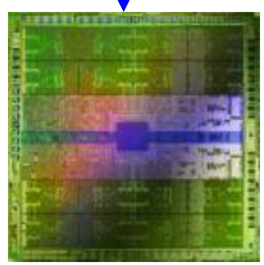
Processing in Memories



Storage, Sensors, Memory, Interconnect
with Processing Capabilities
+ New Paradigms for Computation

Heterogeneous
Processors and
Accelerators





ML/AI, Vision, Genomics,
Biological Signal Processing

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Enable fundamentally better computers

https://safari.ethz.ch/safari-newsletter-july-2024/

Major Current Research Topics

- Fundamentally Robust (Secure/Reliable/Safe) Architectures

- Fundamentally Energy-Efficient Architectures
  - Memory-centric (Data-centric) Architectures

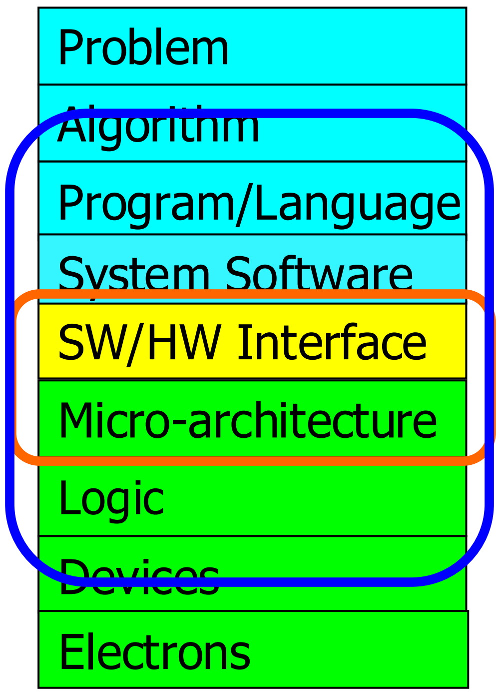
- Fundamentally Low-Latency and Predictable Architectures

- Fundamentally Intelligent and Evolving Architectures
  - ML/AI-Assisted (Data-driven) and Data-aware Architectures

- Architectures for ML/AI, Genomics, Medicine, Health, …

The Transformation Hierarchy

Computer Architecture
(expanded view)



Computer Architecture
(narrow view)

Approach: Cross Layer Design

To achieve the highest efficiency, performance, robustness:

we must take the expanded view

of computer architecture

- Problem
- Algorithm
- Program/Language
- System Software
- SW/HW Interface
- Micro-architecture
- Logic
- Devices
- Electrons

Co-design across the hierarchy: Algorithms to devices

Specialize as much as possible within the design goals

Broad research spanning applications, systems, software, logic, circuits with architecture at the center

Principle: Teaching and Research

...

Teaching drives Research

Research drives Teaching

...

Accessing All Our Courses

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Home | People | Courses | News | Research | Publications | Tools | Work with us | Contact us

researchers and practitioners, including leading companies. Many students and universities without access to state-of-the-art computer architecture classes benefit from our online classes (see our YouTube channels here).

Spring 2025:
Fundamentals of Computer Architecture
Digital Design and Computer Architecture
Seminar in Computer Architecture
SAFARI **Project & Seminars** courses

Fall 2024:
Computer Architecture
Seminar in Computer Architecture
SAFARI **Project & Seminars** courses

Spring 2024:
Digital Design and Computer Architecture
Seminar in Computer Architecture
SAFARI **Project & Seminars** courses

Fall 2023:
Computer Architecture
Seminar in Computer Architecture
SAFARI **Project & Seminars** courses

Spring 2023:
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Fall 2022:

Onur Mutlu Lectures
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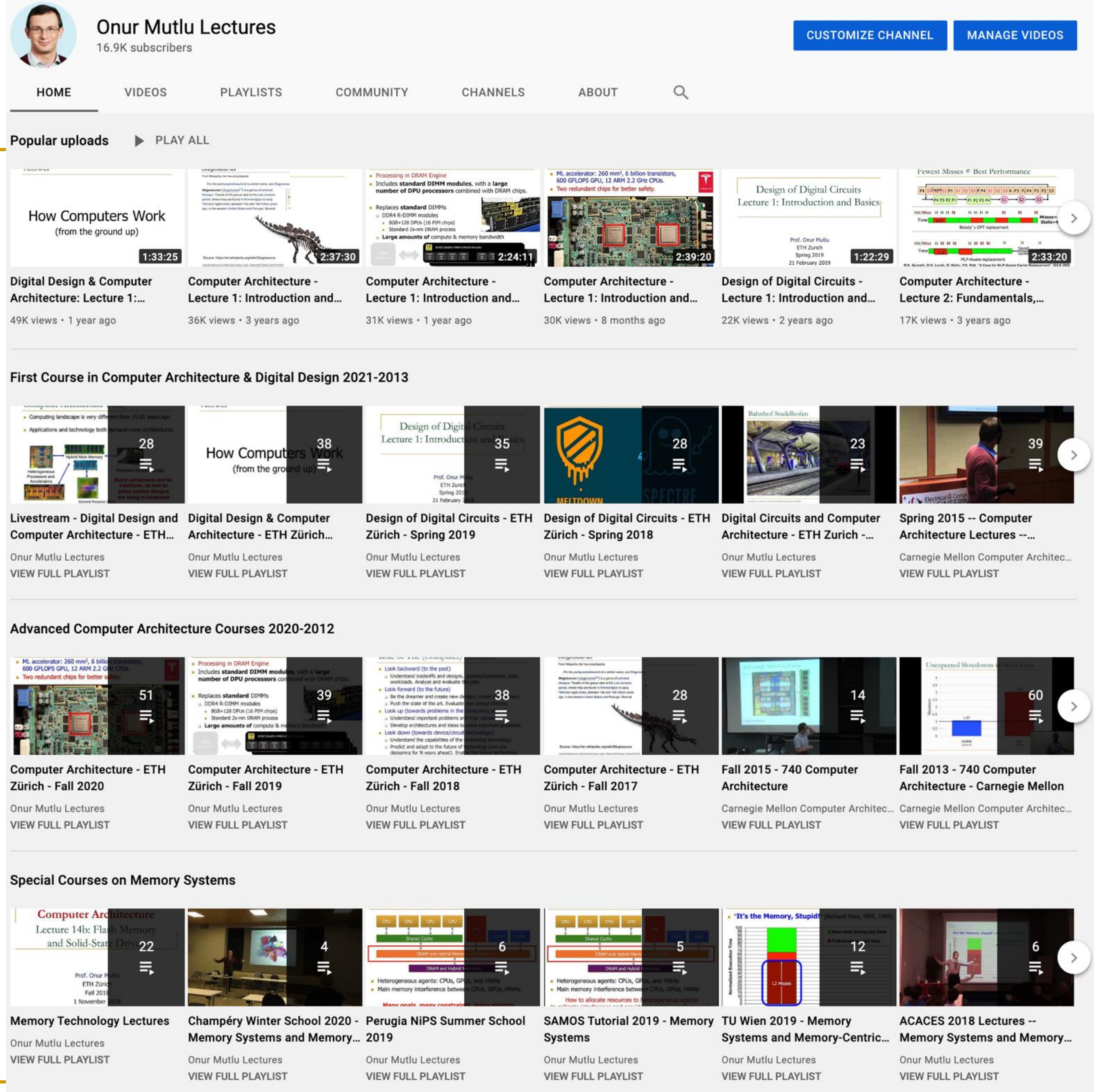


https://www.youtube.com/onurmutlulectures

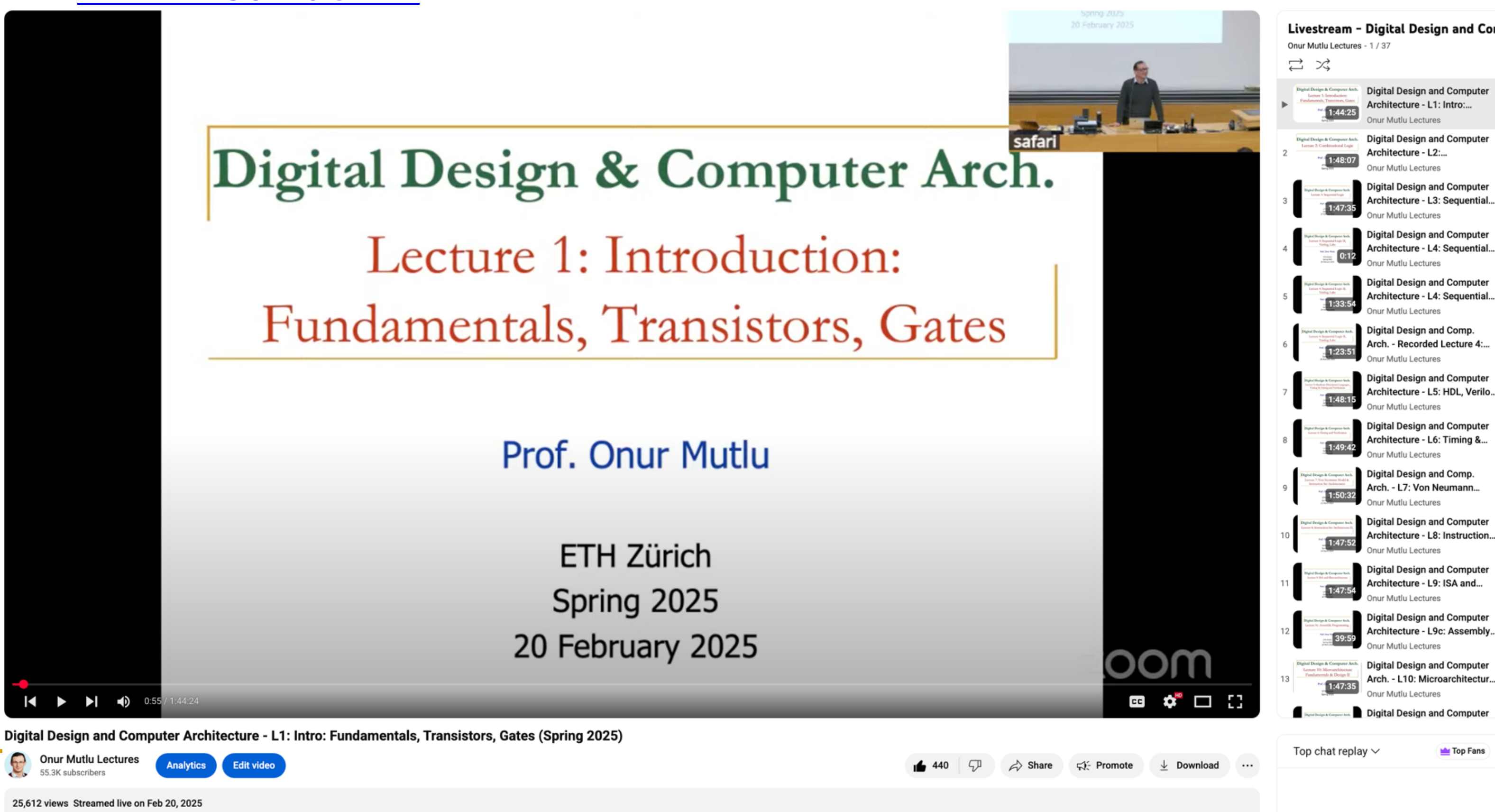
Our Major Courses & Lectures

- **First Digital Design & Computer Architecture Course (DDCA)**
  - Freshman-level (CS & ECE): Digital Design and Computer Architecture
  - Spring 2025 Livestream Edition: https://www.youtube.com/watch?v=ubhxKNlOlRg&list=PL5Q2soXY2Zi9Eo29LMgKVcaydS7V1zZW3&index=1
- **Fundamentals of Computer Architecture Course (FoCA)**
  - Junior/Senior-level (ECE & CS): Computer Architecture
  - Spring 2025 Livestream Edition: https://www.youtube.com/watch?v=uKgMFj1eQQc&list=PL5Q2soXY2Zi_ZMtqz1r-GHm-zzuE1QfIg&index=1
- **Advanced Computer Architecture Course (ACA)**
  - Senior/Masters/PhD-level (CS & ECE): Computer Architecture
  - Fall 2024 Livestream Edition: https://www.youtube.com/watch?v=ziMRjDlLEwo&list=PL5Q2soXY2Zi-LfDdGgWyLcTSqzm6a26wD&index=1
- **Seminar in Computer Architecture (SCA)**
  - Fall 2024 Livestream Edition: https://www.youtube.com/watch?v=SamRIUuT1FA&list=PL5Q2soXY2Zi-B6hFQA2y8lisIlSOYMHL1&index=1

https://www.youtube.com/onurmutlulectures

Links for Last Year's Materials (DDCA)

- Digital Design & Computer Architecture Course (Spring 2025):
  - **https://safari.ethz.ch/ddca/spring2025**
  - **https://www.youtube.com/onurmutlulectures**
  - https://www.youtube.com/watch?v=ubhxKNlOlRg&list=PL5Q2soXY2Zi9Eo29LMgKVcaydS7V1zZW3&index=1



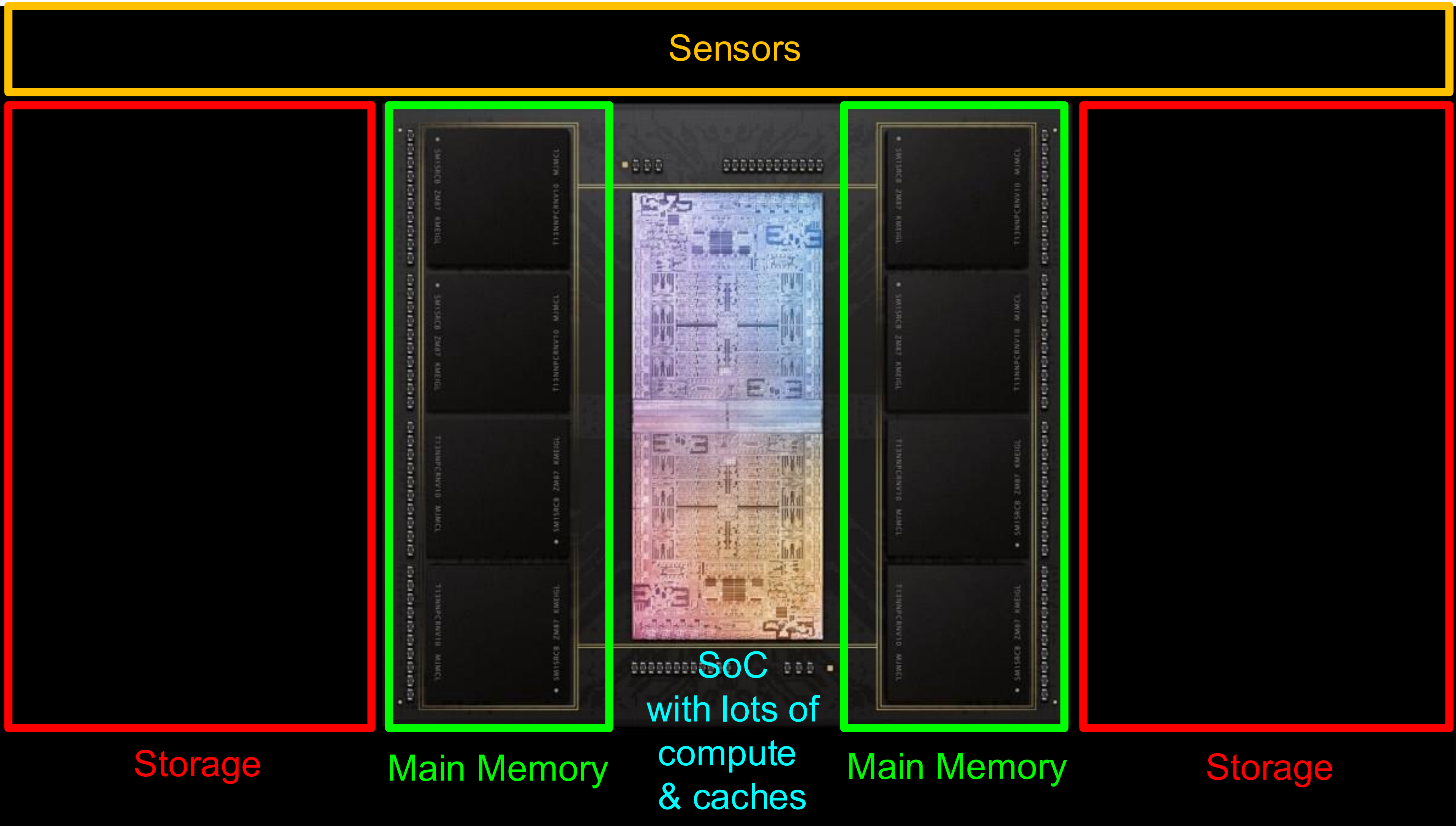
Basic Goals & Structure of DDCA

What Will We Learn in This Course?

How Computers Work

(from the ground up)

We Will Study How Something Like This Works



Apple M1 Ultra System (2022)

https://www.gsmarena.com/apple_announces_m1_ultra_with_20core_cpu_and_64core_gpu-news-53481.php

Major High-Level Goals of This Course

- In Digital Design & Computer Architecture

- Understand the basics
- Understand the principles (of design)
- Understand the precedents

- Based on such understanding:
  - learn how a modern computer works underneath
  - evaluate tradeoffs of different designs and ideas
  - implement a principled design (a simple microprocessor)
  - learn to systematically debug increasingly complex systems
  - Hopefully enable you to develop novel, out-of-the-box designs

- The focus is on basics, principles, precedents, and how to use them to create/implement good designs

Why These Goals?

- Because you are here for a Computer Science degree!

- Regardless of your future direction, learning the principles of digital design & computer architecture will be useful to
  - design better hardware
  - design better software
  - design better systems
  - make better tradeoffs in design
  - understand why computers behave the way they do
  - solve problems better
  - think "in parallel"
  - think critically
  - …

DDCA Course Components

- **Lectures** (understanding concepts)
- **Readings** (reinforcing & going deeper)
- **Homeworks** (problem solving preparation)
- **Labs** (hands-on experience in some concepts)
- **Exam** (test of understanding)
- **Extra Credit Assignments** (fundamental and simple)

- In all, you have freedom to adapt to your learning style

- My advice: Focus on learning & scholarship & understanding

https://safari.ethz.ch/ddca/spring2026/

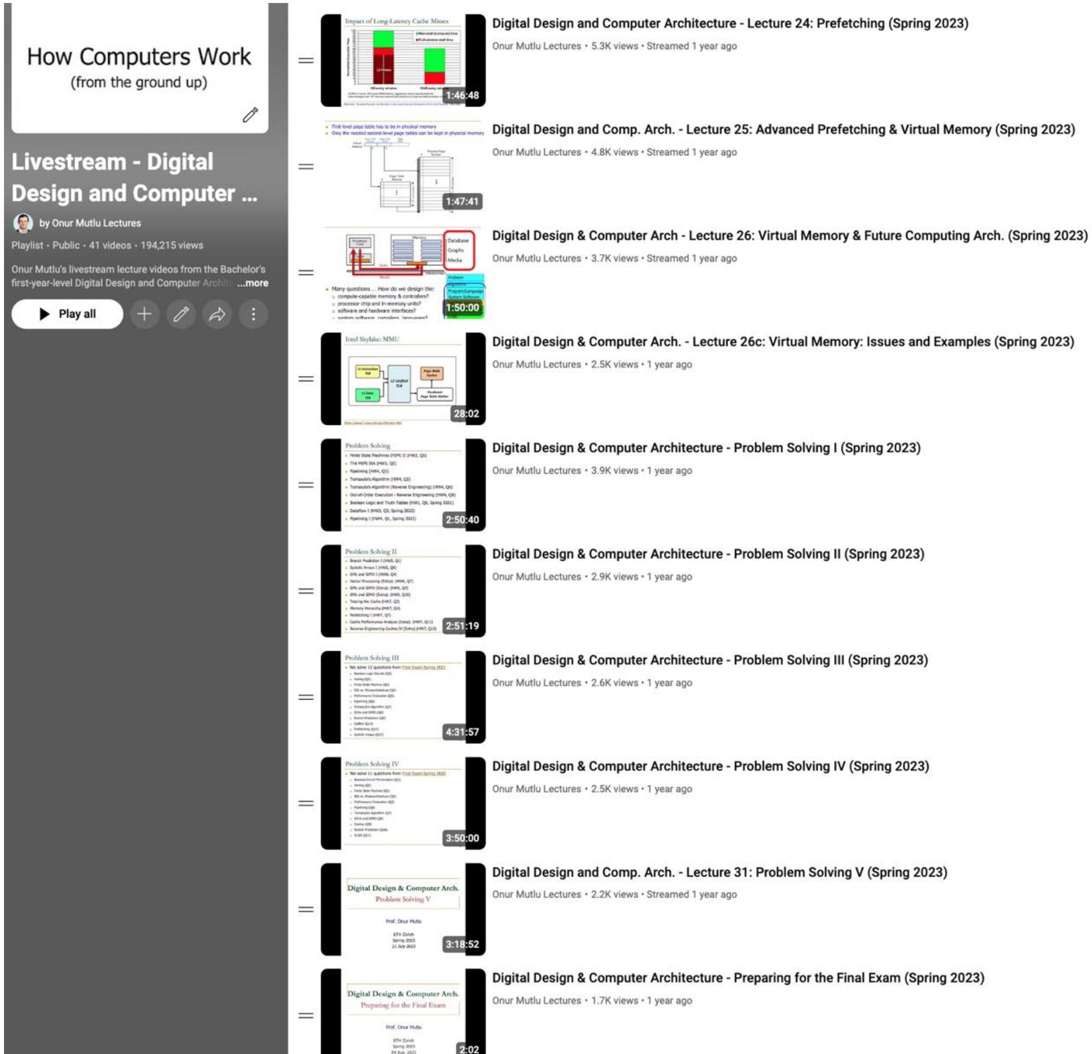
Learning & Exam

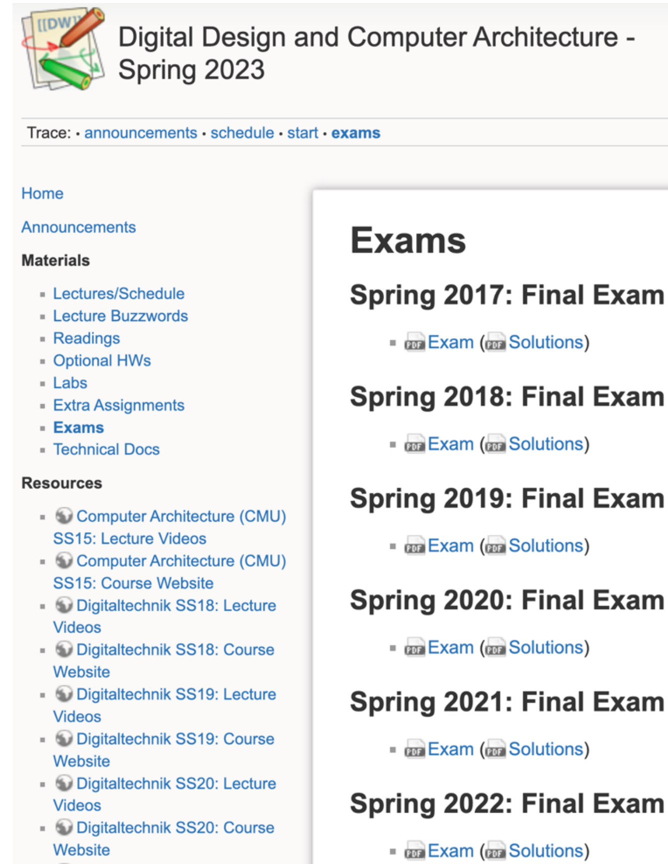
- We will enable you to learn + prepare you for the exam

- My suggestions:
  - focus on understanding, learning, mastering the material
    - lectures, readings, labs, HWs all enable this and prepare you
  - reinforce problem solving skills with homeworks
  - do **not** worry about the exam while listening to lectures
    - most of you will pass this course (historically >80%)

- We will release a lot of material to help you with the exam
  - Problem solving sessions
  - Exam guidance
  - All past exams (and basic solutions) are already online

Problem Solving & Exam Review Sessions





https://safari.ethz.ch/digitaltechnik/spring2023

https://safari.ethz.ch/ddca/spring2025/

https://www.youtube.com/playlist?list=PL5Q2soXY2Zi-EImKxYYY1SZuGiOAOBKaf
https://www.youtube.com/playlist?list=PL5Q2soXY2Zi9Eo29LMgKVcaydS7V1zZW3

Summary

- Learning is for life (never ends)

- Exam study is until you pass (ends, hopefully August 2026)

Focus on

learning and scholarship

How to Approach This Course

Learning experience

Long-term tradeoff analysis

Critical thinking & decision making

How to Approach This Course

Find and choose
the learning style
that works best for you

What Will We Learn in This Course?

What Will We Learn in This Course?

How Computers Work

(from the ground up)

What Will We Learn in This Course?

And Why We Care

Why Do We Have Computers?

Why Do We Do Computing?

Why Do We Do Computing?

To Solve Problems

Answer Reworded

To Gain Insight

Hamming, "Numerical Methods for Scientists and Engineers," 1962.

Answer Extended

To Enable

a Better Life & Future

How Does a Computer Solve Problems?

How Does a Computer Solve Problems?

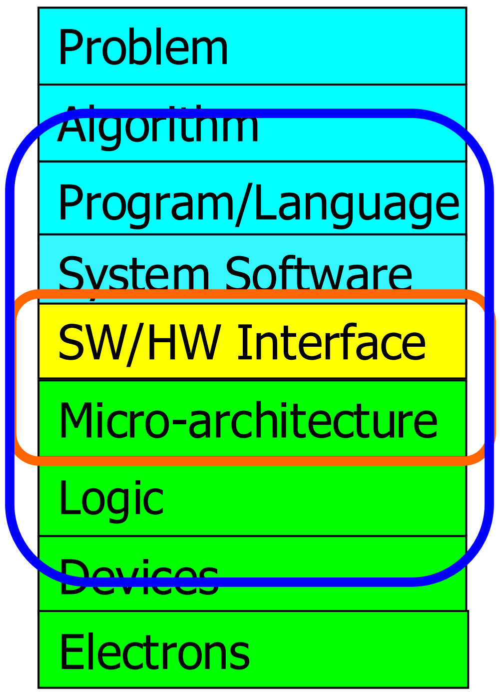
Orchestrating Electrons

In today's dominant technologies

How Do Problems Get Solved by Electrons?

The Transformation Hierarchy

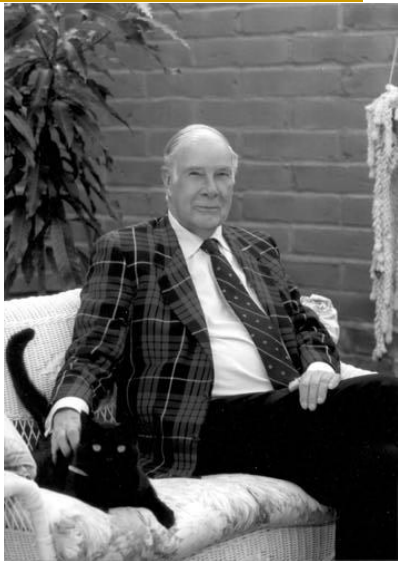
Computer Architecture (expanded view)



Computer Architecture (narrow view)

Levels of Transformation

"The purpose of computing is [to gain] insight" (*Richard Hamming*)
We gain and generate insight by solving problems
How do we ensure problems are solved by electrons?



Algorithm

Step-by-step procedure that is **guaranteed to terminate** where **each step is precisely stated** and **can be carried out by a computer**

- **Finiteness**
- **Definiteness**
- **Effective computability**

Many algorithms for the same problem

Problem
Algorithm
Program/Language
System Software (VM, OS, MM)
ISA (Architecture)
Microarchitecture
Logic
Devices
Electrons

ISA
(Instruction Set Architecture)

Interface/contract between SW and HW.

What the programmer assumes hardware will satisfy.

Microarchitecture
An implementation of the ISA

Digital logic circuits
Building blocks of micro-arch (e.g., gates)

Computer Architecture

- is the science and art of designing computing platforms (hardware, interface, system SW, and programming model)

- to achieve a set of design goals
  - E.g., highest performance on earth on workloads X, Y, Z
  - E.g., longest battery life at a form factor that fits in your pocket with cost < $$$ CHF
  - E.g., best average performance across all known workloads at the best performance/cost ratio
  - …

  - Designing a supercomputer is different from designing a smartphone → But, many fundamental principles are similar

Different Platforms, Different Goals

Source: http://www.sia-online.org (semiconductor industry association)

Different Platforms, Different Goals

Source: https://iq.intel.com/5-awesome-uses-for-drone-technology/

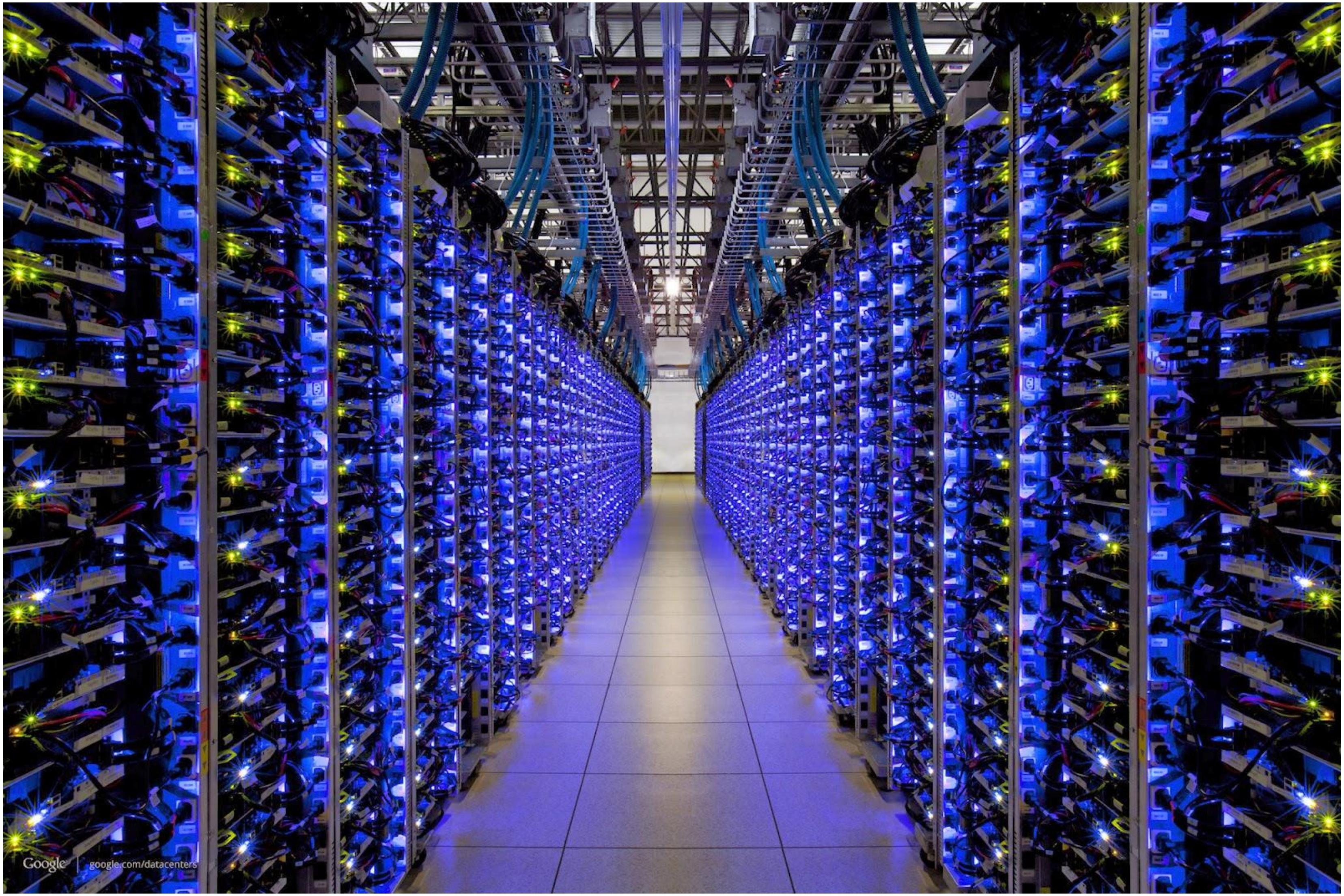
Different Platforms, Different Goals

Source: https://taxistartup.com/wp-content/uploads/2015/03/UK-Self-Driving-Cars.jpg

Different Platforms, Different Goals

Source: http://sm.pcmag.com/pcmag_uk/photo/g/google-self-driving-car-the-guts/google-self-driving-car-the-guts_dwx8.jpg

Different Platforms, Different Goals



Source: http://datacentervoice.com/wp-content/uploads/2015/10/data-center.jpg

Different Platforms, Different Goals

Source: https://fossbytes.com/wp-content/uploads/2015/06/Supercomputer-TIANHE2-china.jpg

Different Platforms, Different Goals

Source: https://www.itmagazine.ch/artikel/72401/Fugaku_Der_schnellste_Supercomputer_der_Welt.html

Different Platforms, Different Goals

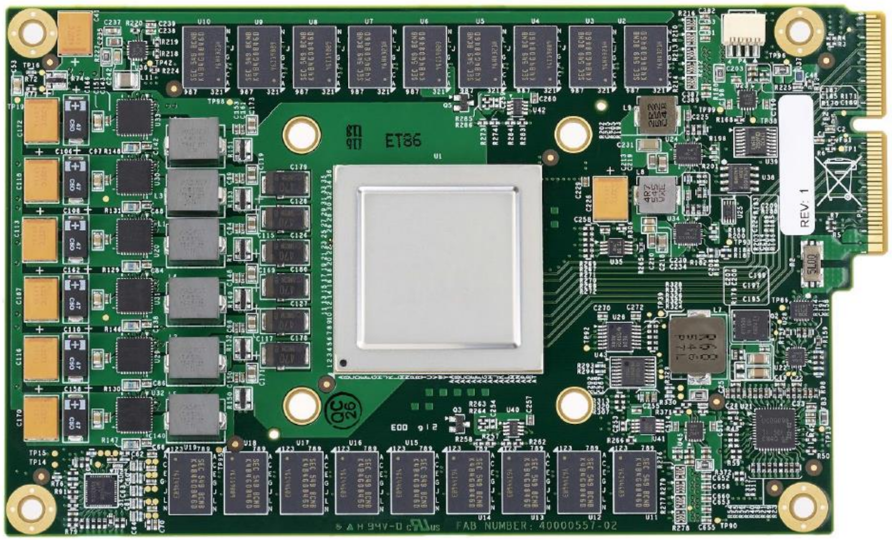


Figure 3. TPU Printed Circuit Board. It can be inserted in the slot for an SATA disk in a server, but the card uses PCIe Gen3 x16.

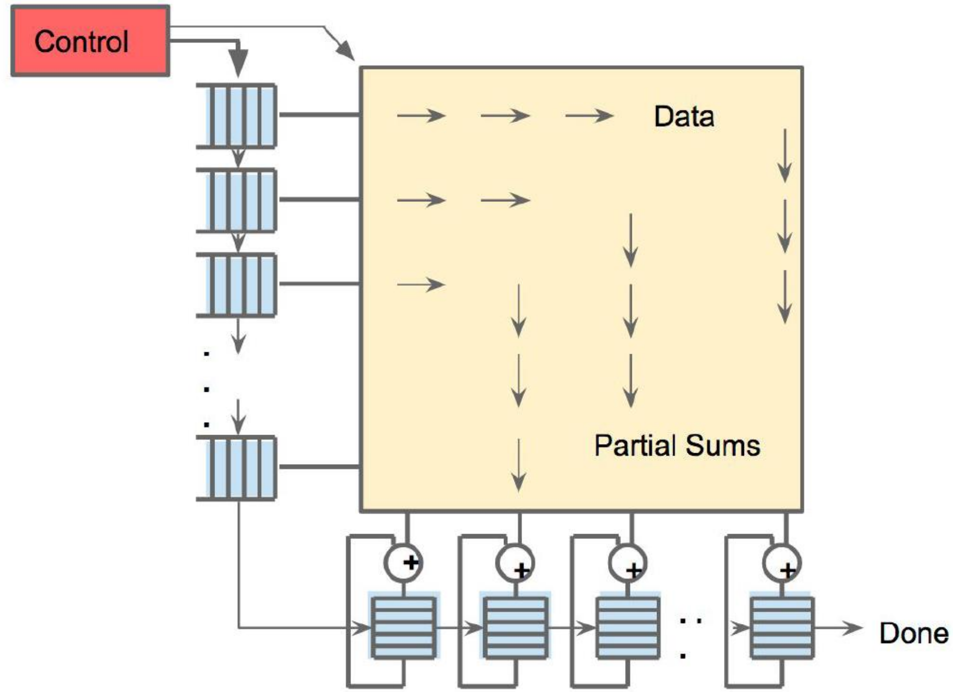
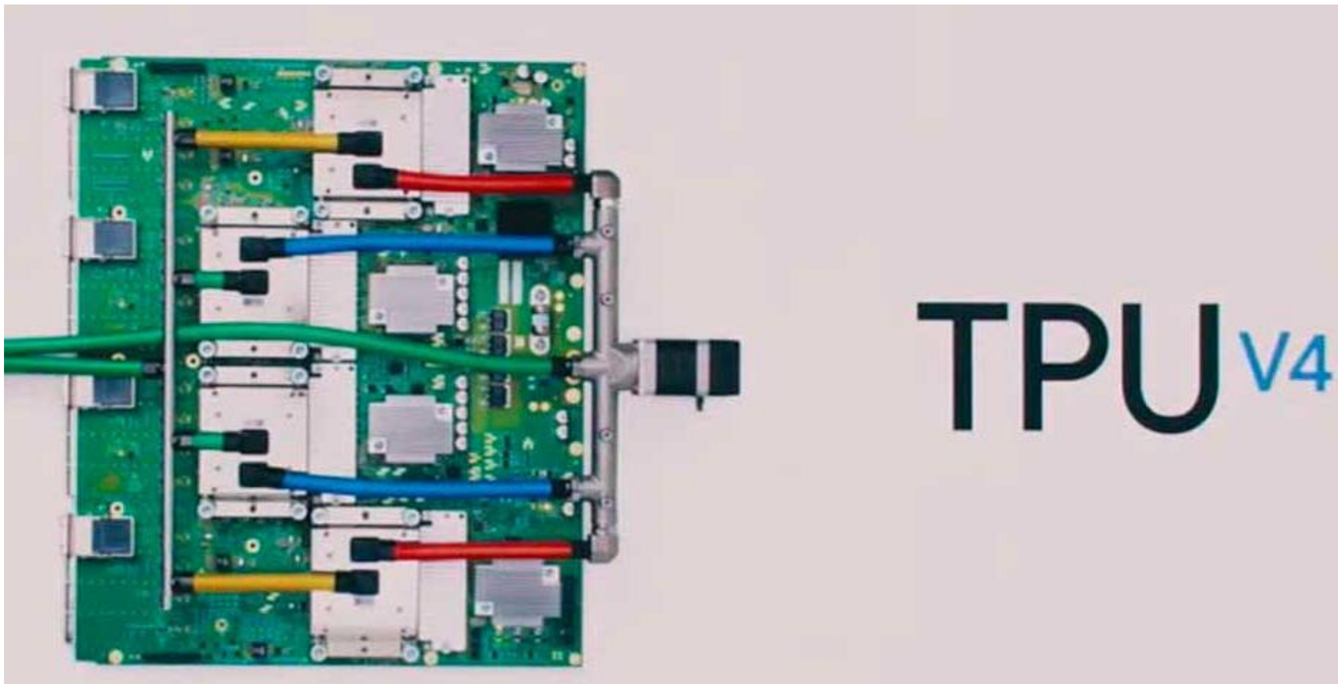


Figure 4. Systolic data flow of the Matrix Multiply Unit. Software has the illusion that each 256B input is read at once, and they instantly update one location of each of 256 accumulator RAMs.

Jouppi et al., "In-Datacenter Performance Analysis of a Tensor Processing Unit", ISCA 2017.

Different Platforms, Different Goals



New ML applications (vs. TPU3):
- Computer vision
- Natural Language Processing (NLP)
- Recommender system
- Reinforcement learning that plays Go

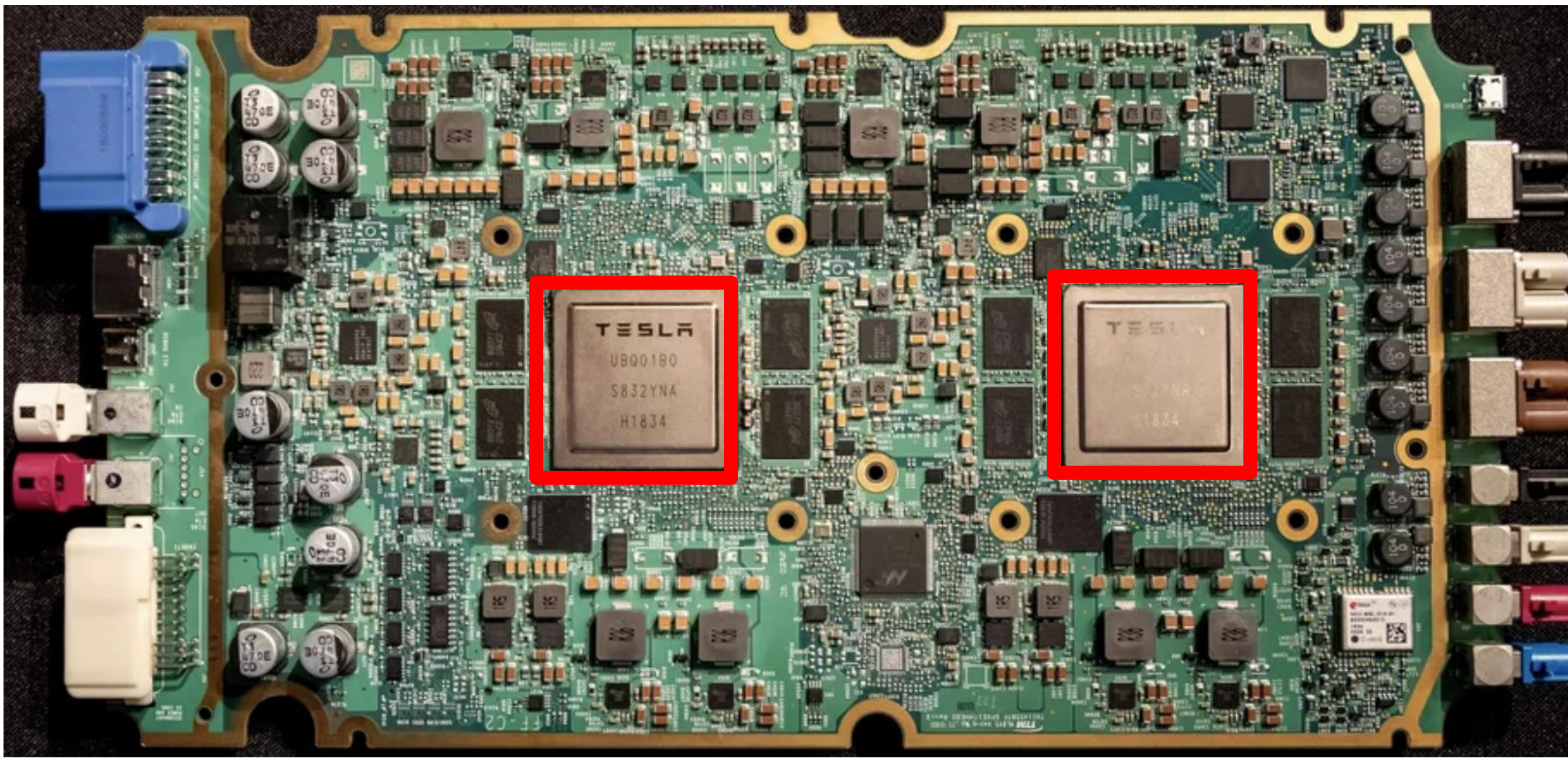
250 TFLOPS per chip in 2021
vs 90 TFLOPS in TPU3

1 ExaFLOPS per board

https://spectrum.ieee.org/tech-talk/computing/hardware/heres-how-googles-tpu-v4-ai-chip-stacked-up-in-training-tests

Different Platforms, Different Goals

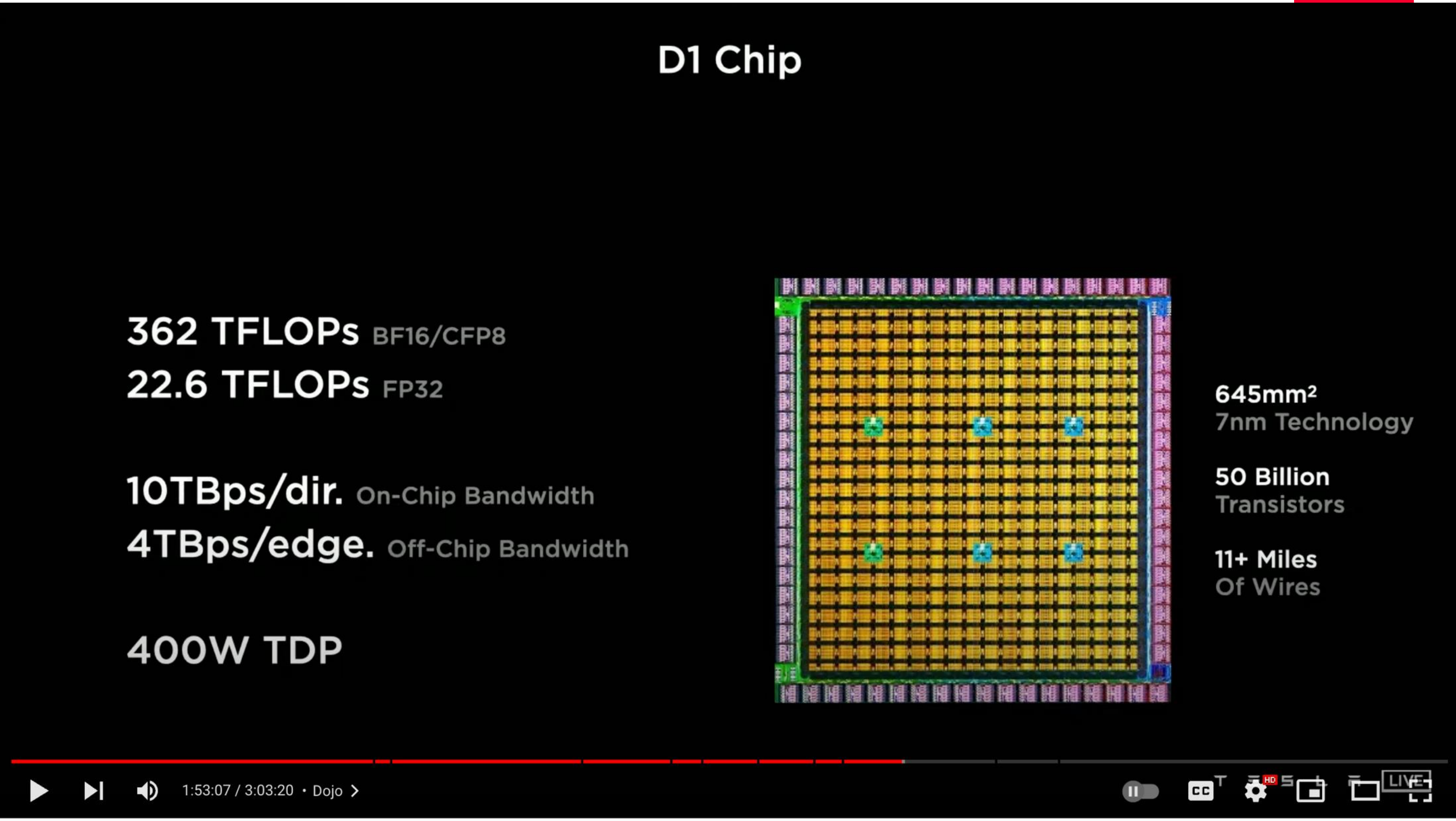
- ML accelerator: 260 mm^2, 6 billion transistors, 600 GFLOPS GPU, 12 ARM 2.2 GHz CPUs.
- Two redundant chips for better safety.



https://www.youtube.com/watch?v=j0z4FweCy4M

Different Platforms, Different Goals

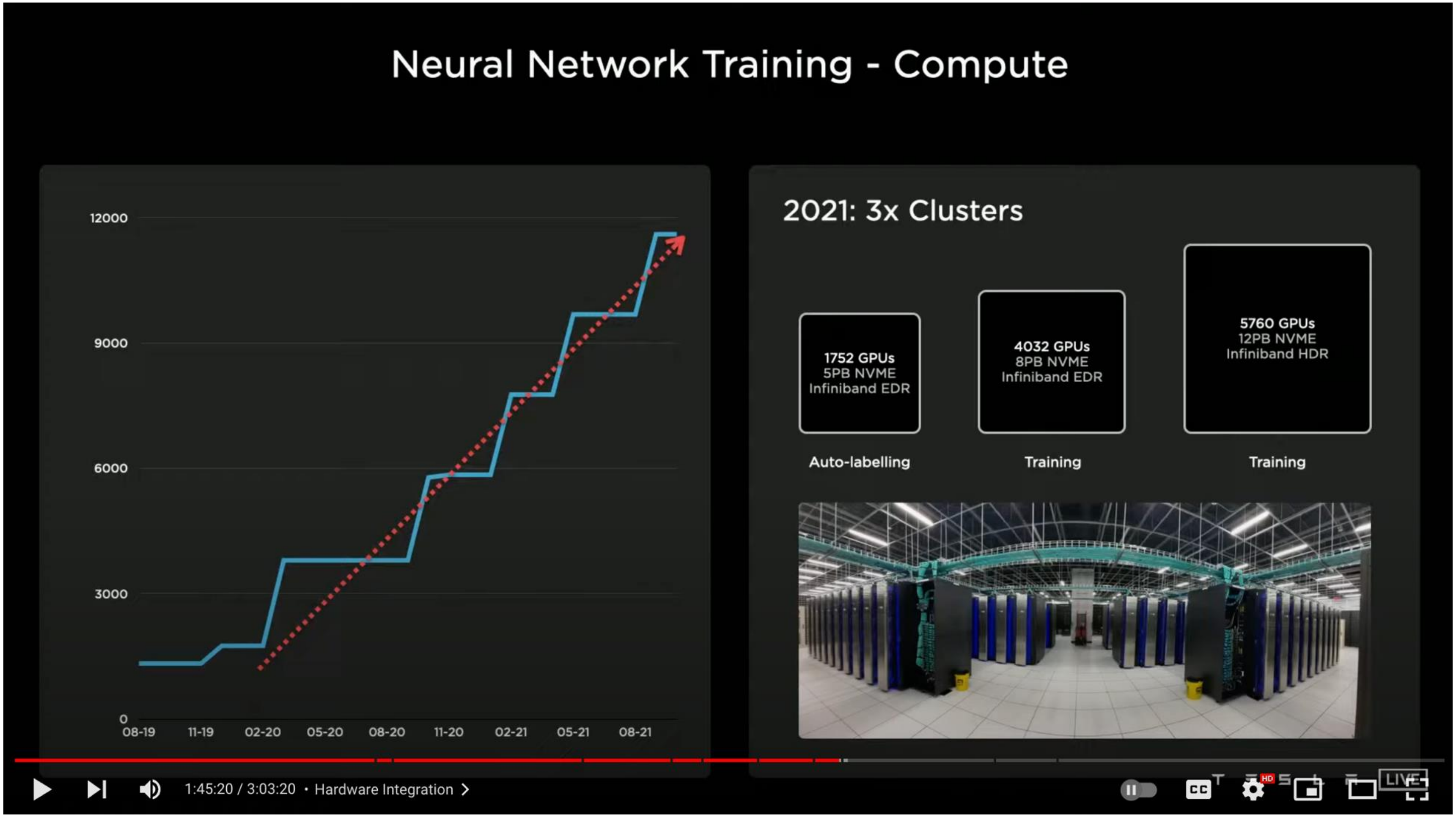
- Tesla Dojo Chip & System



https://www.youtube.com/watch?v=j0z4FweCy4M&t=6340s

Different Platforms, Different Goals

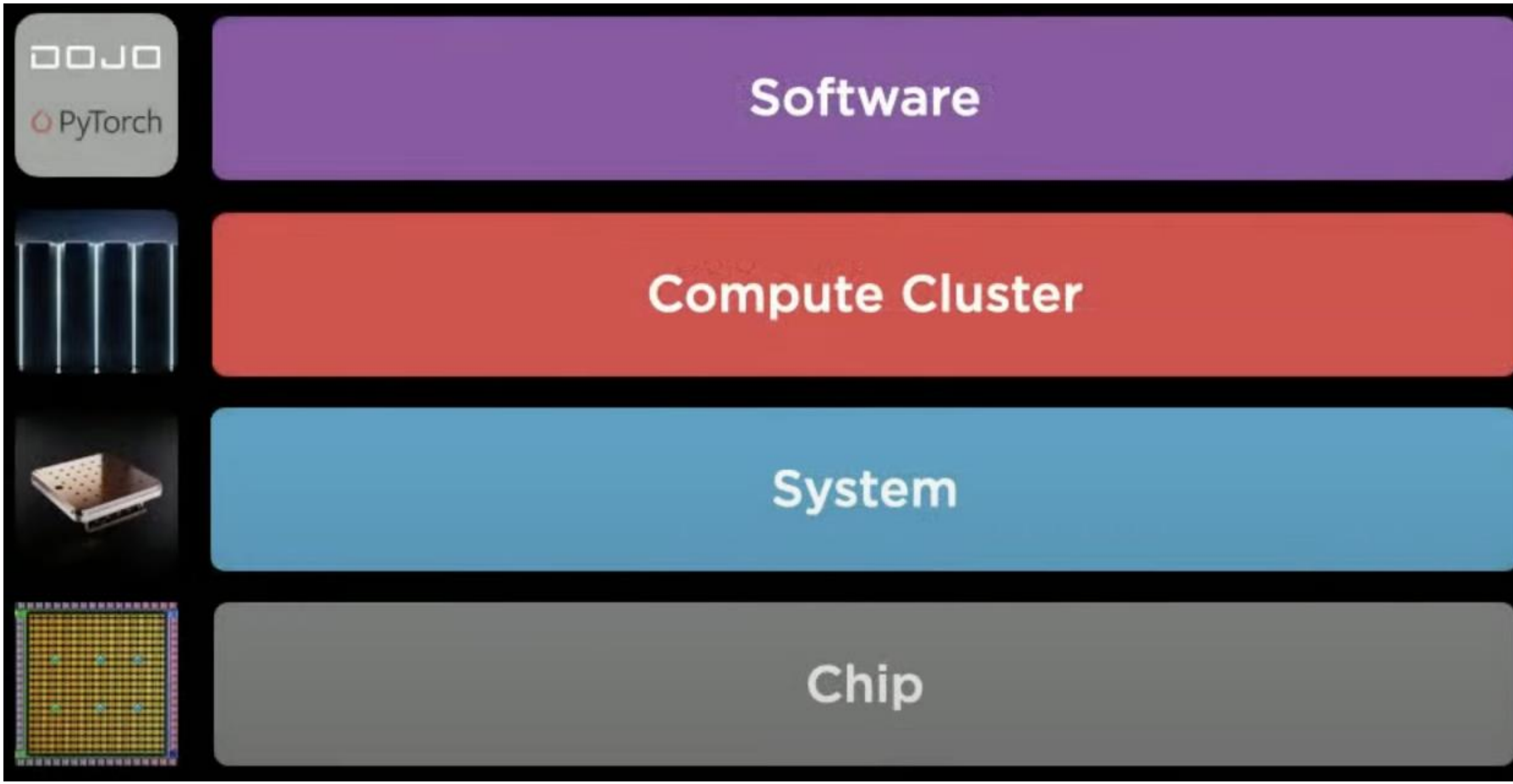
- Tesla Dojo Chip & System



https://www.youtube.com/watch?v=j0z4FweCy4M&t=6340s

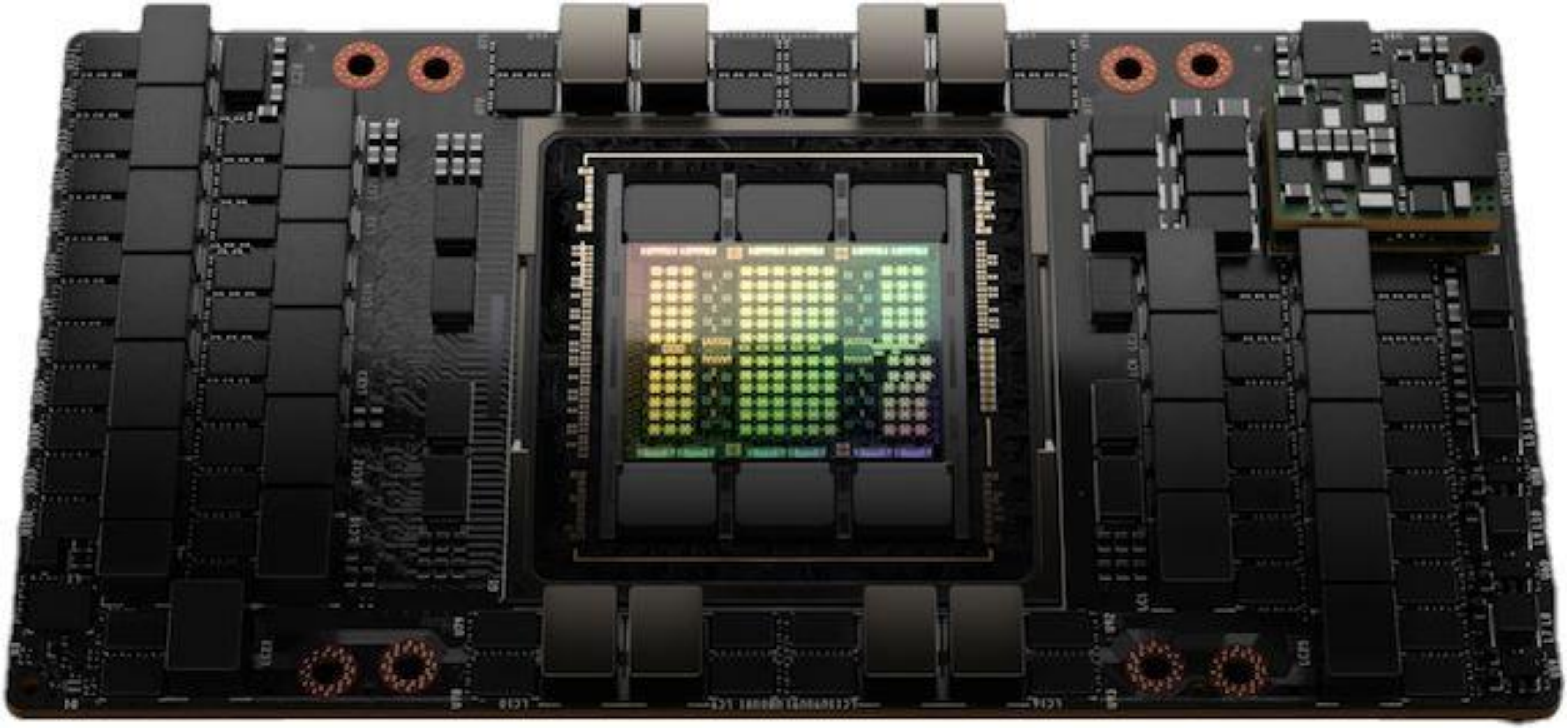
Different Platforms, Different Goals

- Tesla Dojo Chip & System

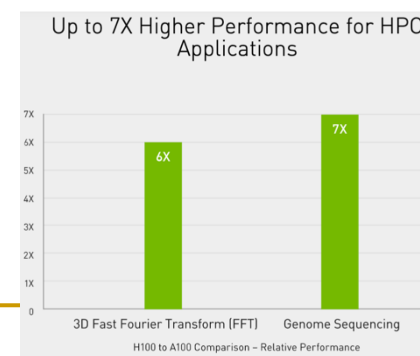


https://www.youtube.com/watch?v=j0z4FweCy4M&t=6340s

Different Platforms, Different Goals

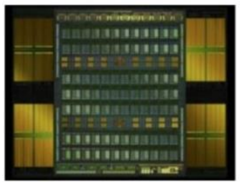


NVIDIA is claiming a **7x improvement** in dynamic programming algorithm (**DPX instructions**) performance on a single H100 versus naïve execution on an A100.

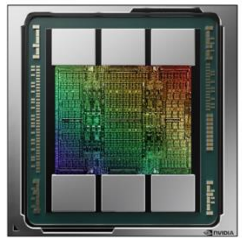


https://www.nvidia.com/en-us/data-center/h100/

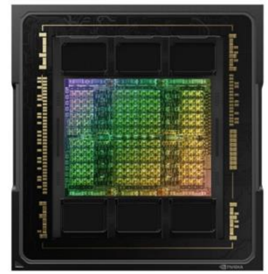
Evolution of Recent GPUs (I)



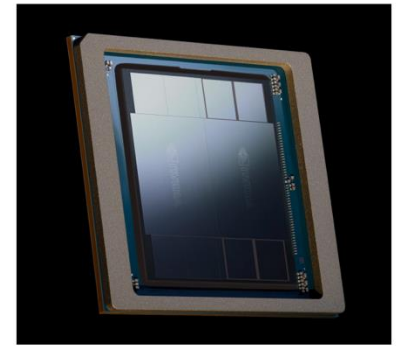
Volta
>21 billion transistors
815mm^2
TSMC 12nm FFN



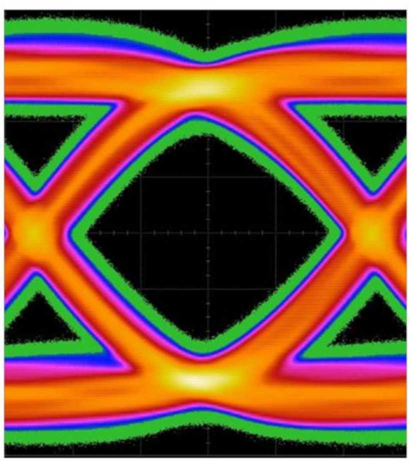
Ampere
>54 billion transistors
826 mm^2
TSMC N7



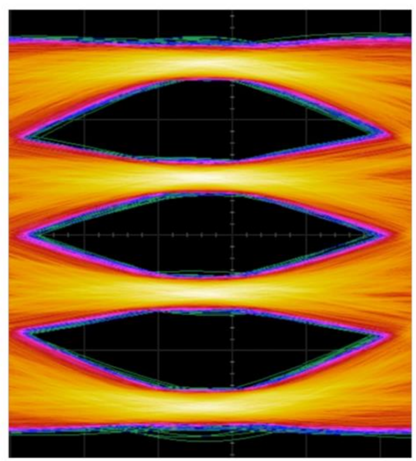
Hopper
>80 billion transistors
814 mm^2
TSMC 4N



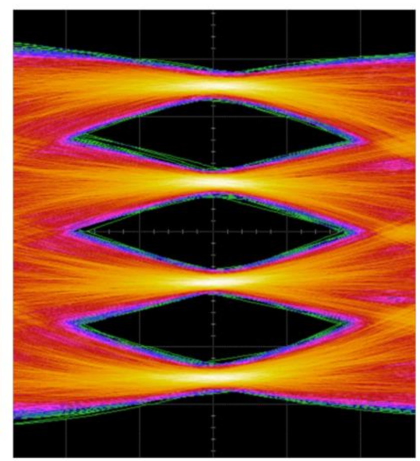
Blackwell
>208 billion transistors
>1600 mm^2
TSMC 4NP



Ampere | NVLink3
12 NVLinks | 50GB/s each
x4@50Gbps-NRZ
600GB/s total



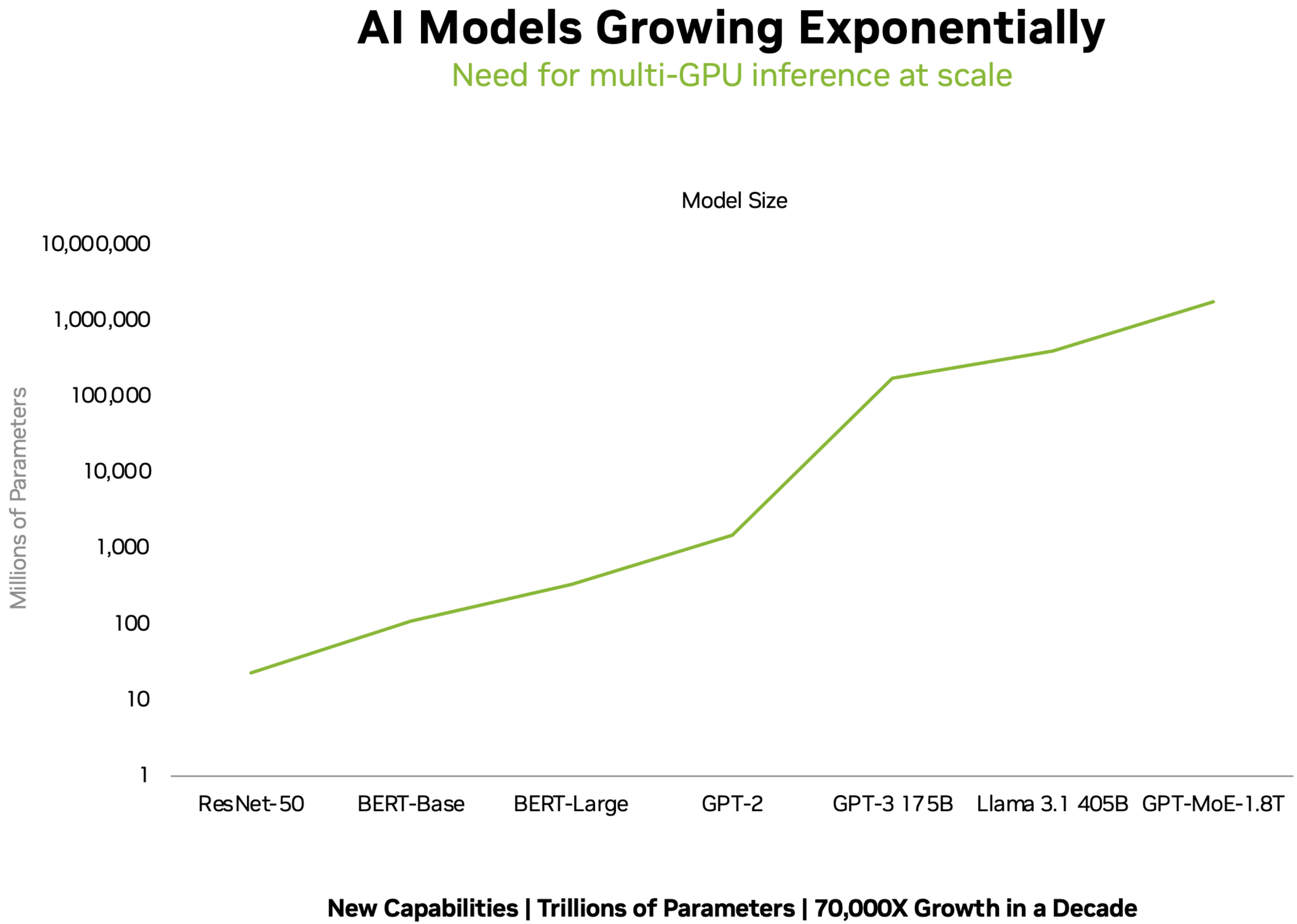
Hopper | NVLink4
18 NVLinks | 50GB/s each
x2@100Gbps-PAM4
900GB/s total



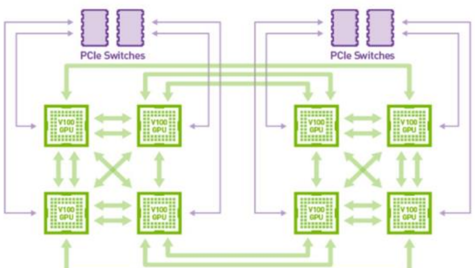
Blackwell | NVLink5
18 NVLinks | 100GB/s each
x2@200Gbps-PAM4
1800GB/s total

NVIDIA Blackwell Platform: Advancing Generative AI and Accelerated Computing, Hot Chips 2024

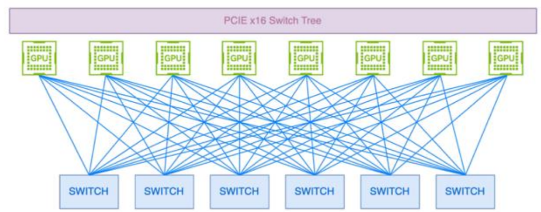
Multiple GPUs to Tackle Large Workloads



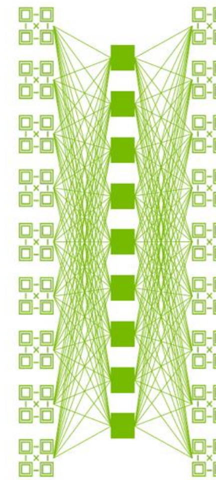
Evolution of Recent GPUs (II)



2016
Hybrid Cube Mesh NVLink technology



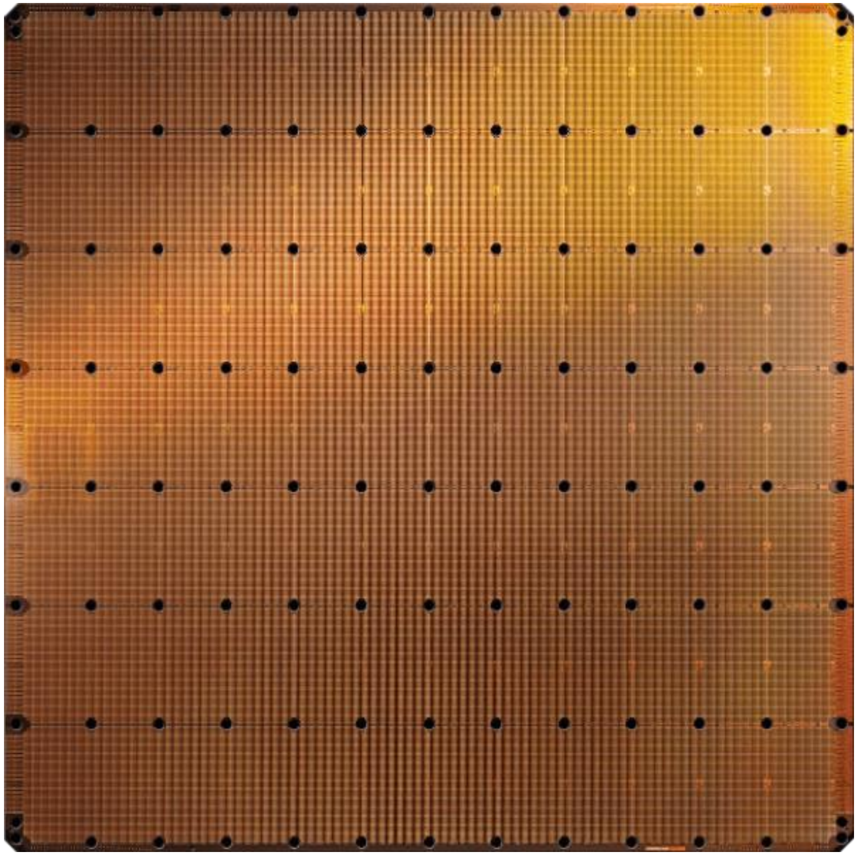
2022
3rd Gen NVLink Switch
All-to-all connection among NVLink domain of 8 GPU



2024
4th Gen NVLink Switch Chip
All-to-all connection among NVLink domain of 72 GPU

NVIDIA Blackwell Platform: Advancing Generative AI and Accelerated Computing, Hot Chips 2024

Different Platforms, Different Goals



- The largest ML accelerator chip (2021)
- 850,000 cores

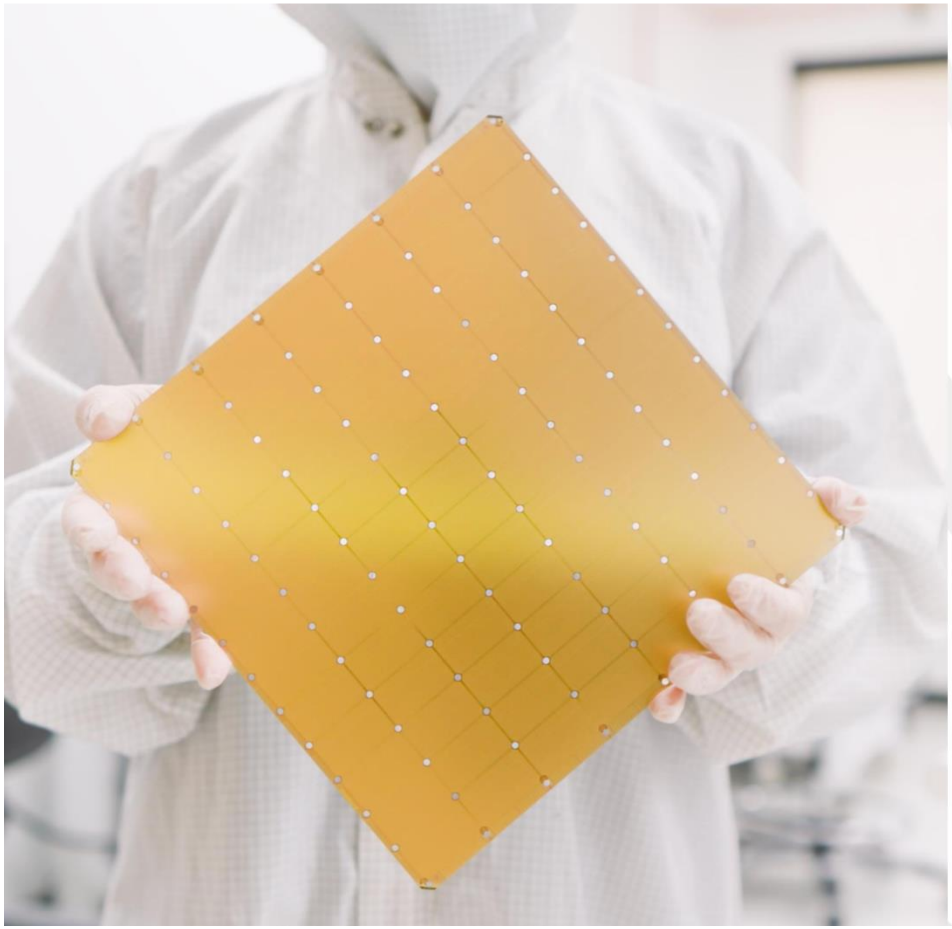
Cerebras WSE-2
2.6 Trillion transistors
46,225 mm^2

Largest GPU
54.2 Billion transistors
826 mm^2
NVIDIA Ampere GA100

https://www.anandtech.com/show/14758/hot-chips-31-live-blogs-cerebras-wafer-scale-deep-learning

https://www.cerebras.net/cerebras-wafer-scale-engine-why-we-need-big-chips-for-deep-learning/

Cerebras's Wafer Scale Engine-3 (2023)



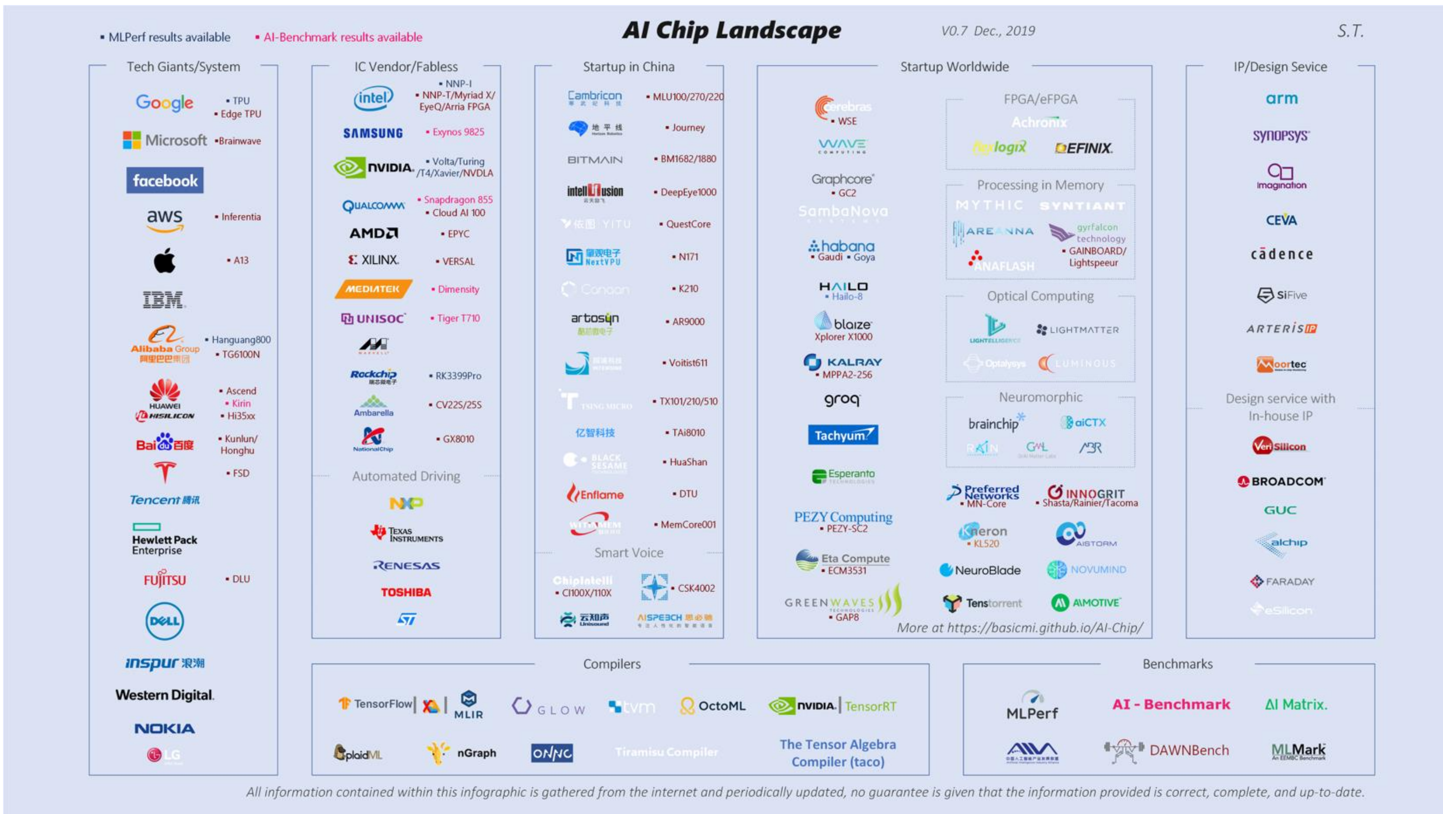
Cerebras Wafer-Scale Engine

The largest chip ever produced

- **46,225 mm²** silicon
- **4 trillion** transistors
- **900,000** AI cores
- **125 Petaflops** of AI compute
- **44 Gigabytes** of on-chip memory
- **21 PByte/s** memory bandwidth
- **214 Pbit/s** fabric bandwidth
- **5nm** TSMC process

Many (Other) (AI/ML) Chips

- Many Companies (including Startups) Building Their Own Chips…



- **Many More to Come…**

Different Platforms, Different Goals

Mohammed Alser, Zülal Bingöl, Damla Senol Cali, Jeremie Kim, Saugata Ghose, Can Alkan, Onur Mutlu
"Accelerating Genome Analysis: A Primer on an Ongoing Journey" IEEE Micro, August 2020.

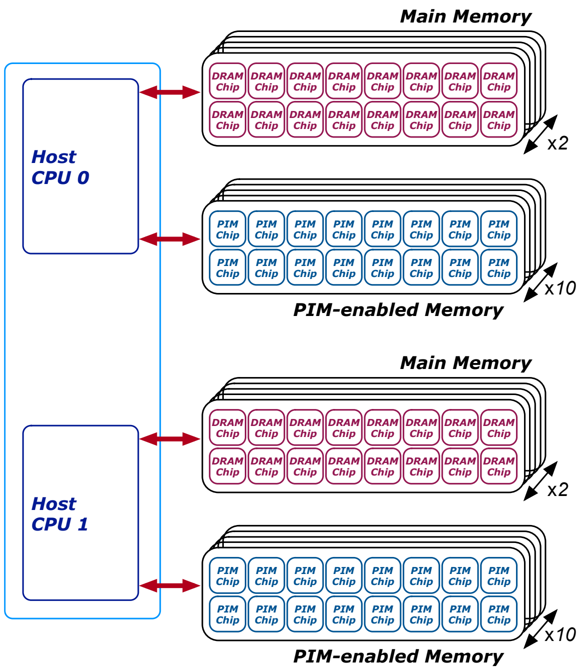
MinION from ONT

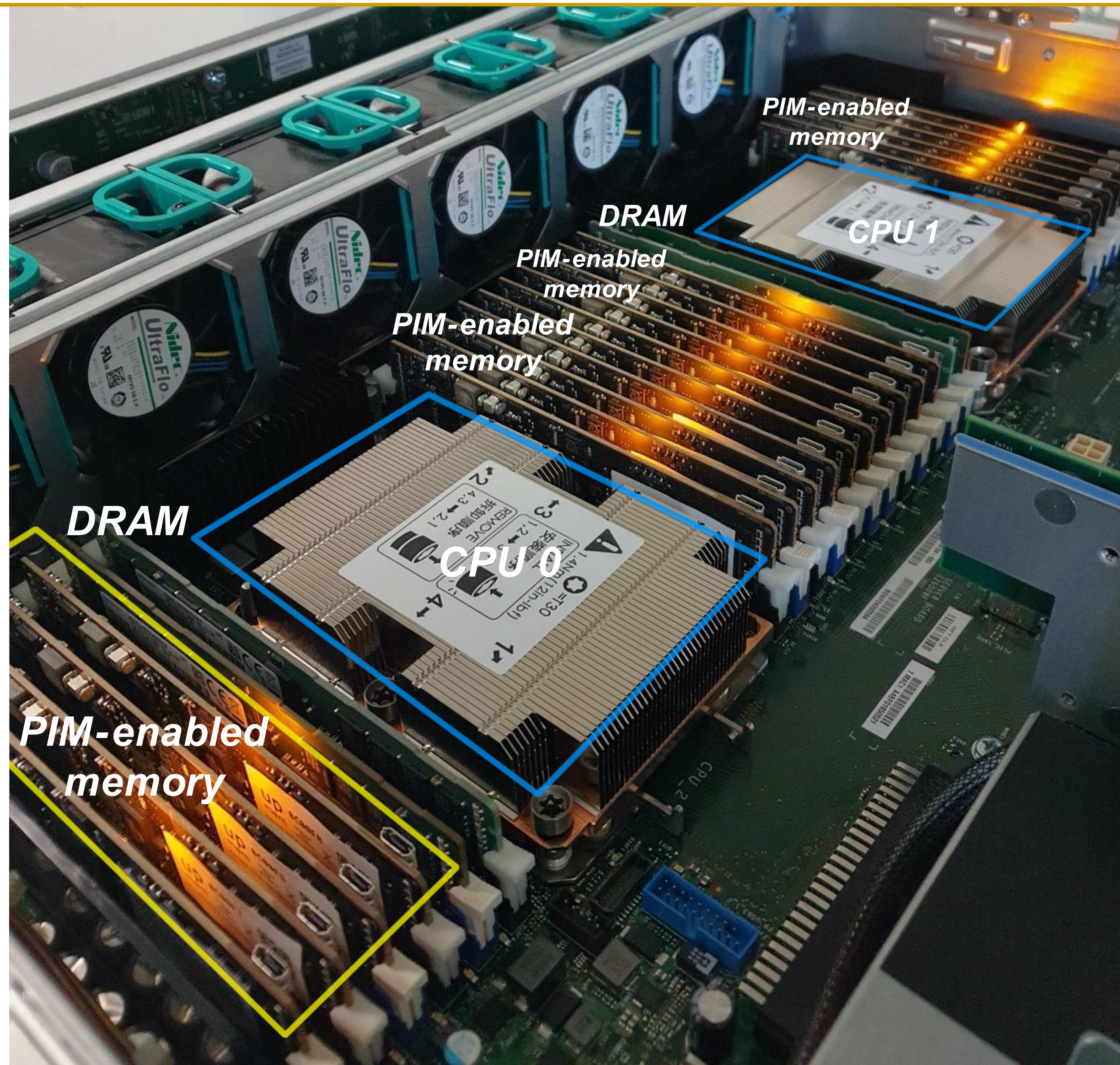
Accelerating Genome Analysis: A Primer on an Ongoing Journey
Sept.-Oct. 2020, pp. 65-75, vol. 40
DOI Bookmark: 10.1109/MM.2020.3013728

FPGA-Based Near-Memory Acceleration of Modern Data-Intensive Applications
July-Aug. 2021, pp. 39-48, vol. 41
DOI Bookmark: 10.1109/MM.2021.3088396

SmidgION from ONT

Different Platforms, Different Goals





Benchmarking a New Paradigm: An Experimental Analysis of a Real Processing-in-Memory Architecture

JUAN GÓMEZ-LUNA, ETH Zürich, Switzerland
IZZAT EL HAJJ, American University of Beirut, Lebanon
IVAN FERNANDEZ, ETH Zürich, Switzerland and University of Malaga, Spain
CHRISTINA GIANNOULA, ETH Zürich, Switzerland and NTUA, Greece
GERALDO F. OLIVEIRA, ETH Zürich, Switzerland
ONUR MUTLU, ETH Zürich, Switzerland

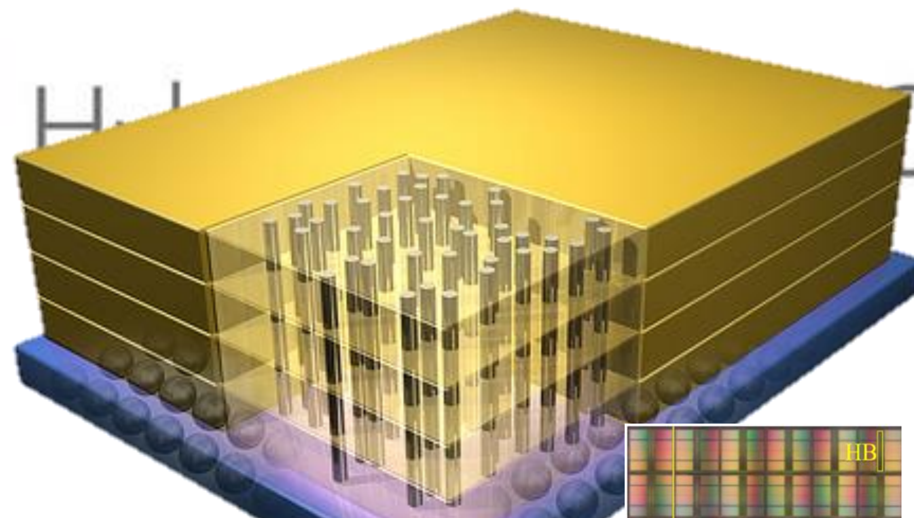
Many modern workloads, such as neural networks, databases, and graph processing, are fundamentally memory-bound. For such workloads, the data movement between main memory and CPU cores imposes a significant overhead in terms of both latency and energy. A major reason is that this communication happens through a narrow bus with high latency and limited bandwidth, and the low data reuse in memory-bound workloads is insufficient to amortize the cost of main memory access. Fundamentally addressing this *data movement bottleneck* requires a paradigm where the memory system assumes an active role in computing by integrating processing capabilities. This paradigm is known as *processing-in-memory* (*PIM*).

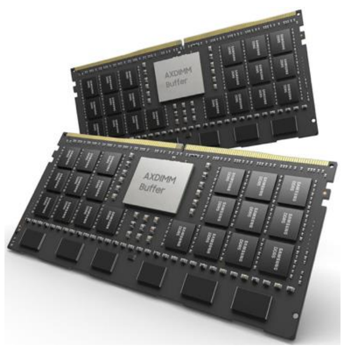
Recent research explores different forms of PIM architectures, motivated by the emergence of new 3D-stacked memory technologies that integrate memory with a logic layer where processing elements can be easily placed. Past works evaluate these architectures in simulation or, at best, with simplified hardware prototypes. In contrast, the UPMEM company has designed and manufactured the first publicly-available real-world PIM architecture. The UPMEM PIM architecture combines traditional DRAM memory arrays with general-purpose in-order cores, called *DRAM Processing Units* (*DPUs*), integrated in the same chip.

This paper provides the first comprehensive analysis of the first publicly-available real-world PIM architecture. We make two key contributions. First, we conduct an experimental characterization of the UPMEM-based PIM system using microbenchmarks to assess various architecture limits such as compute throughput and memory bandwidth, yielding new insights. Second, we present *PrIM* (*Processing-In-Memory benchmarks*), a benchmark suite of 16 workloads from different application domains (e.g., dense/sparse linear algebra, databases, data analytics, graph processing, neural networks, bioinformatics, image processing), which we identify as memory-bound. We evaluate the performance and scaling characteristics of PrIM benchmarks on the UPMEM PIM architecture, and compare their performance and energy consumption to their state-of-the-art CPU and GPU counterparts. Our extensive evaluation conducted on two real UPMEM-based PIM systems with 640 and 2,556 DPUs provides new insights about suitability of different workloads to the PIM system, programming recommendations for software designers, and suggestions and hints for hardware and architecture designers of future PIM systems.

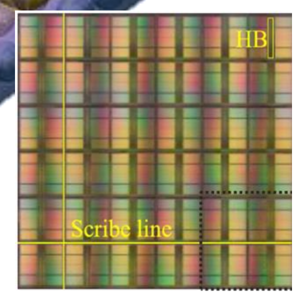
https://arxiv.org/pdf/2105.03814.pdf

Processing-in-Memory Systems (2022)

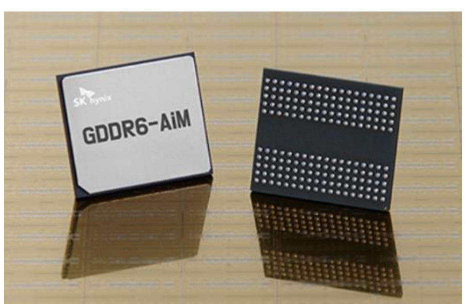




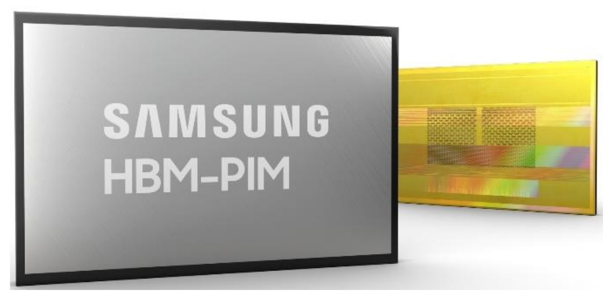
[Samsung 2021]



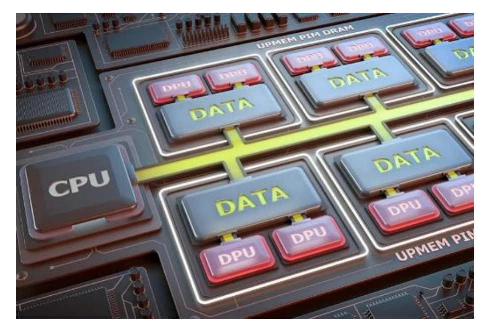
[Alibaba 2022]



[SK Hynix 2022]



[Samsung 2021]

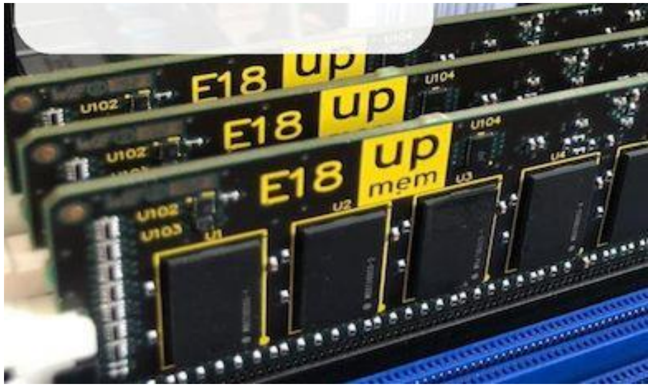


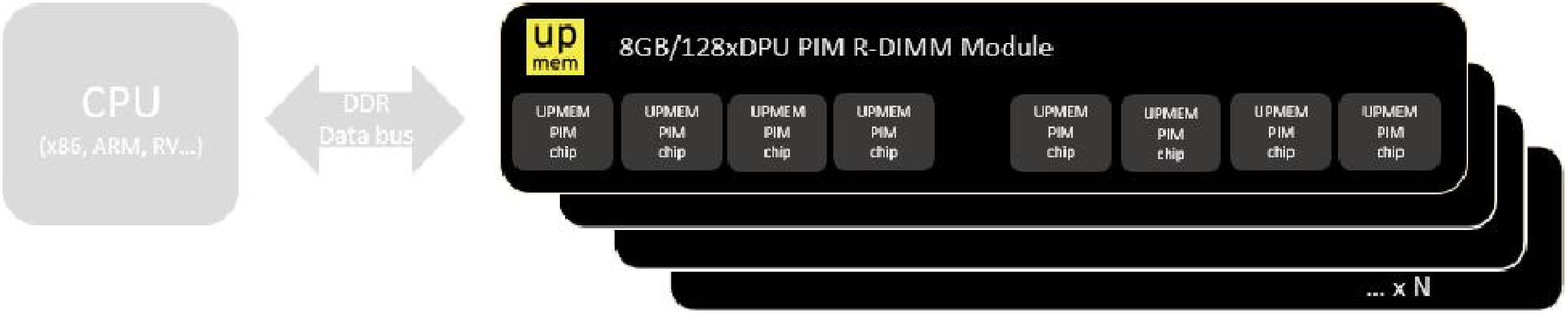
[UPMEM 2019]

And, many other experimental chips and startups

UPMEM Processing-in-DRAM Engine (2019)

- Processing in DRAM Engine
- Includes **standard DIMM modules**, with a **large number of DPU processors** combined with DRAM chips.
- Replaces **standard** DIMMs
  - DDR4 R-DIMM modules
    - 8GB+128 DPUs (16 PIM chips)
    - Standard 2x-nm DRAM process
  - **Large amounts of** compute & memory bandwidth

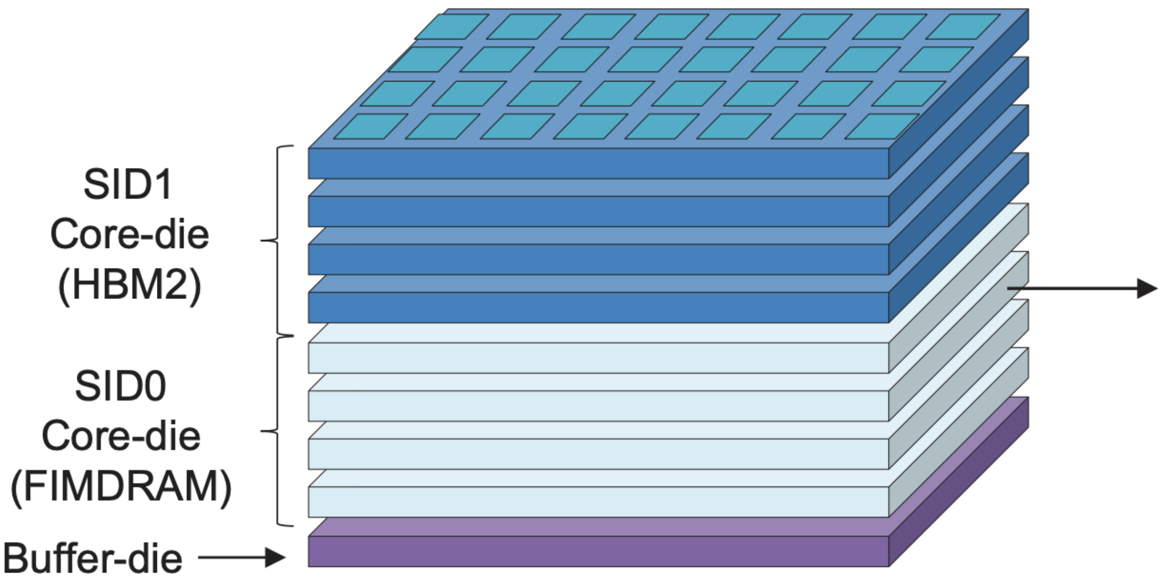




https://www.anandtech.com/show/14750/hot-chips-31-analysis-inmemory-processing-by-upmem
https://www.upmem.com/video-upmem-presenting-its-true-processing-in-memory-solution-hot-chips-2019/

Samsung Function-in-Memory DRAM (2021)

- FIMDRAM based on HBM2



[3D Chip Structure of HBM with FIMDRAM]

Chip Specification
128DQ / 8CH / 16 banks / BL4
32 PCU blocks (1 FIM block/2 banks)
1.2 TFLOPS (4H)
FP16 ADD / Multiply (MUL) / Multiply-Accumulate (MAC) / Multiply-and- Add (MAD)

ISSCC 2021 / SESSION 25 / DRAM / 25.4

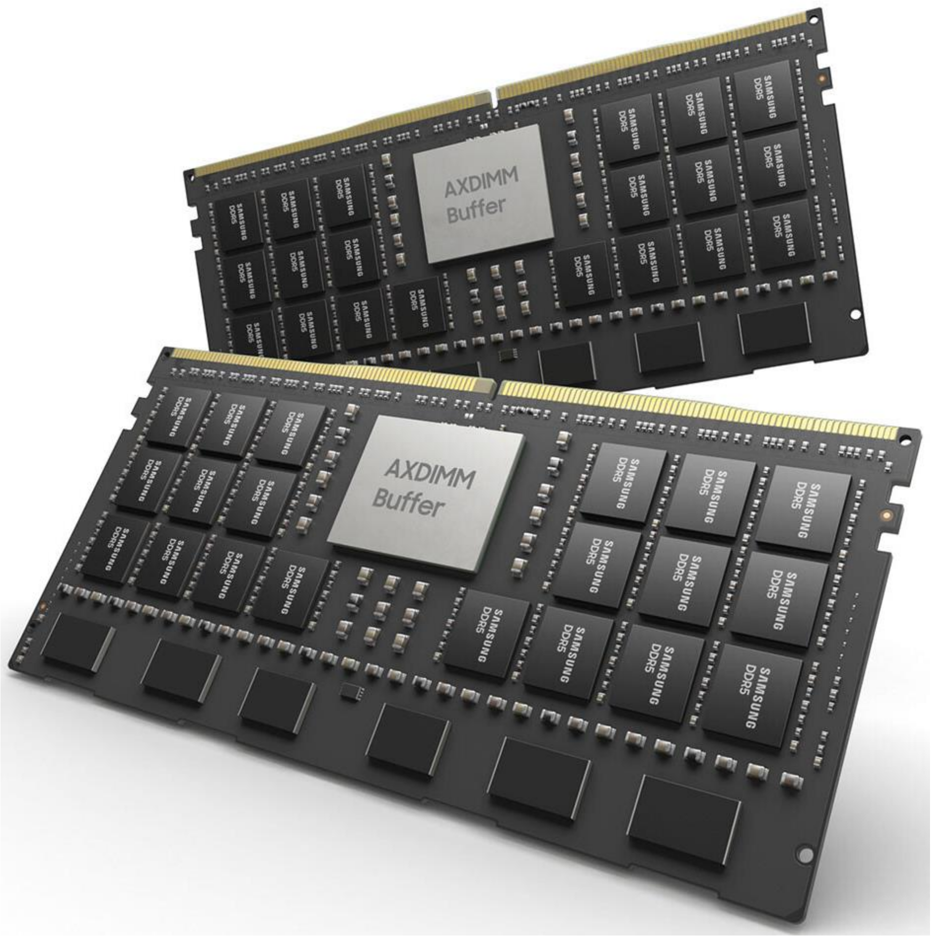
25.4 A 20nm 6GB Function-In-Memory DRAM, Based on HBM2 with a 1.2TFLOPS Programmable Computing Unit Using Bank-Level Parallelism, for Machine Learning Applications

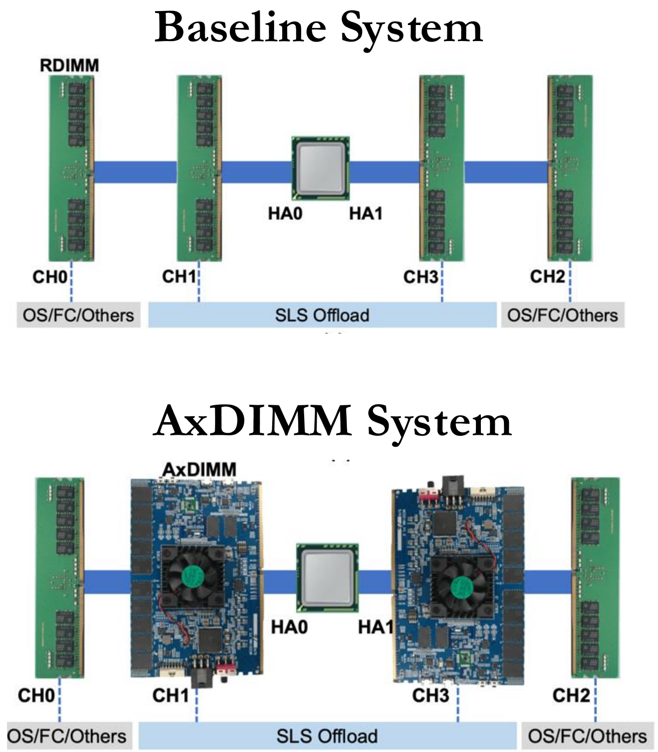
Young-Cheon Kwon[1], Suk Han Lee[1], Jaehoon Lee[1], Sang-Hyuk Kwon[1], Je Min Ryu[1], Jong-Pil Son[1], Seongil O[1], Hak-Soo Yu[1], Haesuk Lee[1], Soo Young Kim[1], Youngmin Cho[1], Jin Guk Kim[1], Jongyoon Choi[1], Hyun-Sung Shin[1], Jin Kim[1], BengSeng Phuah[1], HyoungMin Kim[1], Myeong Jun Song[1], Ahn Choi[1], Daeho Kim[1], SooYoung Kim[1], Eun-Bong Kim[1], David Wang[2], Shinhaeng Kang[1], Yuhwan Ro[3], Seungwoo Seo[3], JoonHo Song[3], Jaeyoun Youn[1], Kyomin Sohn[1], Nam Sung Kim[1]

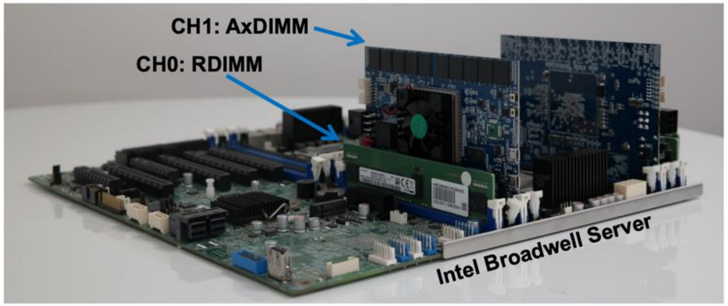
[1]Samsung Electronics, Hwaseong, Korea
[2]Samsung Electronics, San Jose, CA
[3]Samsung Electronics, Suwon, Korea

Samsung AxDIMM (2021)

- DDRx-PIM
  - DLRM recommendation system







Ke et al. "Near-Memory Processing in Action: Accelerating Personalized Recommendation with AxDIMM", IEEE Micro (2021)

SK Hynix Accelerator-in-Memory (2022)

SKhynix NEWSROOM

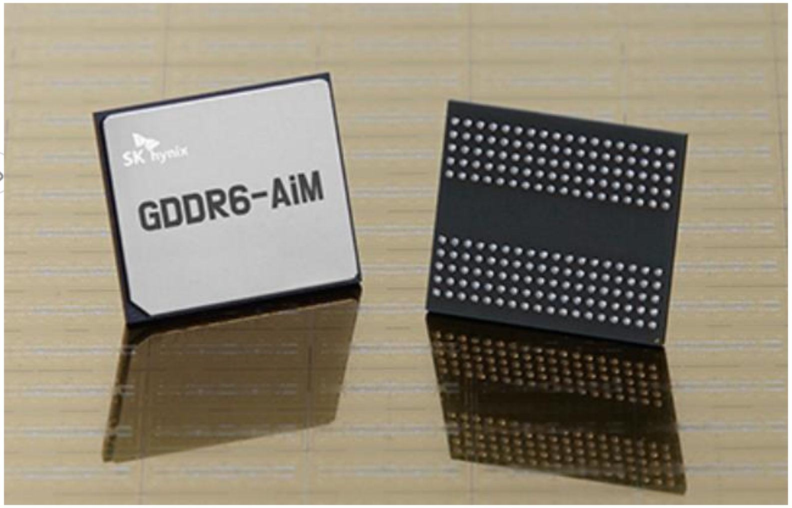
ENG

INSIGHT SK hynix STORY PRESS CENTER MULTIMEDIA

Search

SK hynix Develops PIM, Next-Generation AI Accelerator

February 16, 2022



Seoul, February 16, 2022

SK hynix (or "the Company", www.skhynix.com) announced on February 16 that it has developed PIM*, a next-generation memory chip with computing capabilities.

**PIM(Processing In Memory): A next-generation technology that provides a solution for data congestion issues for AI and big data by adding computational functions to semiconductor memory*

It has been generally accepted that memory chips store data and CPU or GPU, like human brain, process data. SK hynix, following its challenge to such notion and efforts to pursue innovation in the next-generation smart memory, has found a breakthrough solution with the development of the latest technology.

SK hynix plans to showcase its PIM development at the world's most prestigious semiconductor conference, 2022 ISSCC*, in San Francisco at the end of this month. The company expects continued efforts for innovation of this technology to bring the memory-centric computing, in which semiconductor memory plays a central role, a step closer to the reality in devices such as smartphones.

**ISSCC: The International Solid-State Circuits Conference will be held virtually from Feb. 20 to Feb. 24 this year with a theme of "Intelligent Silicon for a Sustainable World"*

For the first product that adopts the PIM technology, SK hynix has developed a sample of GDDR6-AiM (Accelerator* in memory). The GDDR6-AiM adds computational functions to GDDR6* memory chips, which process data at 16Gbps. A combination of GDDR6-AiM with CPU or GPU instead of a typical DRAM makes certain computation speed 16 times faster. GDDR6-AiM is widely expected to be adopted for machine learning, high-performance computing, and big data computation and storage.

11.1 A 1ynm 1.25V 8Gb, 16Gb/s/pin GDDR6-based Accelerator-in-Memory supporting 1TFLOPS MAC Operation and Various Activation Functions for Deep-Learning Applications

Seongju Lee, SK hynix, Icheon, Korea

In Paper 11.1, SK Hynix describes an 1ynm, GDDR6-based accelerator-in-memory with a command set for deep-learning operation. The 8Gb design achieves a peak throughput of 1TFLOPS with 1GHz MAC operations and supports major activation functions to improve accuracy.

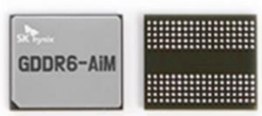
https://news.skhynix.com/sk-hynix-develops-pim-next-generation-ai-accelerator/

SK Hynix Accelerator-in-Memory (2022)

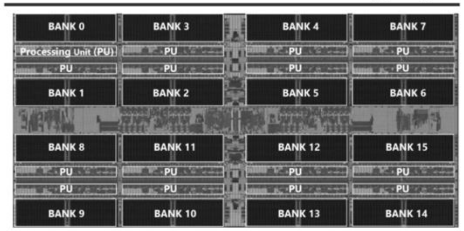
Accelerator-in-Memory: "True All-Bank Parallelism"

SK hynix's First GDDR6-based Processing-in-Memory Product Sample

GDDR6-AiM Package



GDDR6-AiM Die Photograph



GDDR6-AiM Specification* (per die)

(External) Bandwidth**	32 GB/s
Operating Speed	1 GHz
Compute Throughput**	512 GFLOPS
Internal Bandwidth**	512 GB/s
Numeric Precision	BF16

AiMX Card Prototype

AiMX Card Prototype Specification

Host Interface		PCIe Gen3 x8x8 (bifurcated)
Form Factor		FHFL (A100/A30 compatible)
Configuration		2 FPGA*** x 16 AiM package
AiM	Capacity	16 GB
	Bandwidth	170 GB/s (@2.67Gbps****)

(*) S.Lee et al., ISSCC'22
(**) Defined as a peak during burst operations
(***) Xilinx Virtex Ultrascale+ (VU9P)
(****) 1/6 of peak data rate of GDDR6, 16Gbps (or 1TB/s)

AliBaba PIM Recommendation System (2022)

ISSCC 2022 / February 24, 2022 / 8:30 AM

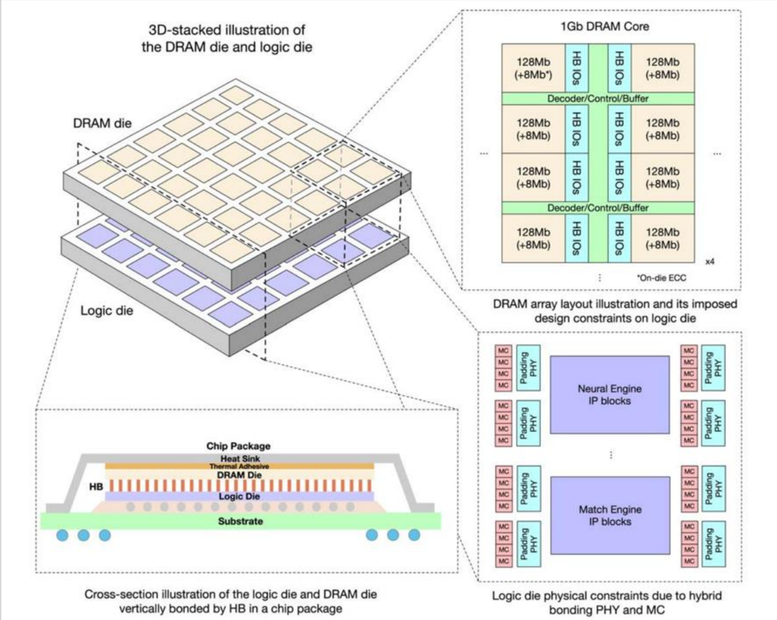
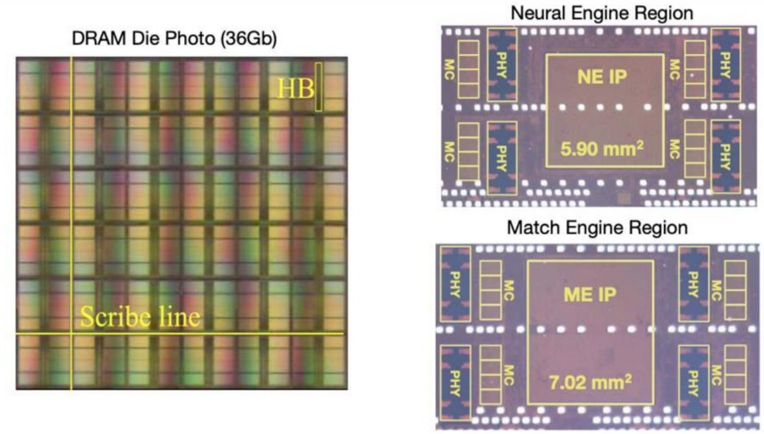


Figure 29.1.2: Illustration of 3D-stacked chip, cross-illustration of package, DRAM array layout and design blocks on logic die.



DRAM Die		
Technology	25nm	
Area	Total*	602.22 mm²
	Neural Engine	32 mm²
	Match Engine	32 mm²
Voltage	1.1 V	
Frequency (max)	150 MHz	
Power	300 mW per 1Gb	
Bandwidth**	153.60 GB/s / 1.38 TB/s	

* Total die size of test chip
** Memory bandwidth of NE and ME / Total bandwidth of DRAM die

Logic Die		
Technology	55nm	
Area	Total*	602.22 mm²
	Neural Engine	5.90 mm²
	Match Engine	7.02 mm²
# of MC	16 per IP	
Voltage	1.2 V	
Frequency	300 MHz	
Power	977.70 mW	
Precision	INT8	

Figure 29.1.7: Die micrographs of DRAM die, NE and ME. Detailed specifications of DRAM die and logic die.

29.1 184QPS/W 64Mb/mm² 3D Logic-to-DRAM Hybrid Bonding with Process-Near-Memory Engine for Recommendation System

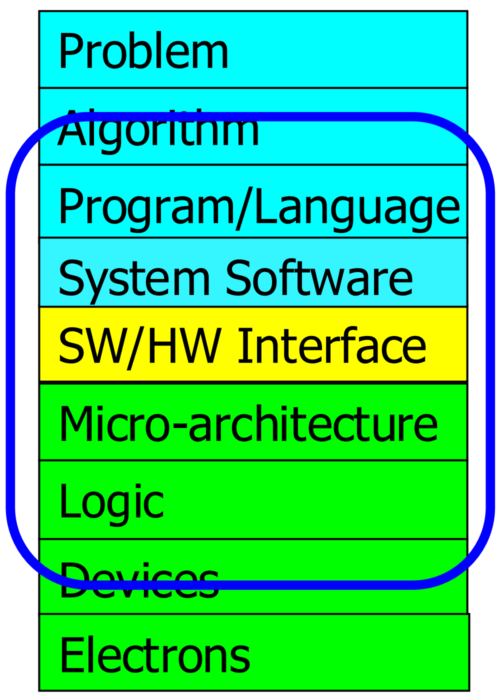
Dimin Niu[1], Shuangchen Li[1], Yuhao Wang[1], Wei Han[1], Zhe Zhang[2], Yijin Guan[2], Tianchan Guan[3], Fei Sun[1], Fei Xue[1], Lide Duan[1], Yuanwei Fang[1], Hongzhong Zheng[1], Xiping Jiang[4], Song Wang[4], Fengguo Zuo[4], Yubing Wang[4], Bing Yu[4], Qiwei Ren[4], Yuan Xie[1]

Axiom

To achieve the highest energy efficiency and performance:

we must take the expanded view

of computer architecture



Co-design across the hierarchy: Algorithms to devices

Specialize as much as possible within the design goals

Why Study Computer Architecture?

What is Computer Architecture?

- The science and art of designing, selecting, and interconnecting hardware components and designing the hardware/software interface to create a computing system that meets functional, performance, energy consumption, cost, and other specific goals.

Why Study Computer Architecture?

- Enable better systems: make computers faster, cheaper, smaller, more reliable, …
  - By exploiting advances and changes in underlying technology/circuits
- Enable new applications
  - Life-like 3D visualization 20 years ago? Virtual reality?
  - Self-driving cars?
  - Personalized genomics? Personalized medicine?
- Enable better solutions to problems
  - Software innovation is built on trends and changes in computer architecture
    - > 50% performance improvement per year has enabled this innovation
- Understand why computers work the way they do

Computer Architecture Today (I)

- Today is a very exciting time to study computer architecture

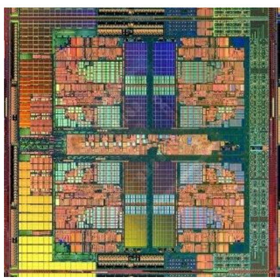
- Industry is in a large paradigm shift (to novel architectures) – many different potential system designs possible

- Many difficult problems *motivating* and *caused by* the shift
  - Huge hunger for data and new data-intensive applications
  - Power/energy/thermal constraints
  - Complexity of design
  - Difficulties in technology scaling
  - Memory bottleneck
  - Reliability problems
  - Programmability problems
  - Security and privacy issues

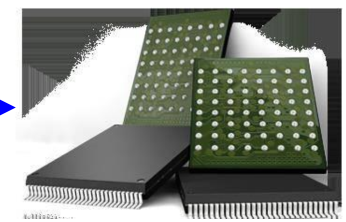
- No clear, definitive answers to these problems

Computer Architecture Today (II)

- Computing landscape is very different from 10-20 years ago
- Applications & technology & requirements all demand novel architectures

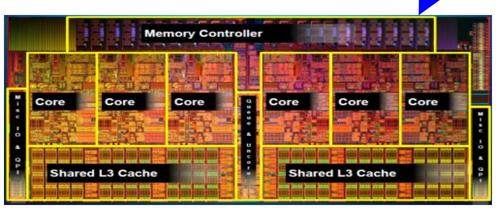


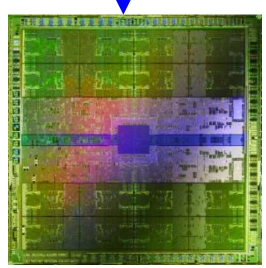
Hybrid Main Memory



Persistent Memory/Storage

Heterogeneous Processors and Accelerators

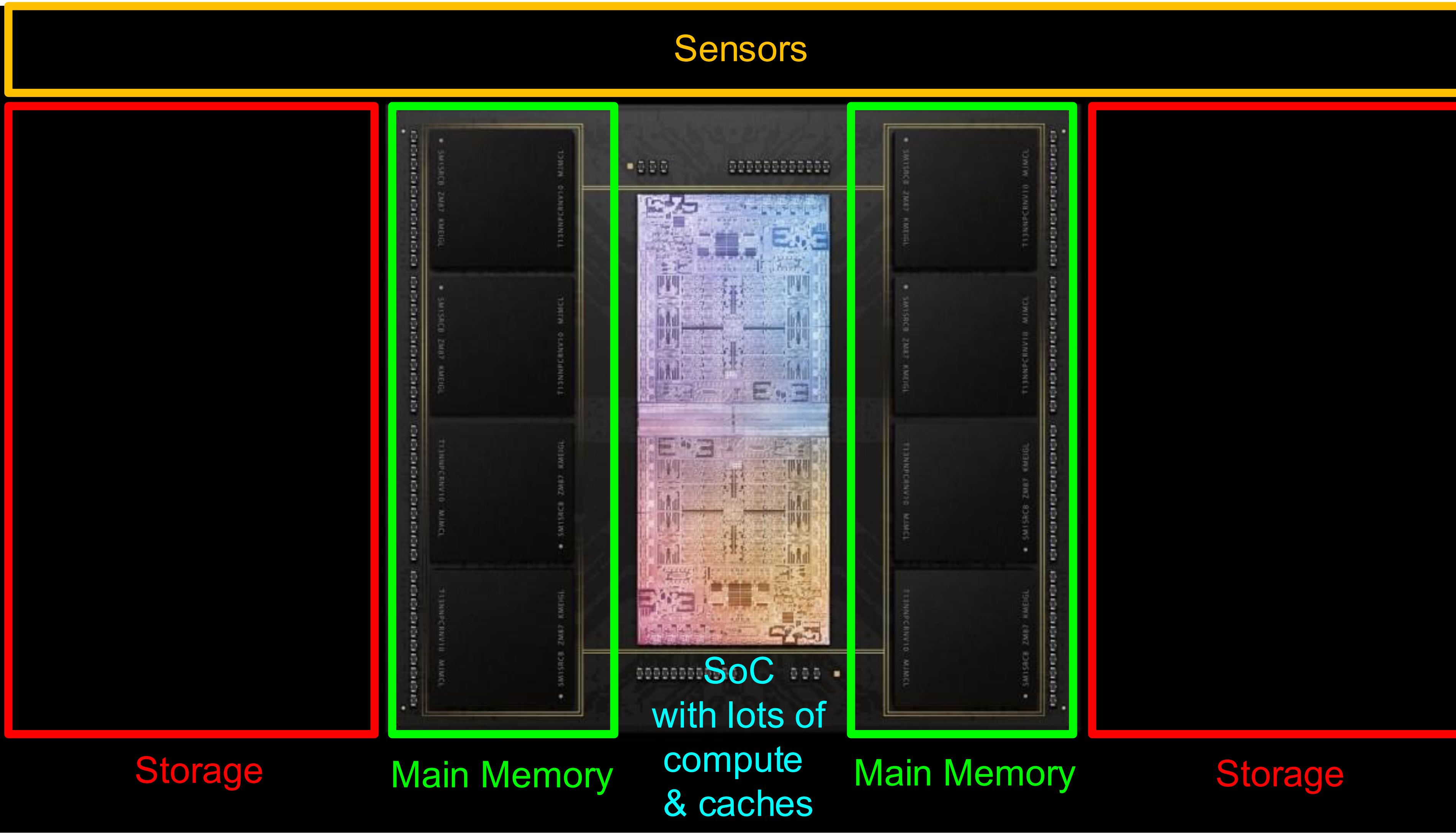




General Purpose GPUs

Every component and its interfaces, as well as entire system designs are being re-examined

An Example System in Your Pocket



Apple M1 Ultra System (2022)

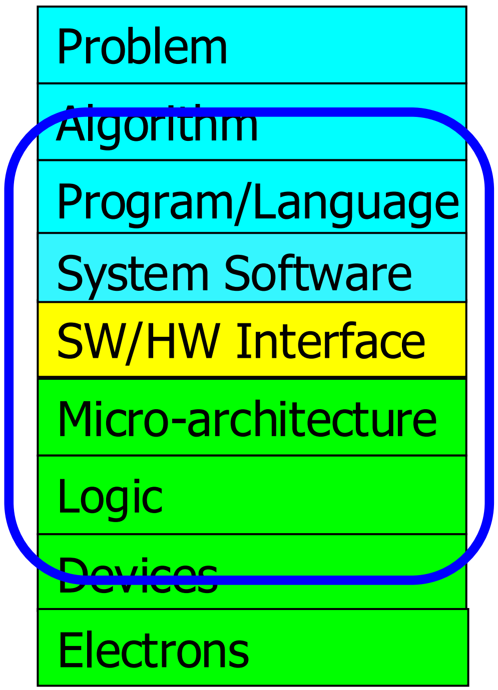
https://www.gsmarena.com/apple_announces_m1_ultra_with_20core_cpu_and_64core_gpu-news-53481.php

Axiom

To achieve the highest energy efficiency and performance:

we must take the expanded view

of computer architecture



Co-design across the hierarchy: Algorithms to devices

Specialize as much as possible within the design goals

Historical: Opportunities at the Bottom

There's Plenty of Room at the Bottom

From Wikipedia, the free encyclopedia

"**There's Plenty of Room at the Bottom: An Invitation to Enter a New Field of Physics**" was a lecture given by physicist Richard Feynman at the annual American Physical Society meeting at Caltech on December 29, 1959.[1] Feynman considered the possibility of direct manipulation of individual atoms as a more powerful form of synthetic chemistry than those used at the time. Although versions of the talk were reprinted in a few popular magazines, it went largely unnoticed and did not inspire the conceptual beginnings of the field. Beginning in the 1980s, nanotechnology advocates cited it to establish the scientific credibility of their work.

https://en.wikipedia.org/wiki/There%27s_Plenty_of_Room_at_the_Bottom

Historical: Opportunities at the Bottom (II)

There's Plenty of Room at the Bottom

From Wikipedia, the free encyclopedia

Feynman considered some ramifications of a general ability to manipulate matter on an atomic scale. He was particularly interested in the possibilities of denser computer circuitry, and microscopes that could see things much smaller than is possible with scanning electron microscopes. These ideas were later realized by the use of the scanning tunneling microscope, the atomic force microscope and other examples of scanning probe microscopy and storage systems such as Millipede, created by researchers at IBM.

Feynman also suggested that it should be possible, in principle, to make nanoscale machines that "arrange the atoms the way we want", and do chemical synthesis by mechanical manipulation.

He also presented the possibility of "swallowing the doctor", an idea that he credited in the essay to his friend and graduate student Albert Hibbs. This concept involved building a tiny, swallowable surgical robot.

https://en.wikipedia.org/wiki/There%27s_Plenty_of_Room_at_the_Bottom

Historical: Opportunities at the Top

REVIEW

There's plenty of room at the Top: What will drive computer performance after Moore's law?

Charles E. Leiserson[1], Neil C. Thompson[1,2,*], Joel S. Emer[1,3], Bradley C. Kuszmaul[1,†], Butler W. Lampson[1,4], ...
+ See all authors and affiliations

Science 05 Jun 2020:
Vol. 368, Issue 6495, eaam9744
DOI: 10.1126/science.aam9744

Much of the improvement in computer performance comes from decades of miniaturization of computer components, a trend that was foreseen by the Nobel Prize–winning physicist Richard Feynman in his 1959 address, "There's Plenty of Room at the Bottom," to the American Physical Society. In 1975, Intel founder Gordon Moore predicted the regularity of this miniaturization trend, now called Moore's law, which, until recently, doubled the number of transistors on computer chips every 2 years.

Unfortunately, semiconductor miniaturization is running out of steam as a viable way to grow computer performance—there isn't much more room at the "Bottom." If growth in computing power stalls, practically all industries will face challenges to their productivity. Nevertheless, opportunities for growth in computing performance will still be available, especially at the "Top" of the computing-technology stack: software, algorithms, and hardware architecture.

https://www.science.org/doi/10.1126/science.aam9744

Axiom, Revisited

There is plenty of room both at the top and at the bottom

but **much more so**

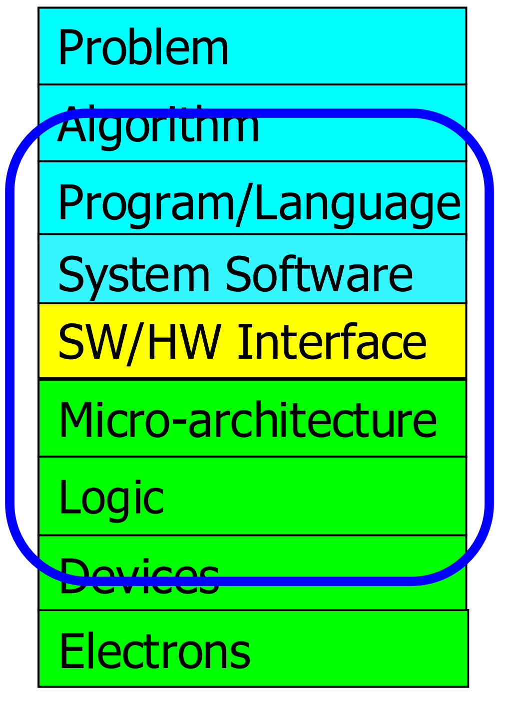
when you

communicate well between and optimize across

the top and the bottom

Hence the Expanded View

Computer Architecture (expanded view)



A Note on Hardware vs. Software

- This course might seem like it is only "Computer Hardware"

- However, you will be much more capable if you master both hardware and software (and the interface between them)
  - Can develop better software if you understand the hardware
  - Can design better hardware if you understand the software
  - Can design a better computing system if you understand both

- This course covers the HW/SW interface and microarchitecture
  - We will focus on tradeoffs and how they affect software

- Recall the example chips & platforms we surveyed

Course Info and Logistics

Brief Self Introduction

- Onur Mutlu
  - Full Professor @ ETH Zurich ITET (INFK), since Sept 2015
  - Strecker Professor @ Carnegie Mellon University ECE (CS), 2009-2016, 2016-…
  - Started the Comp Arch Research Group @ Microsoft Research, 2006-2009
  - Worked @ Google, VMware, Microsoft Research, Intel, AMD
  - PhD in Computer Engineering from University of Texas at Austin in 2006
  - BS in Computer Engineering & Psychology from University of Michigan in 2000
  - https://people.inf.ethz.ch/omutlu/ omutlu@gmail.com

- Research and Teaching in:
  - **Computer architecture, systems, hardware security, bioinformatics**
  - Memory and storage systems
  - Robust & dependable hardware systems: security, safety, predictability, reliability
  - Hardware/software cooperation
  - New computing paradigms; architectures with emerging technologies/devices
  - Architectures for bioinformatics, genomics, health, medicine, AI/ML
  - …

My Co-Instructor

- Dr. Mohammad Sadrosadati
  - Senior Researcher and Lecturer @ SAFARI Research Group, ETHZ
  - PhD in Computer Engineering from Sharif University of Technology in 2019
  - MS in Computer Engineering from Sharif University of Technology in 2014
  - BS in Computer Engineering from Sharif University of Technology in 2012
  - mohammad.sadrosadati@safari.ethz.ch

- Research & Teaching Areas
  - Computer Architecture
  - Memory/Storage Systems
  - Near-Data Processing
  - Heterogeneous System Architecture
  - Bioinformatics
  - Interconnection Network

Our Head Teaching Assistant (I)

- Dr. Konstantina Koliogeorgi
  - Head Teaching Assistant
  - Senior Lecturer & Researcher @ SAFARI
  - PhD from National Technical University of Athens in 2023
  - kkoliogeorgi@safari.ethz.ch

- Research & Teaching Areas
  - Hardware/Software Co-Design
  - Heterogeneous System Architecture
  - Reconfigurable Computing and Architectures
  - Hardware Acceleration
  - Optimized Architectures for Genome analysis
  - High Level Synthesis Tools
  - Design Space Exploration

Our Head Teaching Assistant (II)

- Dr. Zhiheng Yue
  - Head Teaching Assistant
  - Senior Lecturer & Researcher @ SAFARI
  - PhD from Tsinghua University in 2024
  - zhiyue@ethz.ch

- Research & Teaching Areas
  - Processing-in-memory
  - SRAM/DRAM
  - 3D Integration
  - Design technology co-optimization
  - AI Accelerator
  - Physically unclonable function

Our Head Teaching Assistant for Labs (I)

- Ataberk Olgun
  - Head Teaching Assistant (Labs)
  - Senior PhD Student @ SAFARI
  - https://ataberkolgun.com/
  - aolgun@ethz.ch

- Research and Teaching Areas
  - Computer Architecture
  - Memory Systems
  - DRAM
  - Hardware Robustness (Security, Safety, Reliability, Availability)
  - FPGAs
  - Hardware/Software Cooperation

Our Head Teaching Assistant for Labs (II)

- Mayank Kabra
  - Head Teaching Assistant (Labs)
  - PhD Student @ SAFARI
  - kabram@ethz.ch

- Research and Teaching Areas
  - Privacy Enhancing Technologies
  - Memory and Storage-Centric Computing
  - Cryptographic Algorithm-Hardware Codesign
  - Memory Technologies (DRAM, SSD)
  - Computer Architecture

PhD/MSc Assistants

- (Other) Key Assistants and Guest Lecturers
  - Konstantinos Kanellopoulos
  - Nika Mansouri Ghiasi
  - Rakesh Nadig
  - Nisa Bostancı
  - İsmail Emir Yüksel
  - Haocong Luo
  - Andreas Kakolyris
  - Jikun Wang
  - Harshita Gupta
  - Konstantinos Sgouras
  - Harsh Songara
  - Spiros Galanopoulos

Student Assistants

- Maria Makeenkova
- Sezen Burak
- Cremer Paul
- Stadler Nicole
- König Amir
- Rüegg Colin
- Hanak Simon
- von Känel Elena
- Dogan Ebruli
- Drozhilkin Stanislav
- Durrant Joshua
- Schneider Julius
- Sheikh Mohamud Ahmed Hafsa
- Schweizer Constantin
- Pätzold Henrik
- Luchiean Tudor
- Löpfe Noi Finn

Course Info: Lab Assistants (I)

- Tuesday 16-18
  - TBD
- Wednesday 16-18
  - TBD

Course Info: Lab Assistants (II)

- Friday 8-10
  - TBD
- Friday 10-12
  - TBD

If You Need Help

- Post your question on Moodle Q&A Forum
  - https://moodle-app2.let.ethz.ch/course/view.php?id=27943
  - We will create a forum on Moodle for each activity
  - **Preferred** for **technical** questions

- Write an e-mail to:
  - digitaltechnik@lists.inf.ethz.ch
  - The instructor and all assistants will receive this e-mail

- Come to office hours
  - We will provide office locations & Zoom links
  - TBD

Where to Get Up-to-date Course Info?

- Website:
  - **https://safari.ethz.ch/ddca/spring2026/**
  - Lecture slides and videos
  - Readings
  - Lab information
  - Course schedule, handouts, FAQs
  - Software
  - Any other useful information for the course
  - Check frequently for announcements and due dates
  - This is your single point of access to all resources

- Your ETH Email
- Lecturers and Teaching Assistants

Lecture and Lab Times and Policies

- Lectures:
  - Thursday and Fridays, 14:00-16:00
  - YouTube livestream playlist: https://www.youtube.com/playlist?list=PL5Q2soXY2Zi-yo9kK-BKrq11ykNKkVEpd
  - Zoom link provided via Moodle
  - Attendance is for your benefit and is therefore important
  - Some days, we may have guest lectures and exercise sessions

- Lab sessions:
  - See online
  - You should definitely attend the lab sessions
    - In-class evaluation (70%) and mandatory lab reports (30%)
  - Labs will start on March 3rd
  - Lab information and handouts are here:
    - https://safari.ethz.ch/ddca/spring2026/doku.php?id=labs

Lab Organization (I)

- Groups
  - Choose your preferred group in Moodle
    - Form link to be announced on Moodle later today
    - Due 01.03.2026 at 11:59pm
  - Choose your partner
    - Form link to be announced on Moodle later today
    - Due 25.02.2026 at 11:59pm
    - Fill this form ONLY if you have a partner
      - We will assign you a random partner if you do not fill the form

Lab Organization (II)

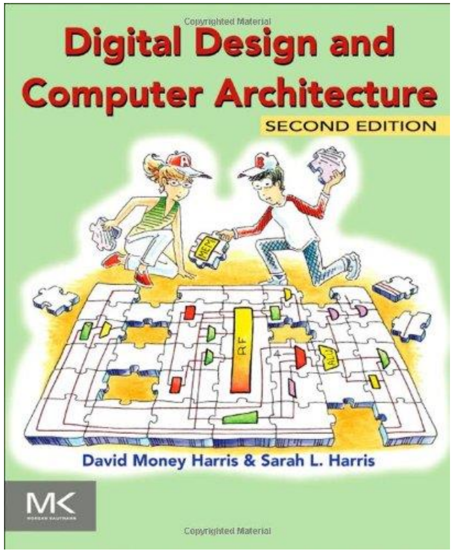
- Lab grades from previous years
  - Form link to be announced on Moodle later today
  - Choose one of the options (due 01.03.2026 at 11:59pm):
    - 1) I will use my lab grades from previous years, and I won't do the labs this year
    - 2) I will use my lab grades from previous years, but I will do the labs this year
    - 3) I won't use my lab grades from previous years. I will do the labs this year

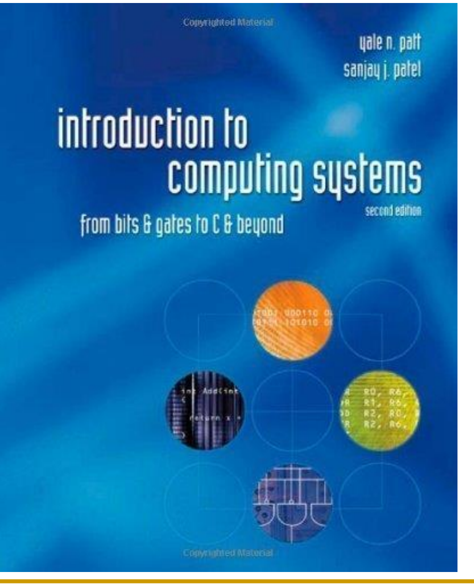
Final Exam

- 180-minute written exam
  - Find examination rules in Course Catalogue
    - https://www.vvz.ethz.ch/Vorlesungsverzeichnis/lerneinheit.view?lerneinheitId=188955&semkez=2025S&ansicht=LEHRVERANSTALTUNGEN&lang=en
  - Also: study the first pages of previous exams
    - https://safari.ethz.ch/digitaltechnik/spring2023/lib/exe/fetch.php?media=ddca-s23-en.pdf
  - Some exam questions are similar to questions in Optional HWs and Past Exams
    - Optional HWs are not graded, but highly recommended to solve
    - Solving past exams could also be useful

Reading Assignments for This/Next Week

- Chapters 1-2 in Harris & Harris





- Chapters 1,2,3 in Patt and Patel

- Supplementary Lecture Slides on Binary Numbers

Reading Assignments for This/Next Week

- This week
  - Introduction
    - P&P Chapters 1 & 2 + H&H Chapter 1
  - Combinational Logic
    - P&P Chapter 3 until 3.3 + H&H Chapter 2

- Next week
  - Hardware Description Languages and Verilog
    - H&H Chapter 4 until 4.3 and 4.5
  - Sequential Logic
    - P&P Chapter 3.4 until end + H&H Chapter 3 in full

- Within 2-3 weeks, we will be done with
  - **P&P Chapters 1-3 + H&H Chapters 1-4**

Extra Credit Assignment: Talk Analysis

- The Story of RowHammer, RowPress & Beyond
- **Watch and analyze this short lecture (30 minutes)**
  - https://www.youtube.com/watch?v=U1EcqXlclKU (June 2024)

- **Assignment – for 1% extra credit**
  - **Write a good 1-page individualized summary (no AI use)**
    - What are your key takeaways? What did you learn?
    - What surprised you about the content presented? What excited you?
    - What do you think solutions should be like?
    - Submit your summary to Moodle – deadline March 21

Future Lectures and Assignments

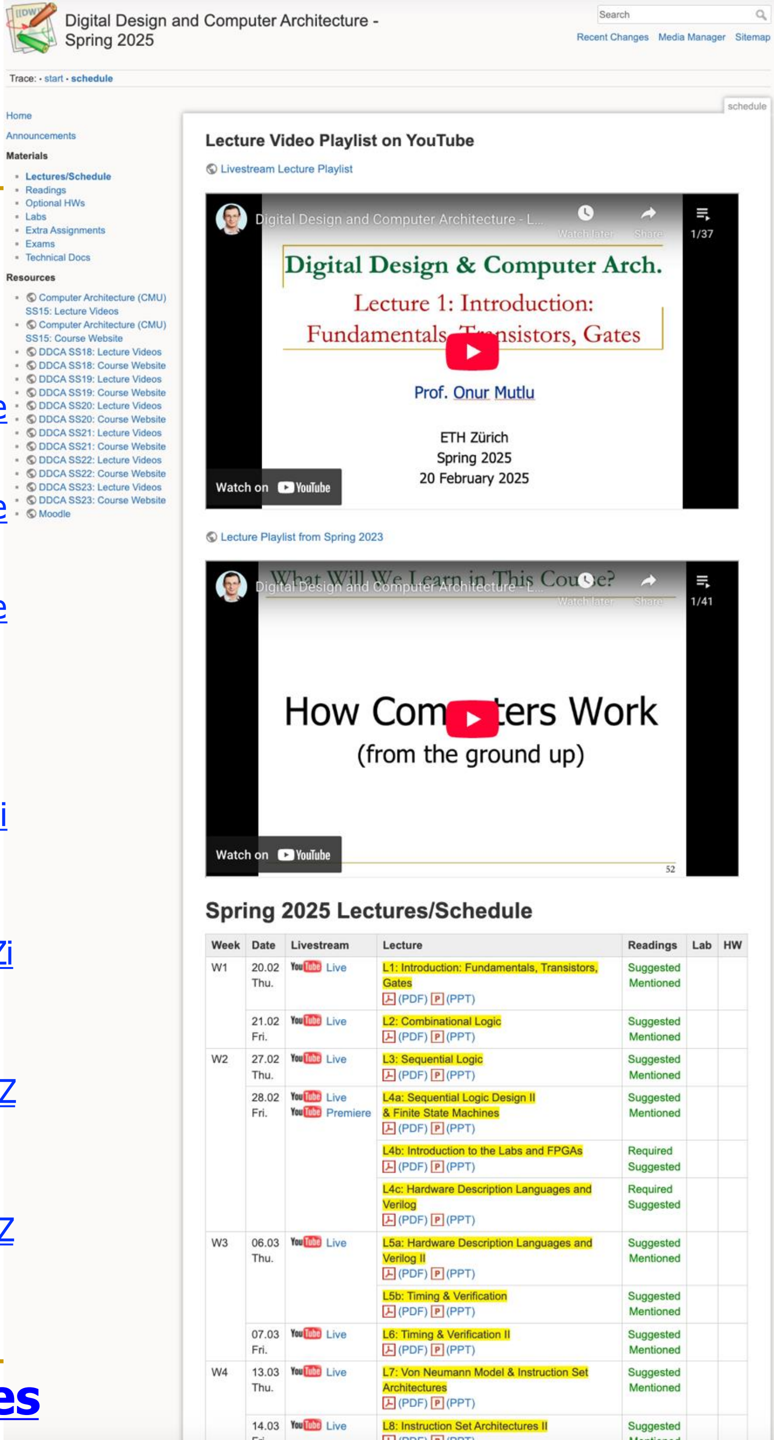
- You can also anticipate (and plan for) future lectures and assignments based on the Spring 2025 schedule:
  - https://safari.ethz.ch/ddca/spring2025/doku.php?id=schedule
  - https://www.youtube.com/watch?v=ubhxKNlOlRg&list=PL5Q2soXY2Zi9Eo29LMgKVcaydS7V1zZW3&index=1

- An example of "Last Time Prediction"
  - Speculative Execution
    - The concept of doing something before knowing it is needed
  - A key concept we will cover in the design of microprocessors

DDCA (2025-2021)

- **[Spring 2025 Edition:](https://safari.ethz.ch/ddca/spring2025/doku.php?id=schedule)**
  - https://safari.ethz.ch/ddca/spring2025/doku.php?id=schedule
- **Spring 2023 Edition:**
  - https://safari.ethz.ch/digitaltechnik/spring2023/doku.php?id=schedule
- **Spring 2022 Edition:**
  - https://safari.ethz.ch/digitaltechnik/spring2022/doku.php?id=schedule
- **Spring 2021 Edition:**
  - https://safari.ethz.ch/digitaltechnik/spring2021/doku.php?id=schedule

- **Youtube Livestream (Spring 2025):**
  - https://www.youtube.com/watch?v=ubhxKNlOlRg&list=PL5Q2soXY2Zi9Eo29LMgKVcaydS7V1zZW3
- **Youtube Livestream (Spring 2023):**
  - https://www.youtube.com/watch?v=VcKjvwD930o&list=PL5Q2soXY2Zi-EImKxYYY1SZuGiOAOBKaf
- **Youtube Livestream (Spring 2022):**
  - https://www.youtube.com/watch?v=cpXdE3HwvK0&list=PL5Q2soXY2Zi97Ya5DEUpMpO2bbAoaG7c6
- **Youtube Livestream (Spring 2021):**
  - https://www.youtube.com/watch?v=LbC0EZY8yw4&list=PL5Q2soXY2Zi_uej3aY39YB5pfW4SJ7LlN



https://www.youtube.com/onurmutlulectures

Fundamental Concepts

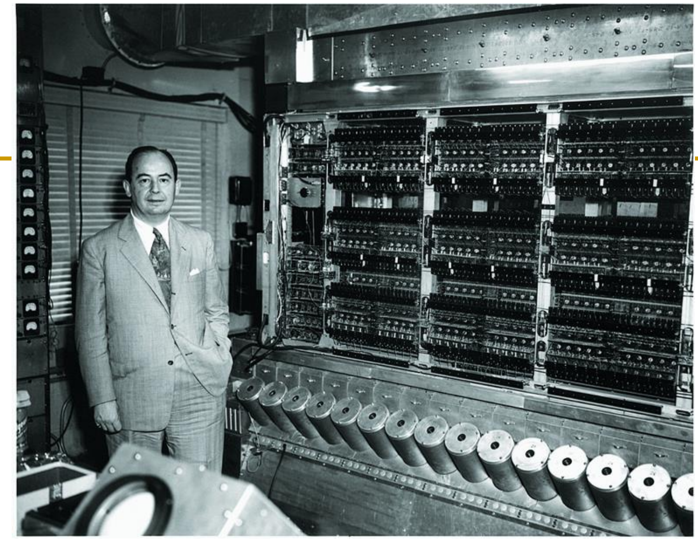
First …

- Let’s understand the fundamentals…

- You can change the world only if you understand it well enough…
  - Especially the basics (fundamentals)
  - Past and present dominant paradigms
  - And, their advantages and shortcomings – tradeoffs
  - And, what remains fundamental across generations
  - And, what techniques you can use and develop to solve problems

What is A Computer?

- Three key components
- Computation
- Communication
- Storage/memory



Burks, Goldstein, von Neumann, "Preliminary discussion of the logical design of an electronic computing instrument," 1946.

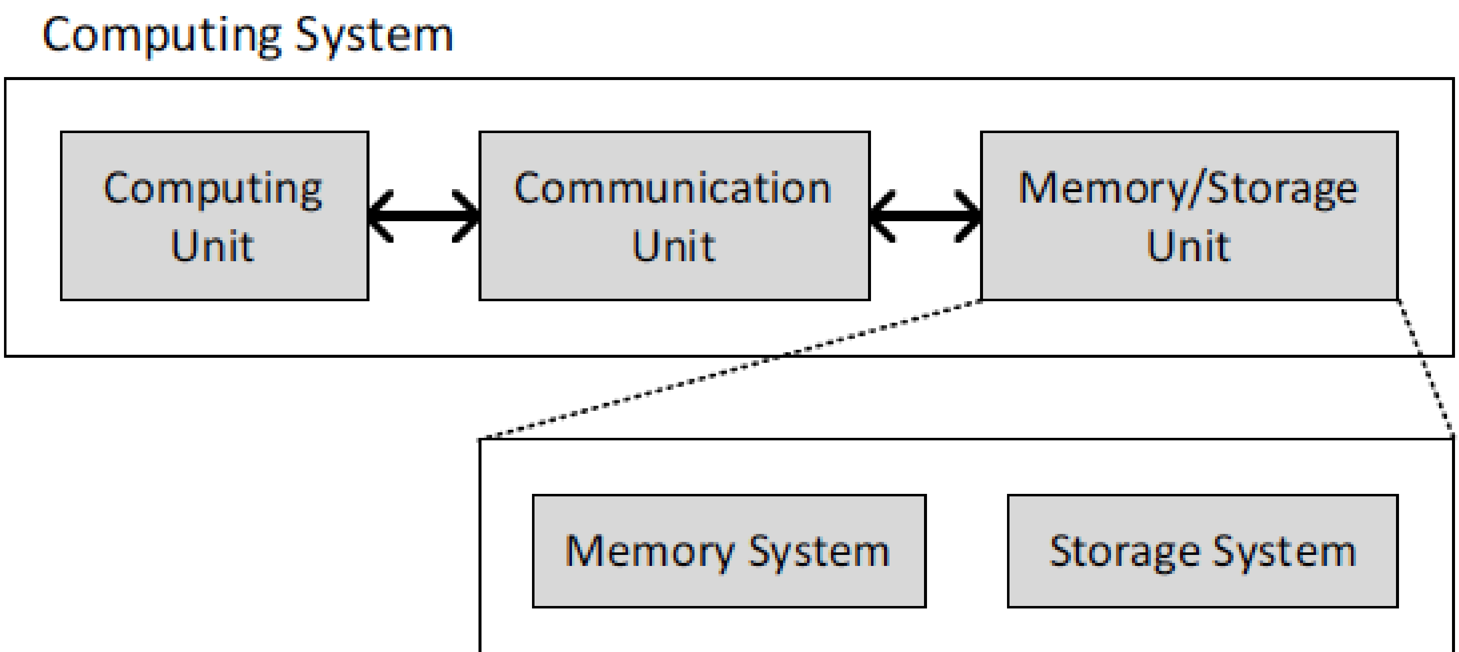


Image source: https://lbsitbytes2010.wordpress.com/2013/03/29/john-von-neumann-roll-no-15/

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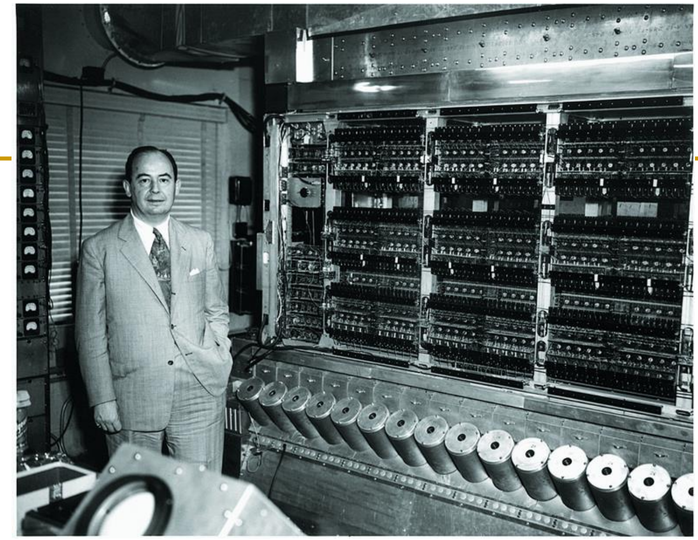

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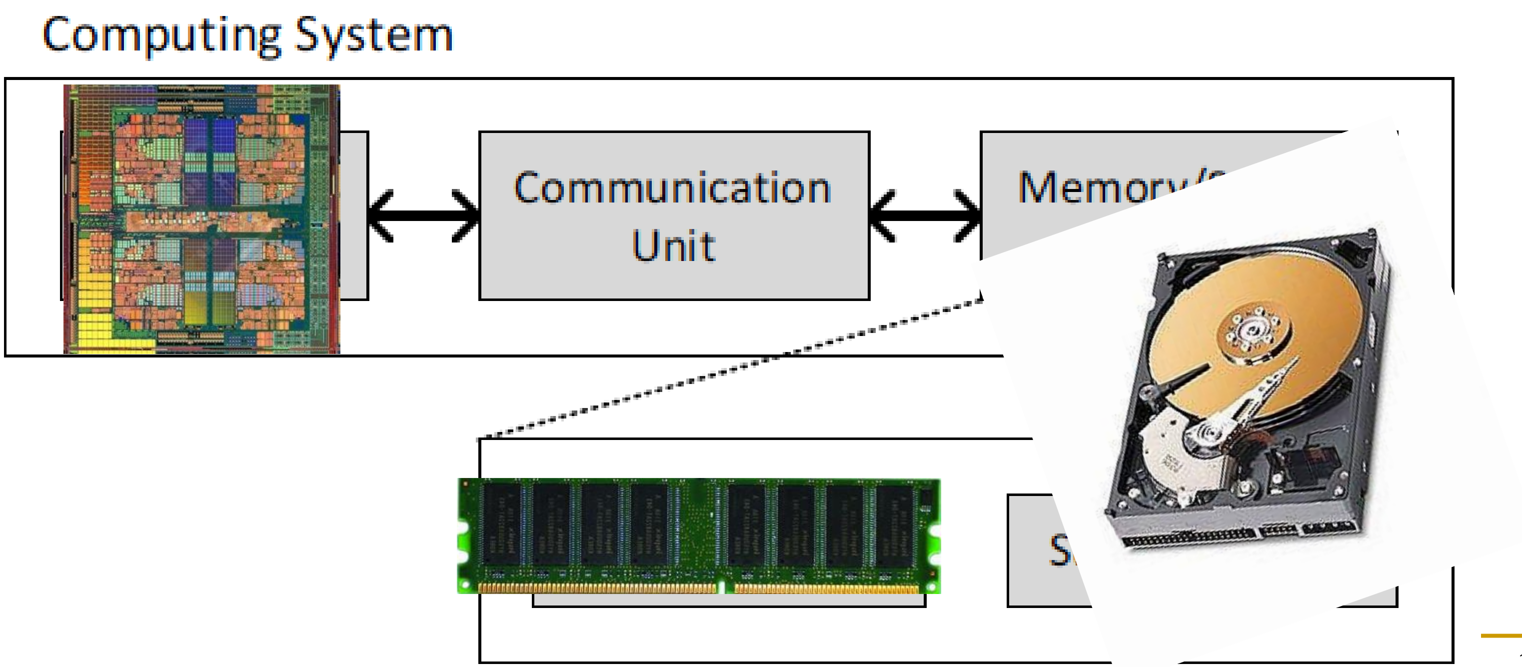
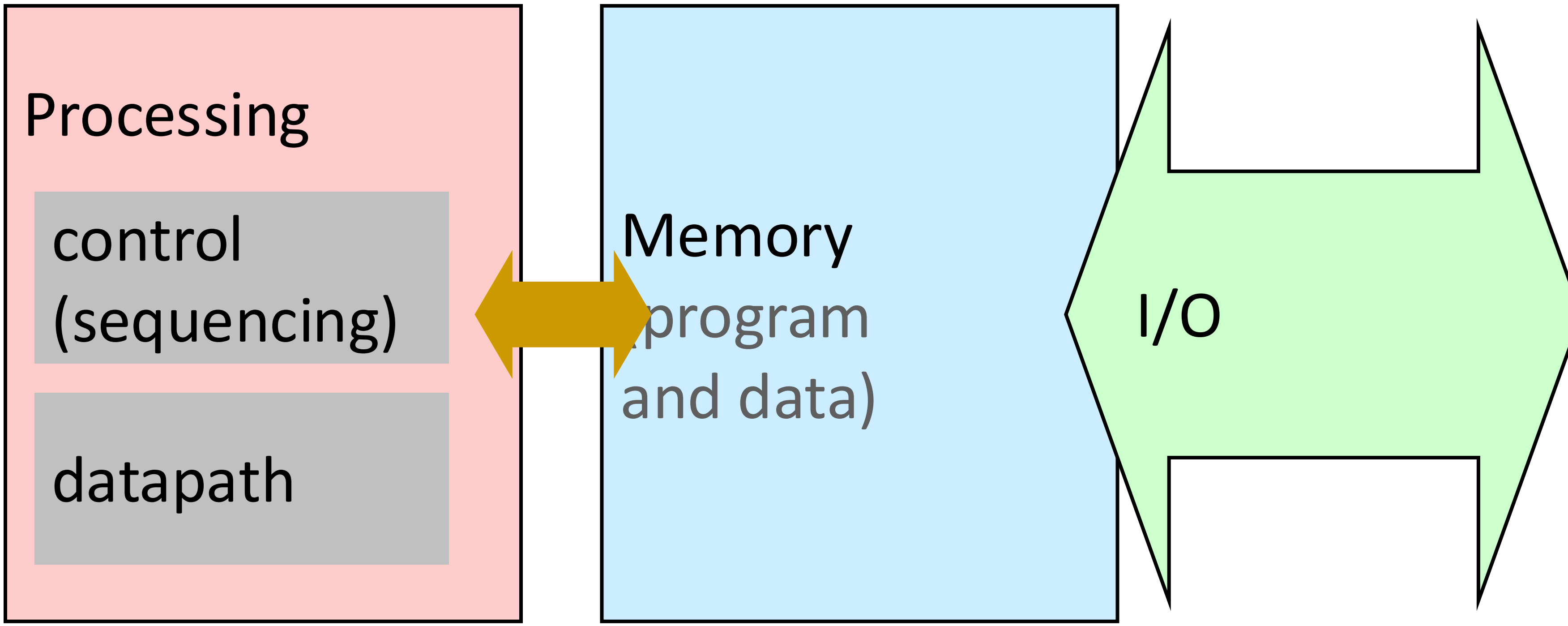


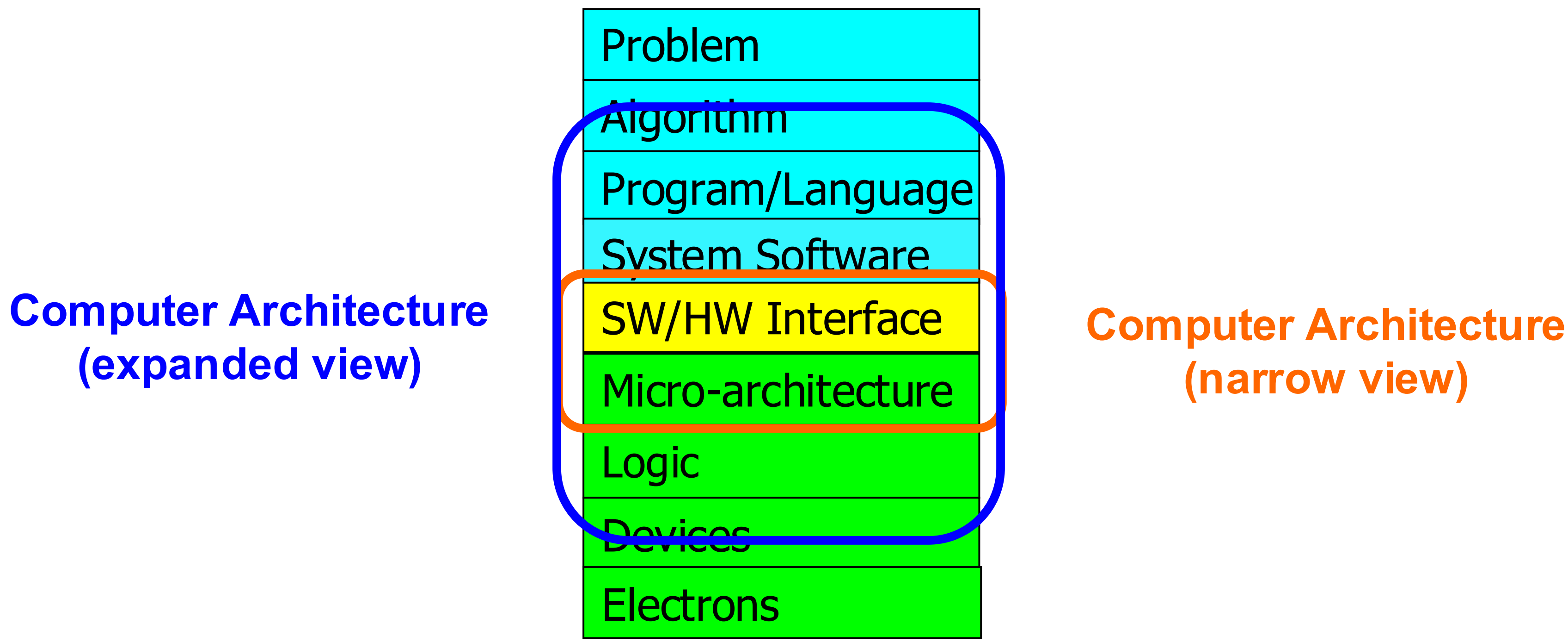
Image source: https://lbsitbytes2010.wordpress.com/2013/03/29/john-von-neumann-roll-no-15/

What is A Computer?

- We will cover all three components

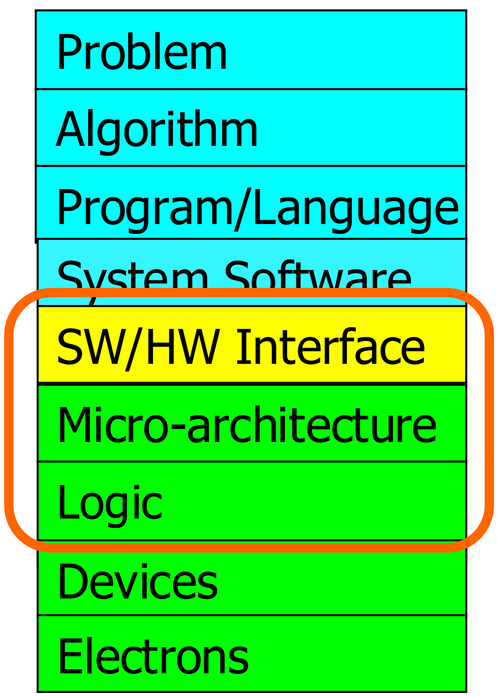


Recall: The Transformation Hierarchy



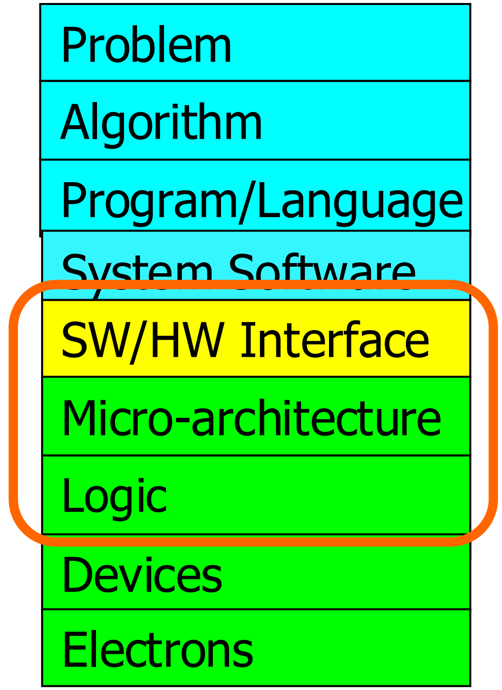
What We Will Cover (I)

- Combinational Logic Design
- Hardware Description Languages (Verilog)
- Sequential Logic Design
- Timing and Verification
- ISA (MIPS and LC3b as examples)
- Assembly Programming



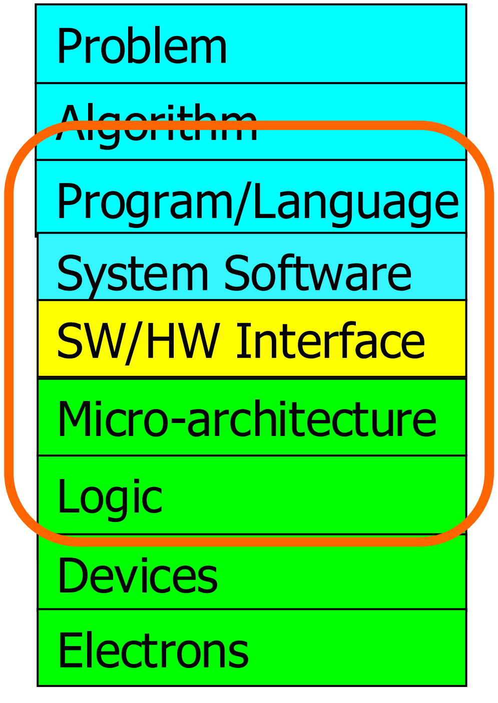
What We Will Cover (II)

- Microarchitecture Fundamentals
- Single-cycle Microarchitectures
- Multi-cycle and Microprogrammed Microarchitectures
- Pipelining
- Issues in Pipelining: Dependence Handling, State Maintenance and Recovery, …
- Branch Prediction
- Out-of-Order Execution
- Superscalar Execution
- Other Paradigms: Dataflow, VLIW, Systolic, SIMD/GPUs, …



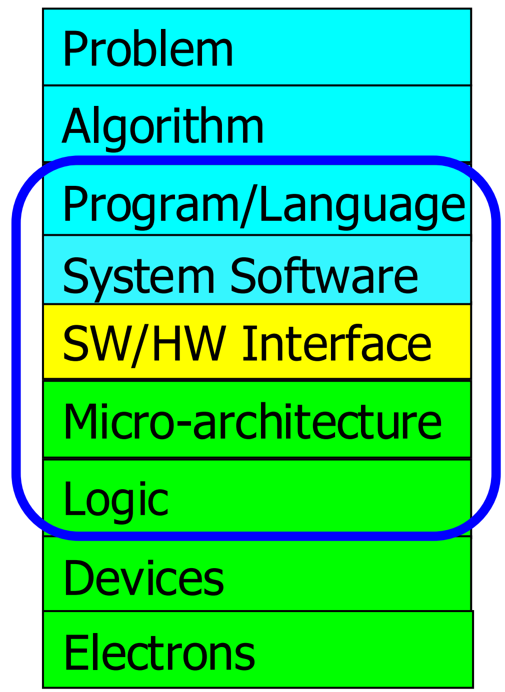
What We Will Cover (III)

- Memory Technology and Organization
- Memory Hierarchy
- Caches
- Multi-Core Caches
- Prefetching
- Virtual Memory



Processing Paradigms We Will Cover

- Pipelining
- Out-of-order execution
- Dataflow (at the ISA level)
- Superscalar Execution
- VLIW
- Decoupled Access-Execute
- Systolic Arrays
- SIMD Processing (Vector & Array)
- GPUs



Combinational Logic Circuits and Design

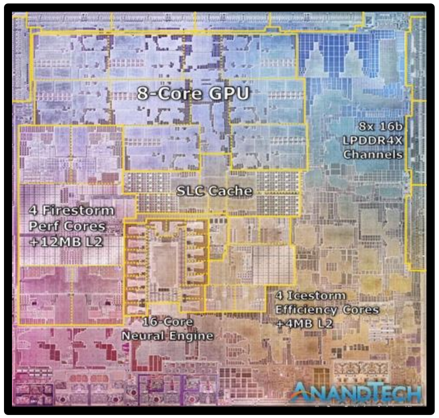
What Will We Learn Today & Tomorrow?

- Basic building blocks of modern computers
  - Transistors
  - Logic gates
- Boolean algebra
- Combinational logic circuits
- How to use Boolean algebra to represent combinational circuits
- Minimizing logic circuits (if time permits)

General Purpose vs. Special Purpose Systems

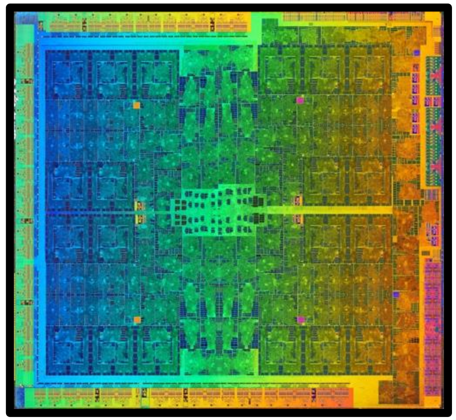
General Purpose

CPUs



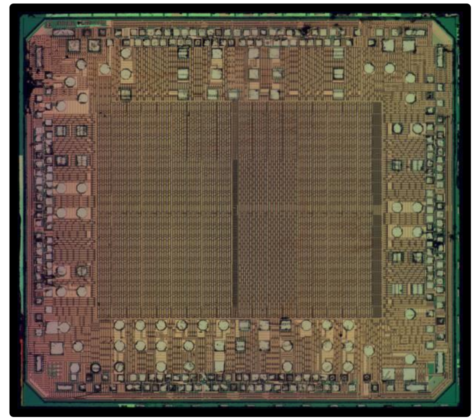
Apple M1

GPUs



Nvidia GTX 1070

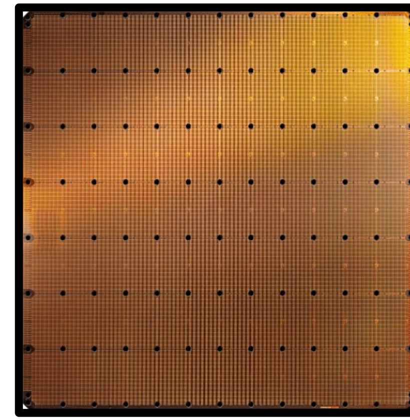
FPGAs



Xilinx Spartan

Special Purpose

ASICs



Cerebras WSE-2

Flexible: Can execute any program
Easy to program & use
Not the best performance & efficiency

Efficient & High performance
(Usually) Difficult to program & use
Inflexible: Limited set of programs

General Purpose vs. Special Purpose Systems

General Purpose

Special Purpose

Flexible: Can work with any bolt

Easy to use

Not the best fit, results or efficiency

Efficient & High performance

(Usually) Difficult to use

Inflexible: Works only for fitting bolts

https://www.ubuy.vn/en/product/2MBUW8M-crescent-8-adjustable-wrench-carded-ac28vs
https://capritools.com/shop/bolt-extractor-wrench-set-metric-8-19-mm/

https://www.vigor-equipment.com/en/combination-wrench-4971

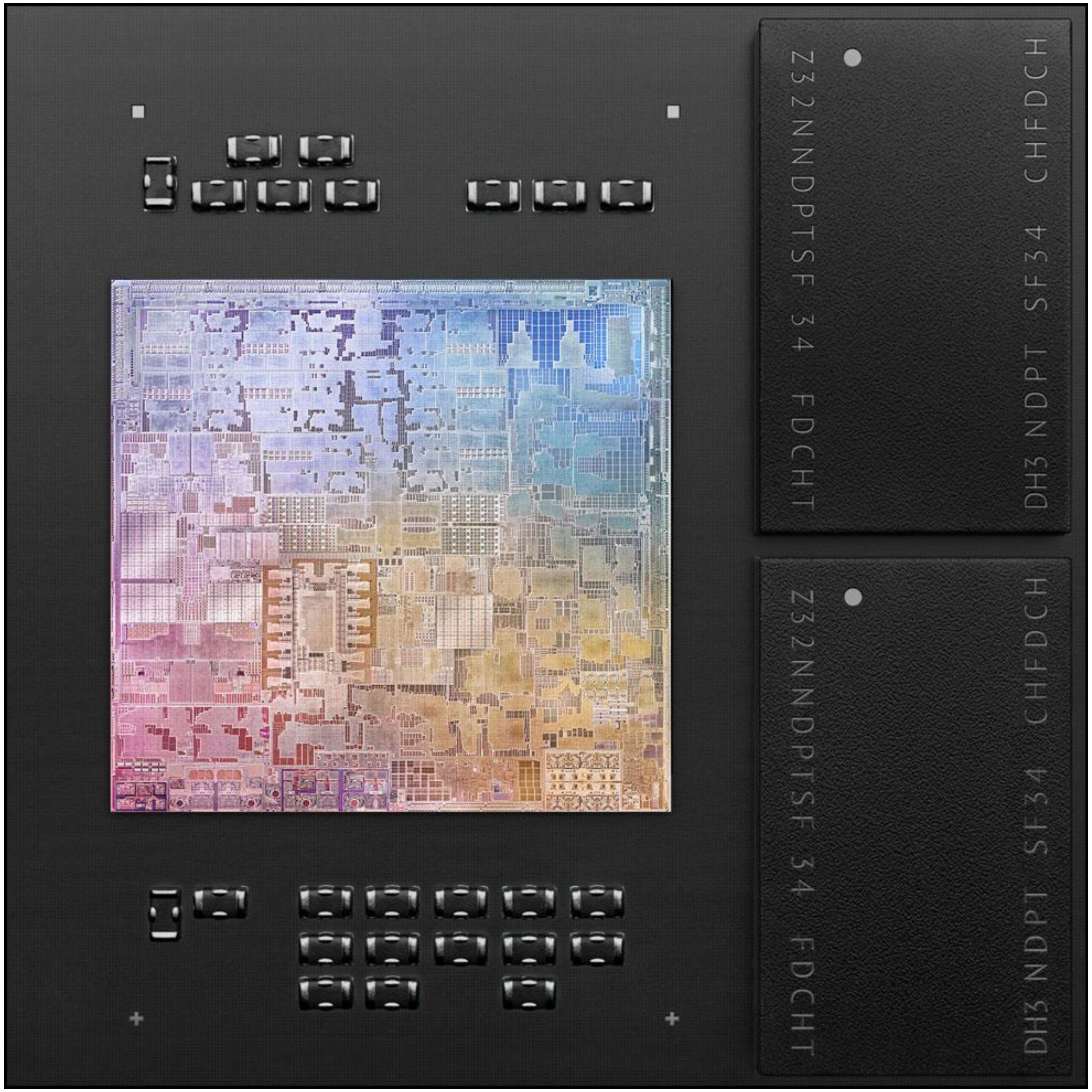
Modern General-Purpose Microprocessors

5-nanometer process

The first personal computer chip built with this cutting-edge technology.

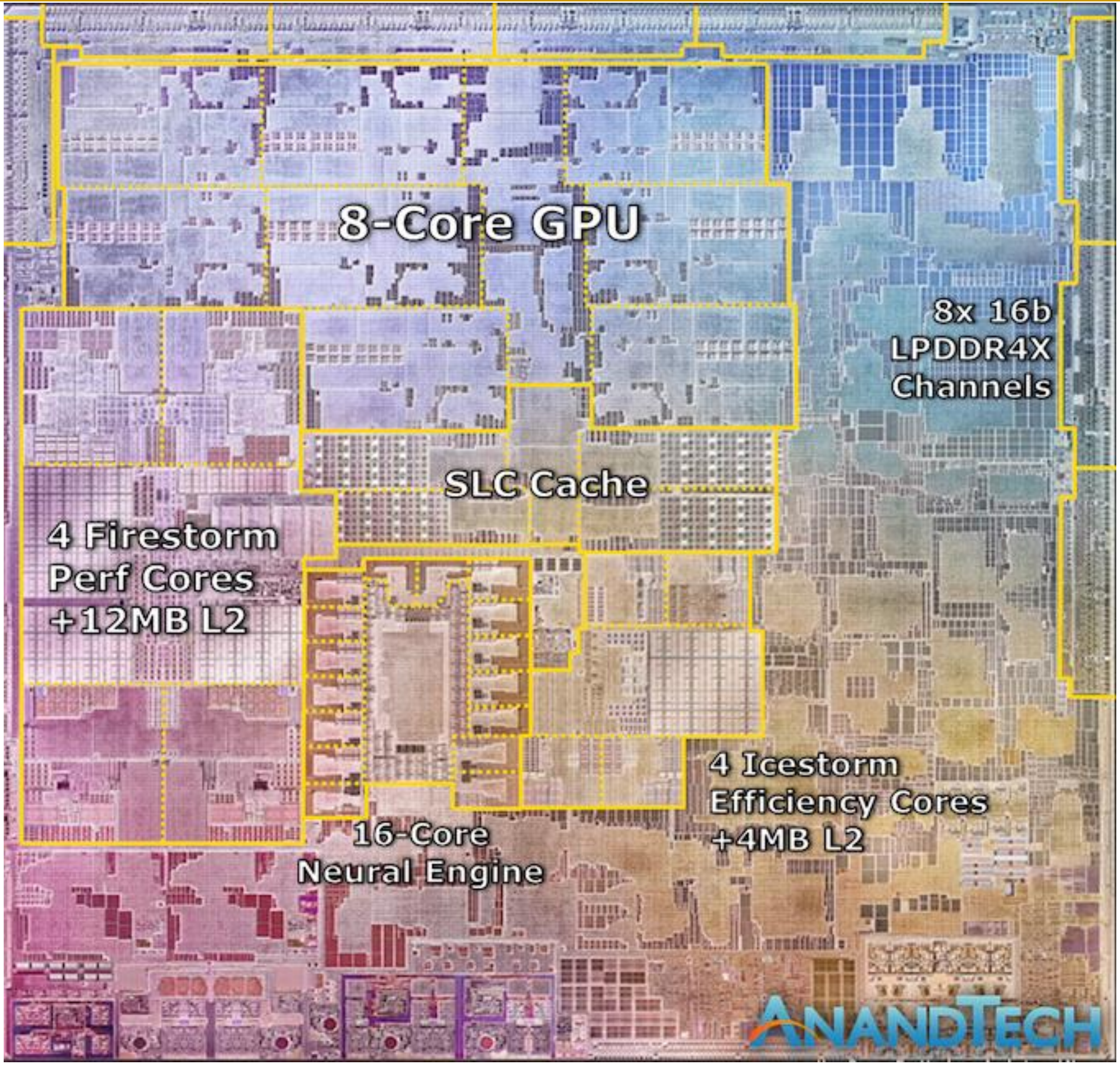
16 billion transistors

The most we've ever put into a single chip.



Source: https://www.apple.com/mac/m1/

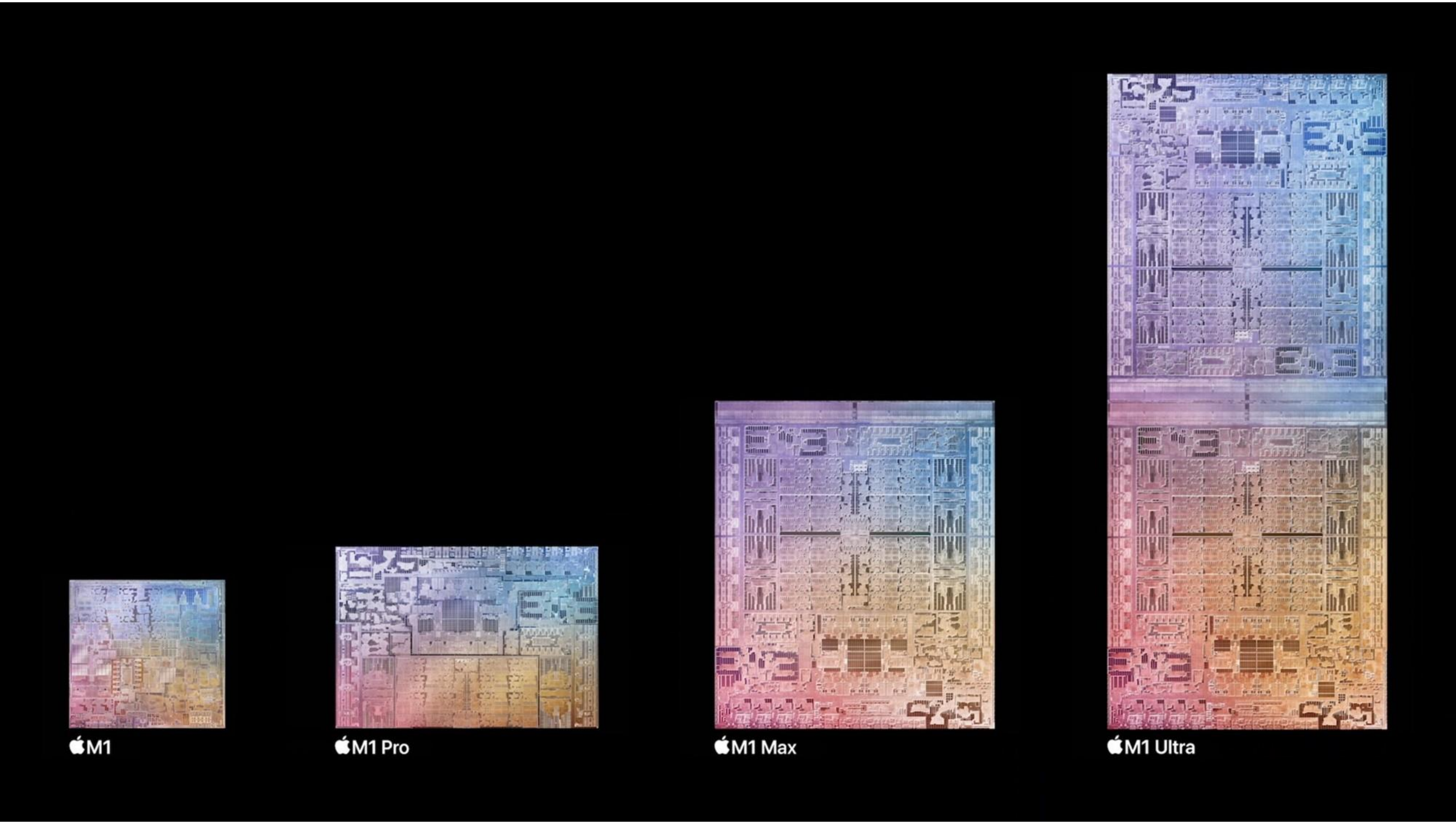
Modern General-Purpose Microprocessors



Apple M1,
2021

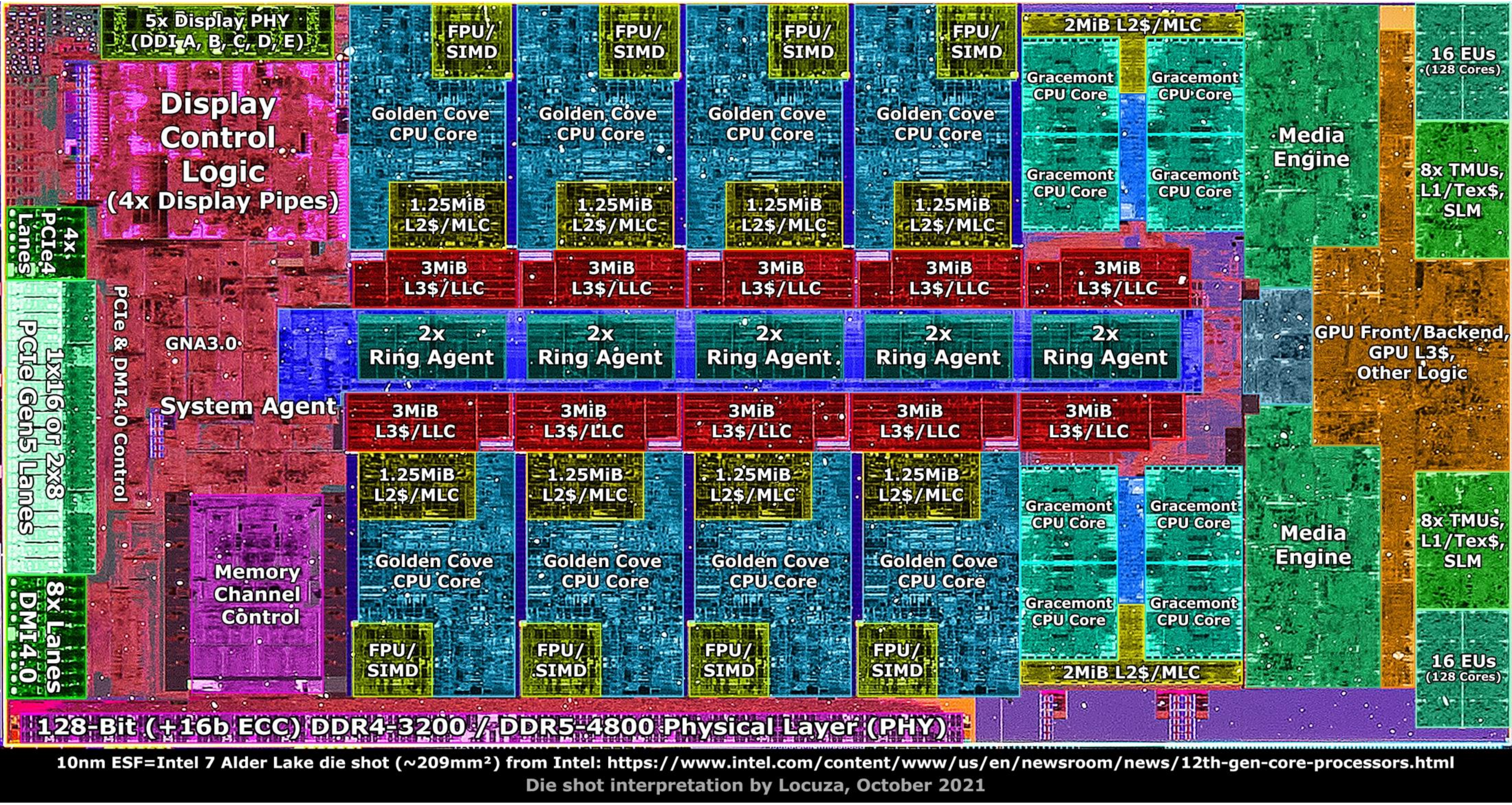
Source: https://www.anandtech.com/show/16252/mac-mini-apple-m1-tested

Apple M1 Ultra (2022)



https://www.theverge.com/2022/3/9/22968611/apple-m1-ultra-gpu-nvidia-rtx-3090-comparison

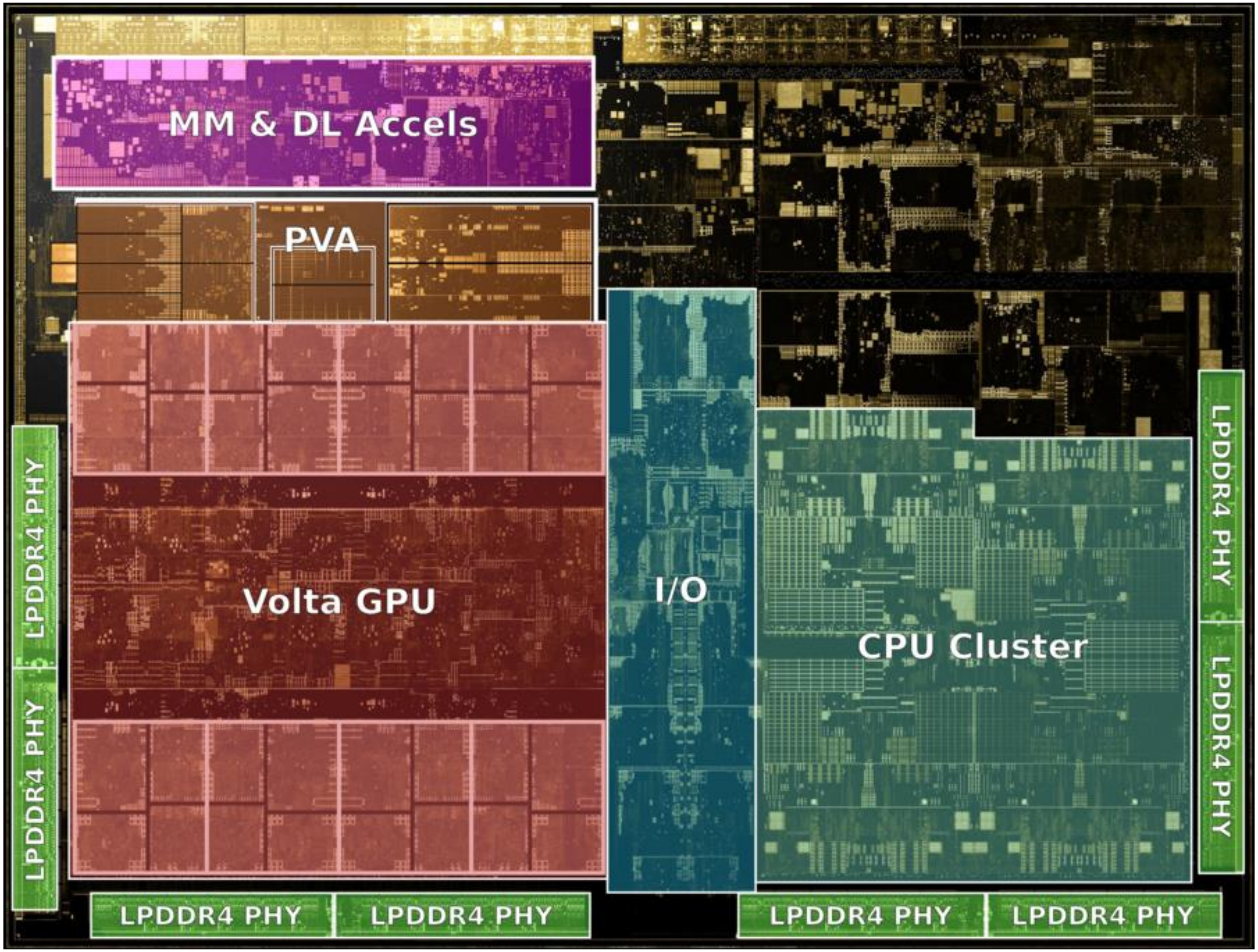
Modern General-Purpose Microprocessors



Intel Alder Lake, 2021

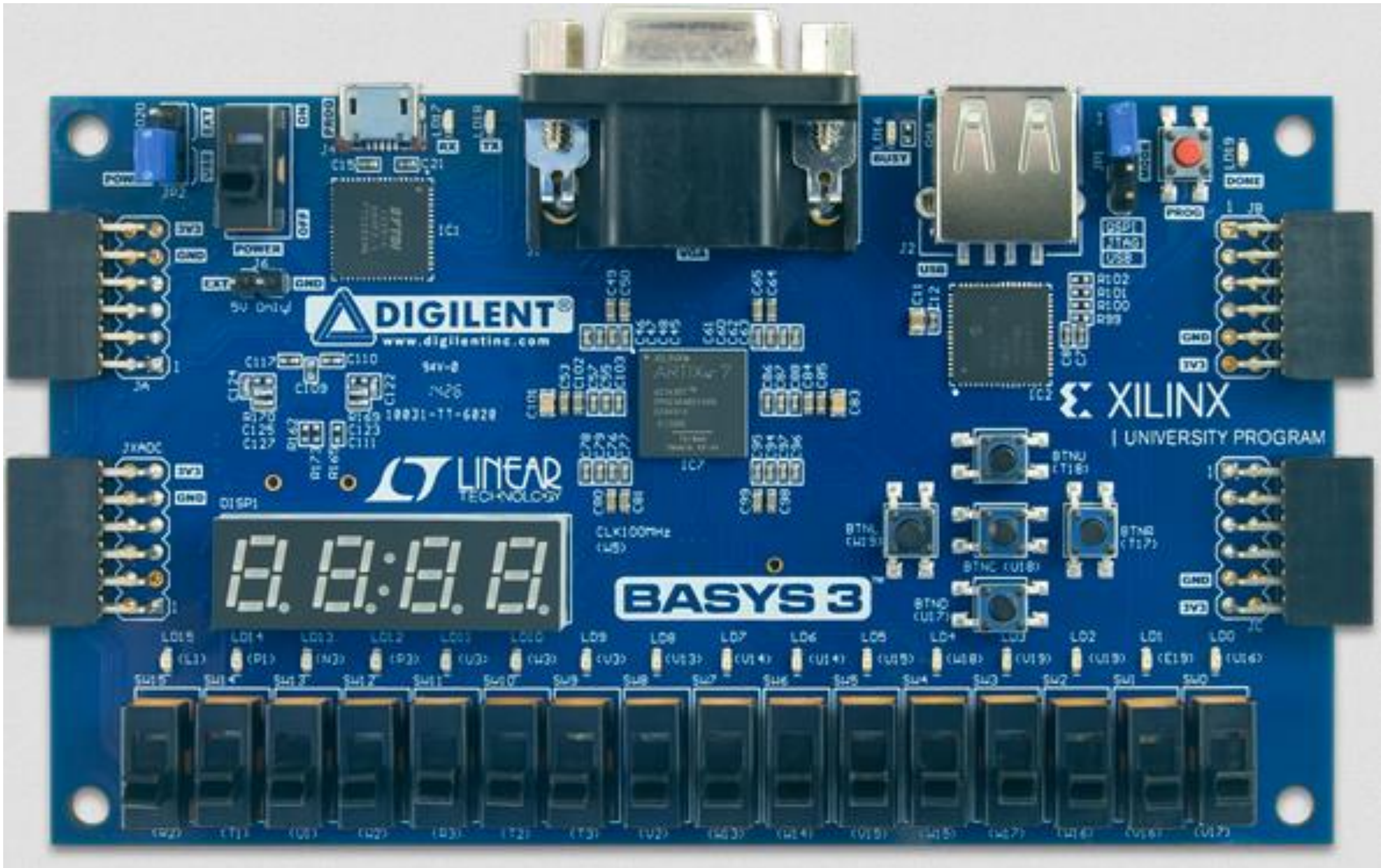
Source: https://twitter.com/Locuza_/status/1454152714930331652

Modern GPUs

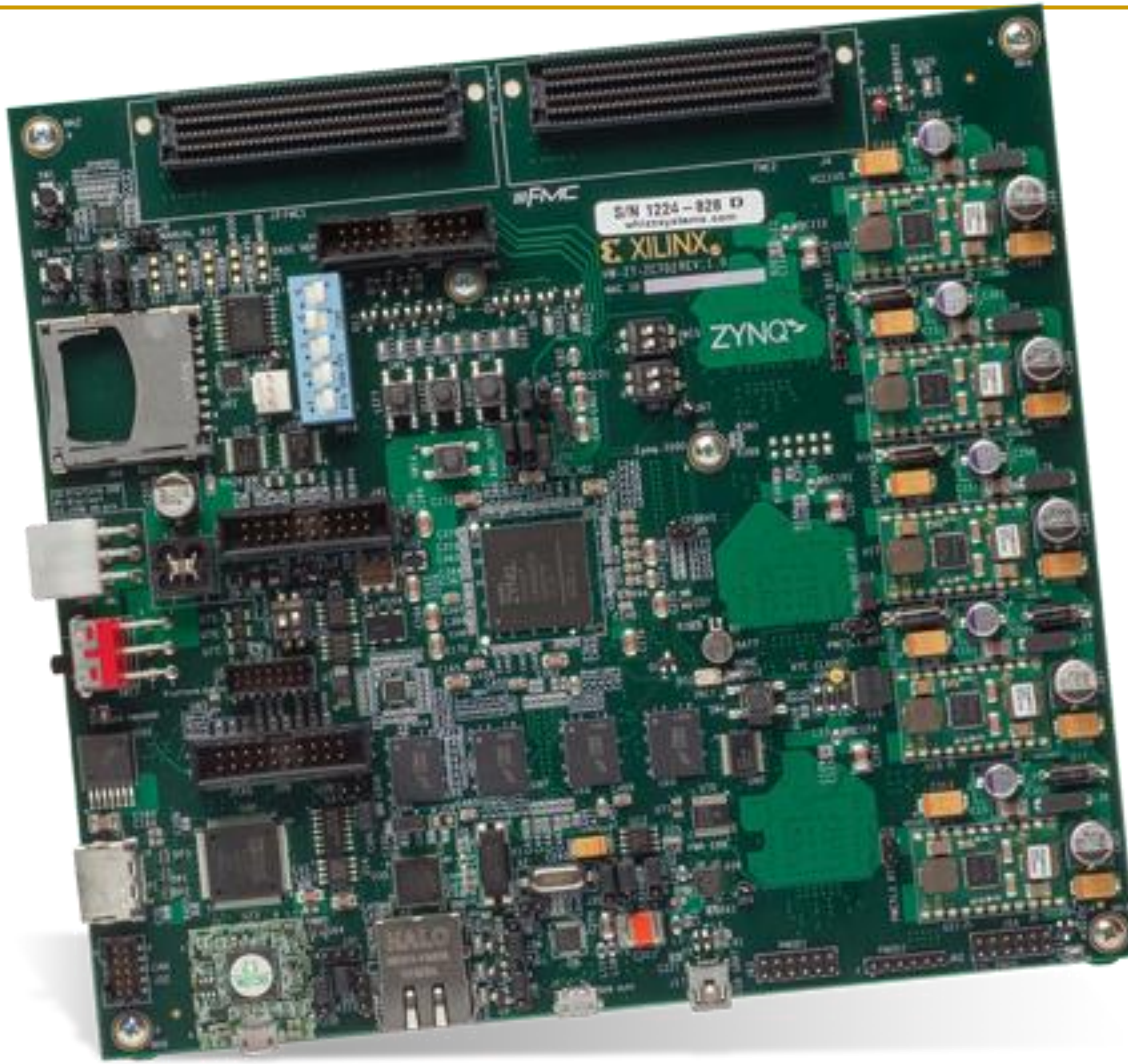


Source: https://en.wikichip.org/wiki/nvidia/tegra/xavier

FPGAs

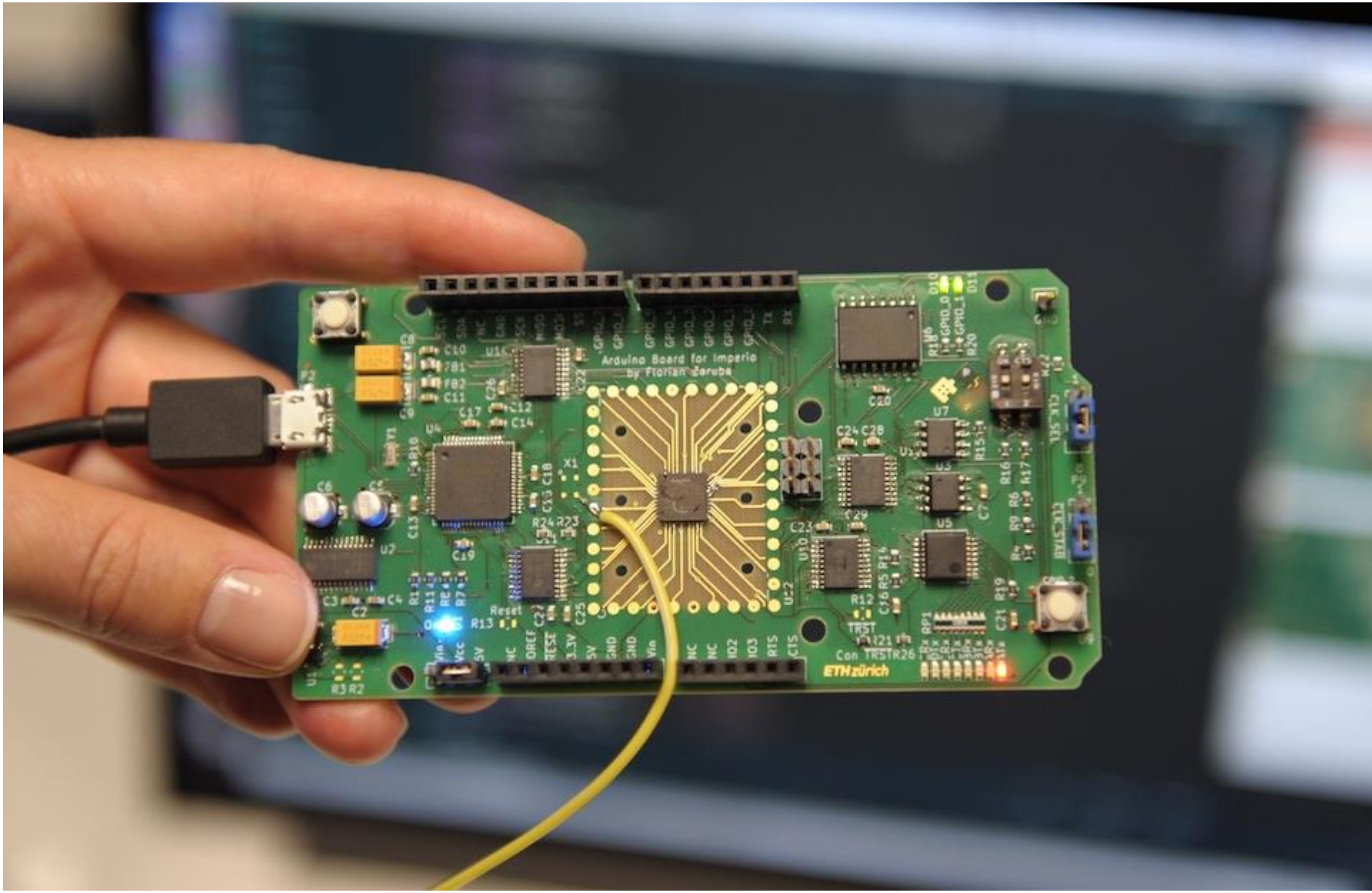

DIGILENT
www.digilentinc.com
XILINX
UNIVERSITY PROGRAM
LINEAR
TECHNOLOGY
BASYS 3

Modern FPGAs



Source: https://www.mouser.ch/new/xilinx/xilinx-zynq-7000-zc702-eval-kit/

Special-Purpose ASICs (App-Specific Integrated Circuits)



Modern Special-Purpose ASICs

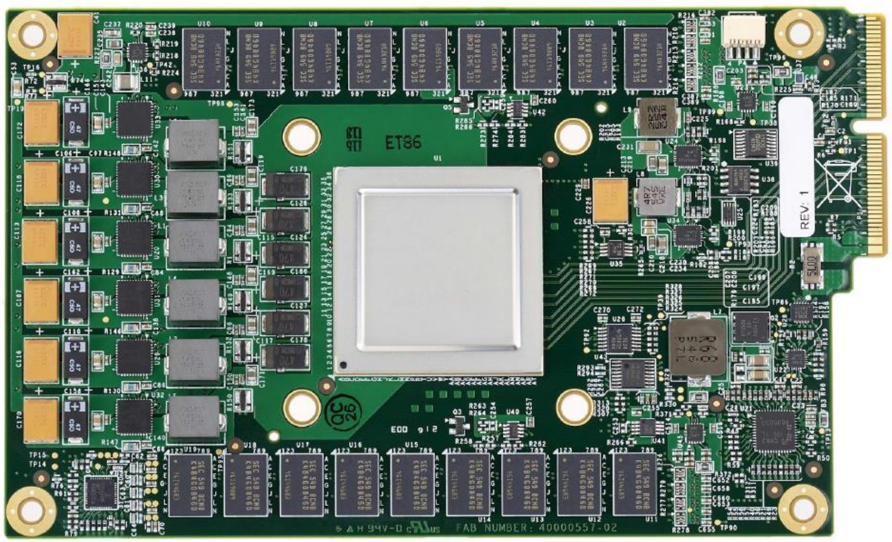


Figure 3. TPU Printed Circuit Board. It can be inserted in the slot for an SATA disk in a server, but the card uses PCIe Gen3 x16.

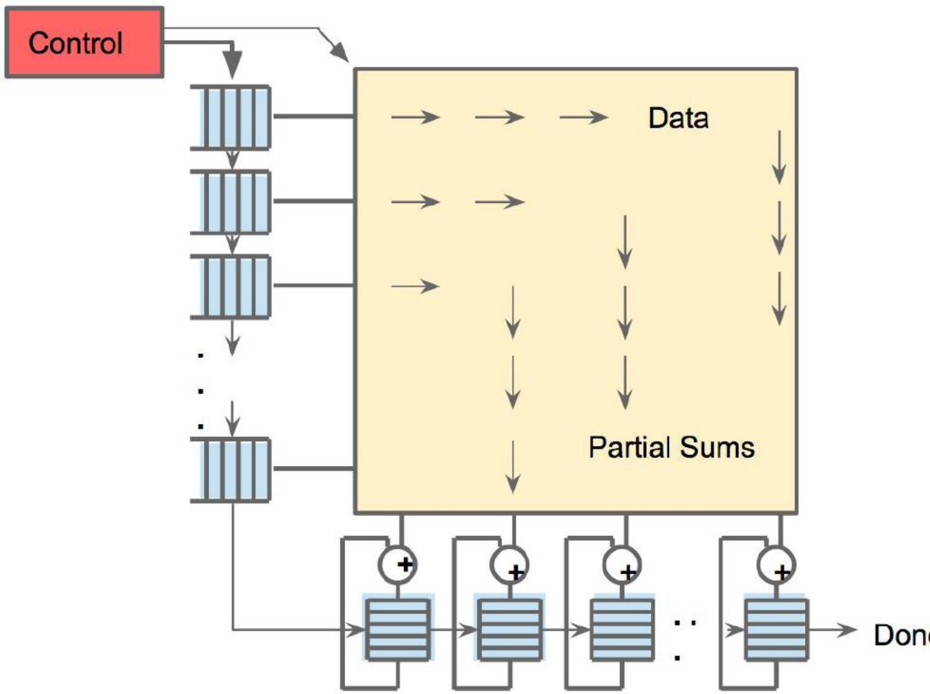
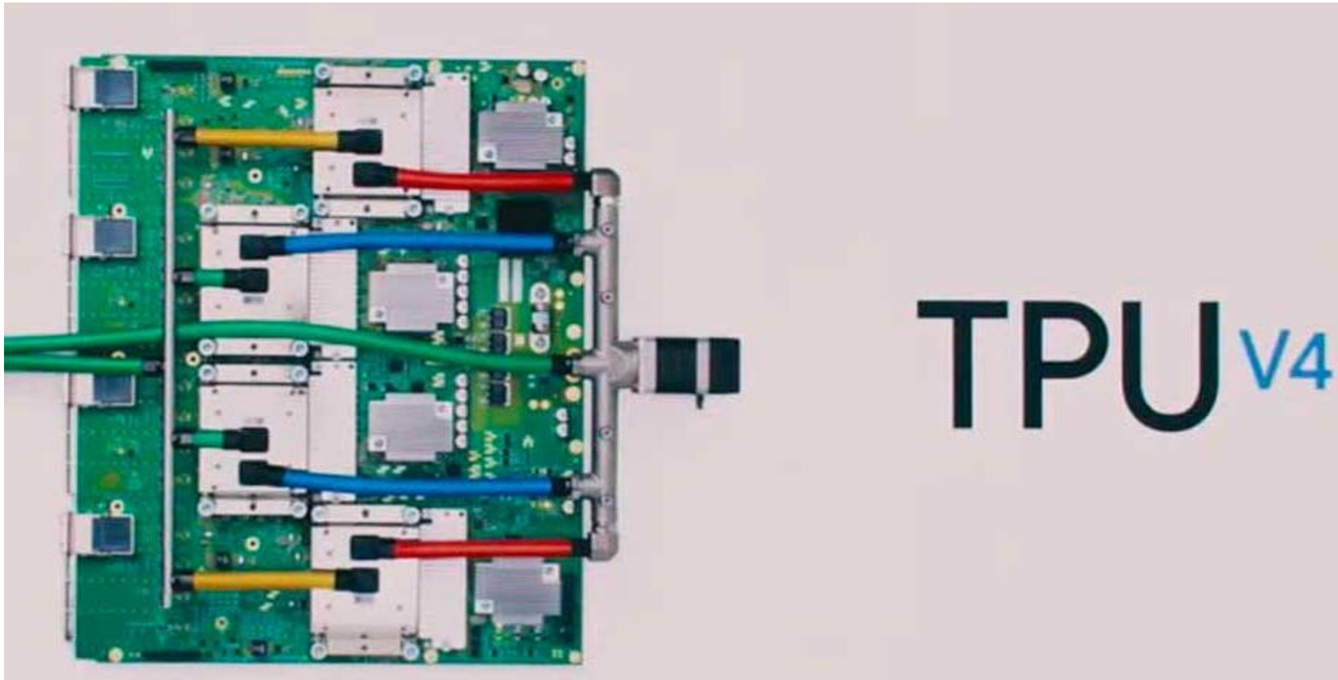


Figure 4. Systolic data flow of the Matrix Multiply Unit. Software has the illusion that each 256B input is read at once, and they instantly update one location of each of 256 accumulator RAMs.

Jouppi et al., "In-Datacenter Performance Analysis of a Tensor Processing Unit", ISCA 2017.

Modern Special-Purpose ASICs



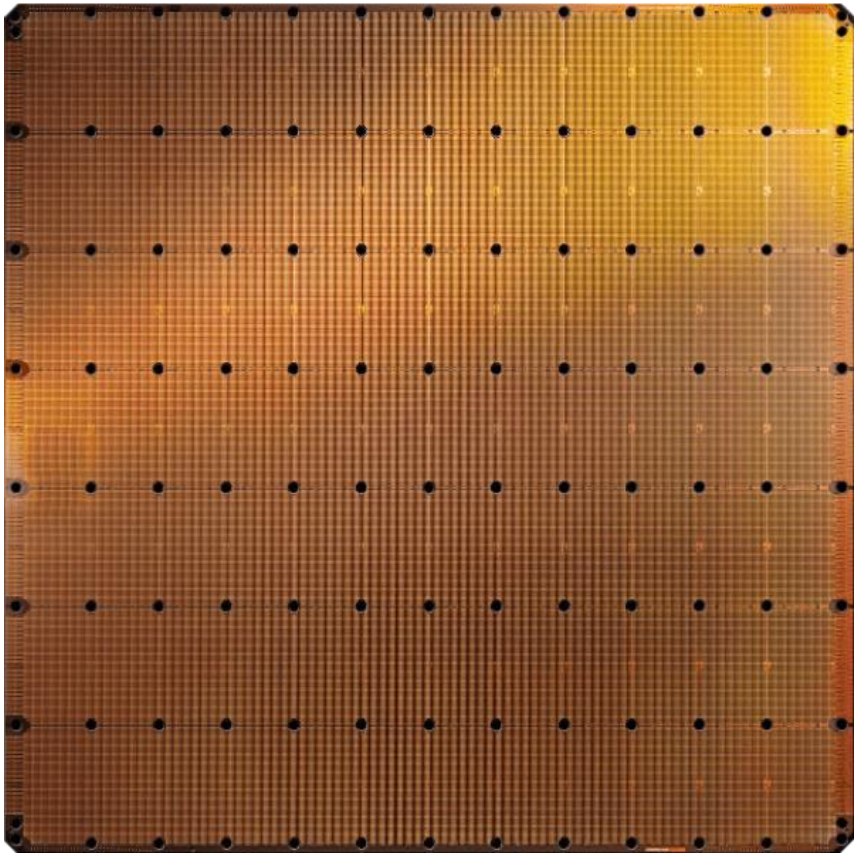
New ML applications (vs. TPU3):
- Computer vision
- Natural Language Processing (NLP)
- Recommender system
- Reinforcement learning that plays Go

250 TFLOPS per chip in 2021
vs 90 TFLOPS in TPU3

1 ExaFLOPS per board

https://spectrum.ieee.org/tech-talk/computing/hardware/heres-how-googles-tpu-v4-ai-chip-stacked-up-in-training-tests

Modern Special-Purpose ASICs



- The largest ML accelerator chip (2021)
- 850,000 cores

Cerebras WSE-2
2.6 Trillion transistors
46,225 mm^2

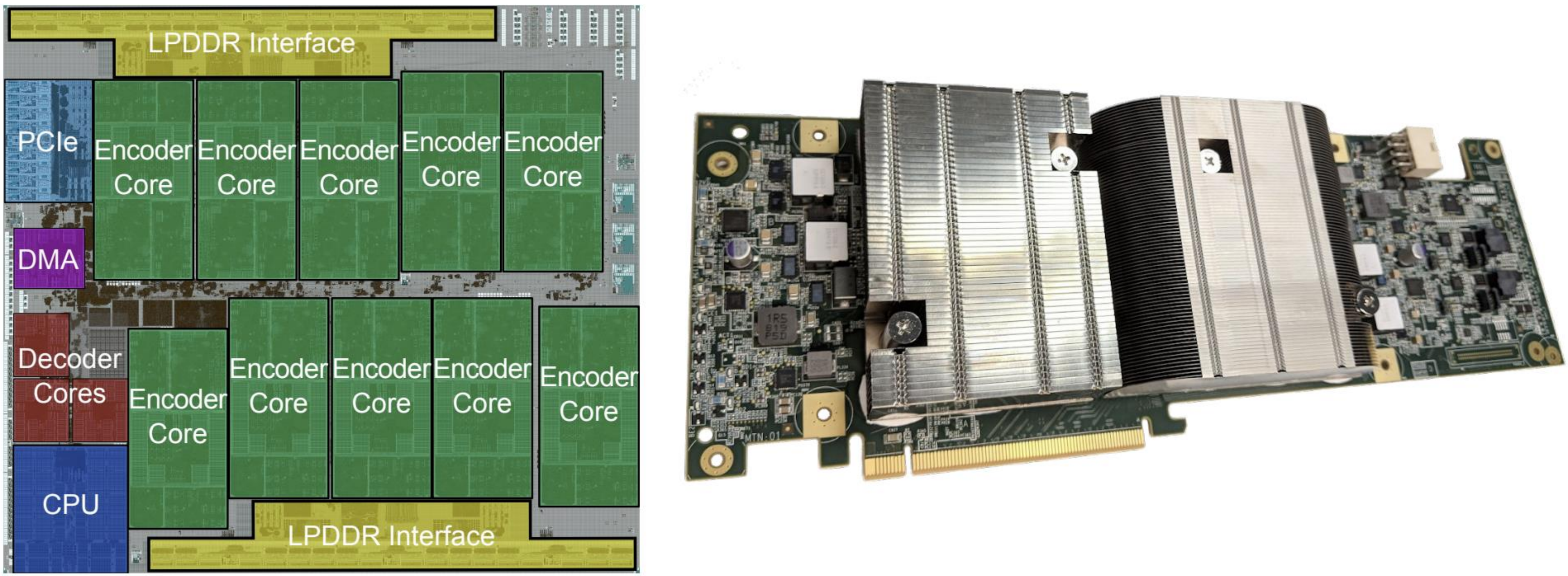
Largest GPU
54.2 Billion transistors
826 mm^2
NVIDIA Ampere GA100

https://www.anandtech.com/show/14758/hot-chips-31-live-blogs-cerebras-wafer-scale-deep-learning

https://www.cerebras.net/cerebras-wafer-scale-engine-why-we-need-big-chips-for-deep-learning/

Modern Special-Purpose ASICs

Warehouse-Scale Video Acceleration: Co-design and Deployment in the Wild



(a) Chip floorplan **(b) Two chips on a PCBA**

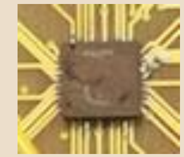
Figure 5: Pictures of the VCU

Source: https://dl.acm.org/doi/pdf/10.1145/3445814.3446723

They All Look the Same

	Microprocessors	FPGAs	ASICs
			
In short:	Common building block of computers	Reconfigurable hardware, flexible	You customize everything

They All Look the Same

	Microprocessors	FPGAs	ASICs
			
In short:	Common building block of computers	Reconfigurable hardware, flexible	You customize everything
Program Development Time	minutes	days	months

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			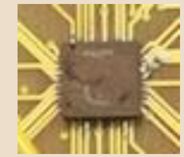
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Performance	o	+	++

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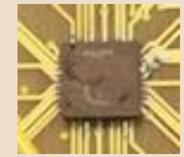
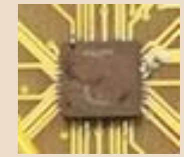
			
In short:	Common building block of computers	Reconfigurable hardware, flexible	You customize everything
Program Development Time	minutes	days	months
Performance	o	+	++
Good for	Ubiquitous Simple to use	Prototyping Small volume	Mass production, Max performance

They All Look (Kind of) the Same

	Microprocessors	FPGAs	ASICs
			
In short:	Common building block of computers	Reconfigurable hardware, flexible	You customize everything
Program Development Time	minutes	days	months
Performance	o	+	++
Good for	Ubiquitous Simple to use	Prototyping Small volume	Mass production, Max performance
Programming	Executable file	Bit file	Design masks
Languages	C/C++/Java/…	Verilog/VHDL	Verilog/VHDL
Main Companies	Intel, ARM, AMD, Apple, NVIDIA	Xilinx, Altera	TSMC, Globalfoundries

Labs: Build A Microprocessor on FPGA

	Microprocessors	FPGAs	
			
In short	Common building block of computers	Reconfigurable hardware, flexible	...nize ...ng
Program Development Time	minutes	days	months
Performance	o	+	++
Good for	Ubiquitous Simple to use	Prototyping	Mass production,
Programming	Executable file		
Languages	C/C++/Java/...	Verilog/VHDL	Verilog/VHDL
Main Companies	Intel, ARM, AMD, Apple, NVIDIA	Xilinx, Altera	TSMC, Globalfoundries

Want to learn how these work

By programming these

Using this language

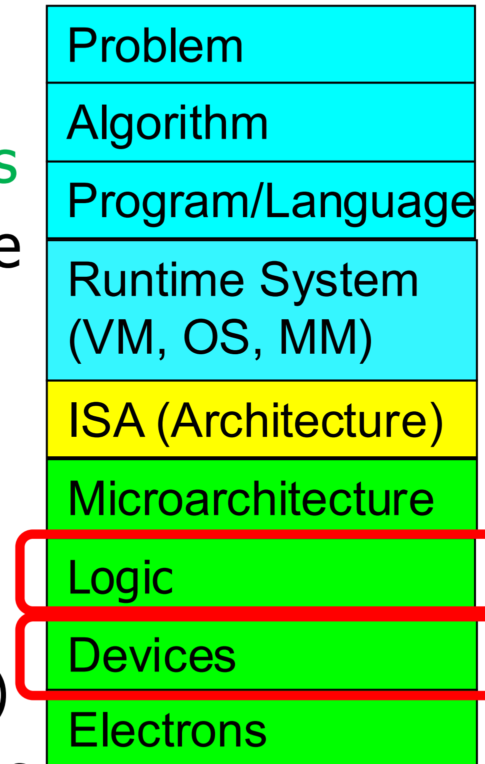
All Computers are Built Upon the Same Building Blocks

Building Blocks of Modern Computers

Transistors

Transistors

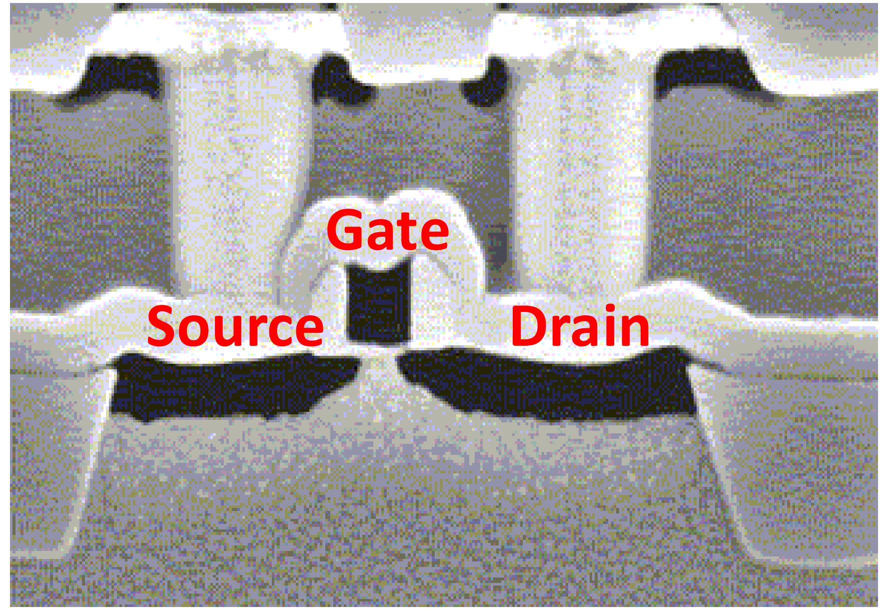
- **Computers are built from very large numbers of very small (and relatively simple) structures: transistors**
  - Intel 4004, in 1971, had 2300 MOS transistors
  - Intel's Pentium IV microprocessor, 2000, was made up of more than 42 **M**illion MOS transistors
  - Apple's M2 Max, offered for sale in 2022, is made up of more than 67 **B**illion MOS transistors
  - Micron's V-NAND flash memory (2022), is made up of more than 5 **T**rillion MOS transistors



- **This lecture**
  - How the MOS transistor works (as logic element)
  - How transistors are connected to form logic gates
  - How logic gates are interconnected to form larger units that are needed to construct a computer

MOS Transistor

- By combining
  - Conductors (**M**etal)
  - Insulators (**O**xide)
  - **S**emiconductors
- We get a Transistor (MOS)



- Why is this useful?
  - We can combine many of these to realize simple logic gates
- The electrical properties of metal-oxide semiconductors are well beyond the scope of what we want to understand in this course
  - They are below our lowest level of abstraction

Sections 1.6 and 1.7 in Harris and Harris provide some lower-level explanations

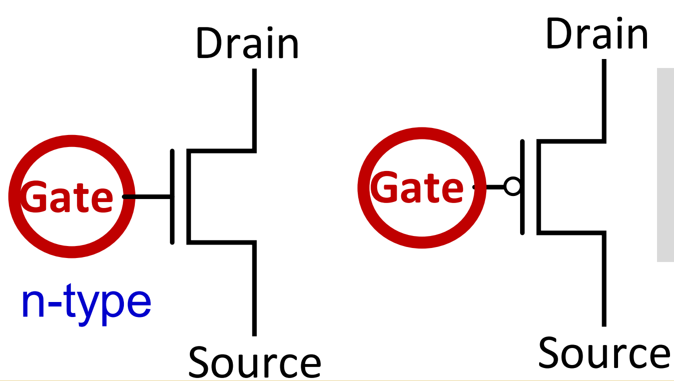
Different Types of MOS Transistors

- There are two types of MOS transistors: n-type and p-type

n-type p-type

- They both operate "logically," very similar to the way wall switches work

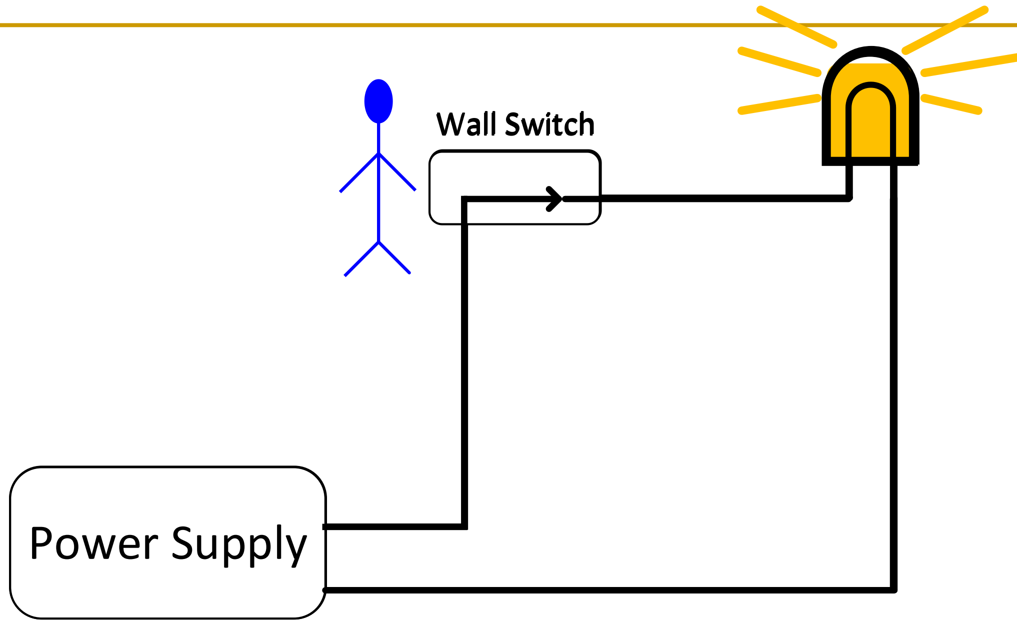
The circuit is closed when the gate is supplied with 3V



The circuit is closed when the gate is supplied with 0V

p-type

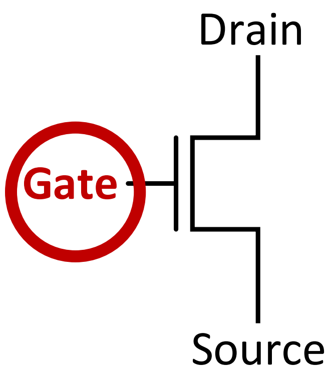
How Does a Wall Switch Work?



- In order for the lamp to glow, electrons must flow
- In order for electrons to flow, there must be a closed circuit from the power supply to the lamp and back to the power supply
- The lamp can be turned on and off by simply manipulating the wall switch to make or break the closed circuit

Transistor Works As a Switch

- Instead of the wall switch, we could use an n-type or a p-type MOS transistor to make or break the closed circuit



Schematic of an n-type MOS transistor

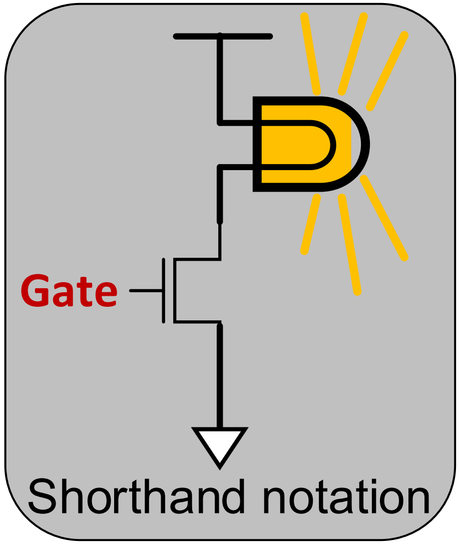
If the gate of an n-type transistor is supplied with a **high** voltage, the connection from source to drain acts like a piece of wire (i.e., the circuit is closed)

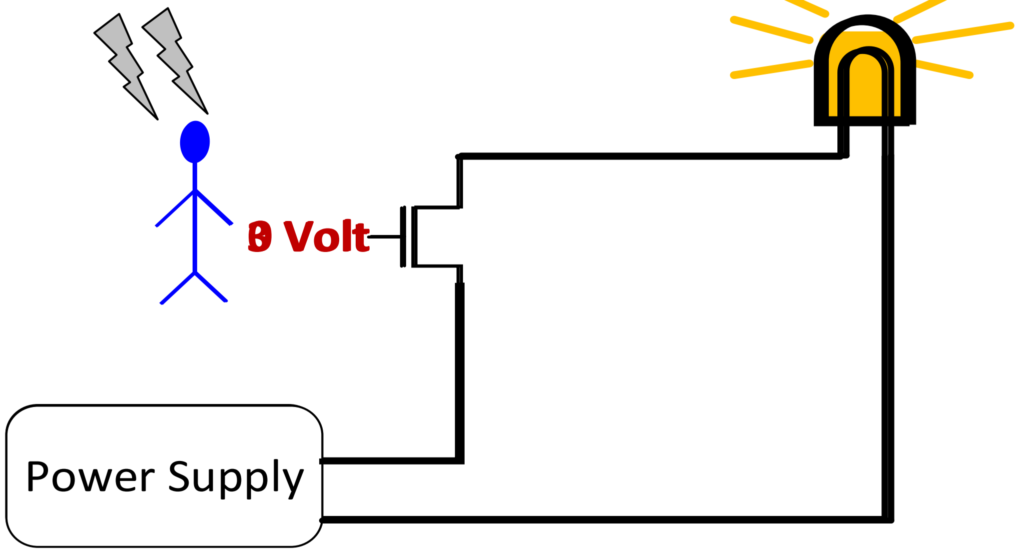
Depending on the technology, high voltage can range from 0.3V to 3V

If the gate of the n-type transistor is supplied with **zero** voltage, the connection between the source and drain is broken (i.e., the circuit is open)

Abstraction: Transistor As a Switch

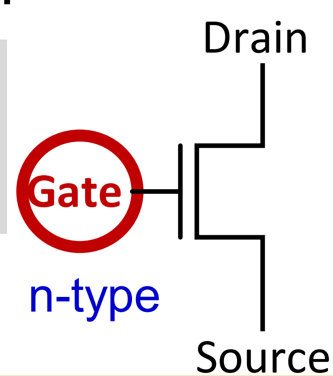
- n-type transistor in a circuit with a battery and a bulb

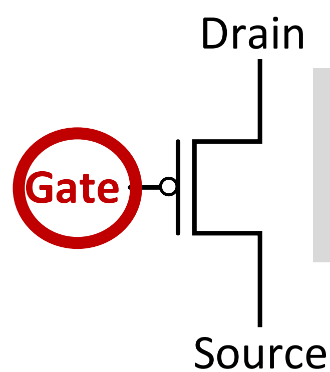




- p-type transistor works in exactly the opposite fashion from n-type transistor

The circuit is closed when the gate is supplied with 3V





The circuit is closed when the gate is supplied with 0V

p-type

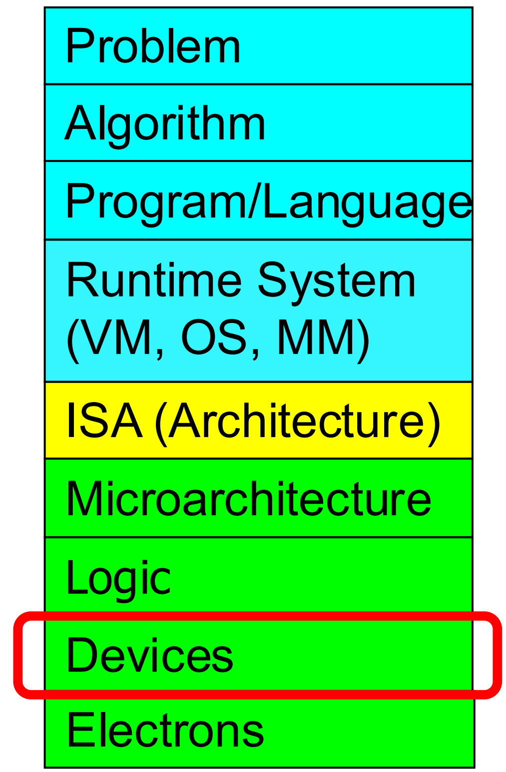
Logic Gates

One Level Higher in the Abstraction

- **Now, we know how a MOS transistor works**
- How do we build logic structures out of MOS transistors?

- We construct basic logical units out of individual MOS transistors

- These logical units are called logic gates
  - They implement simple Boolean functions



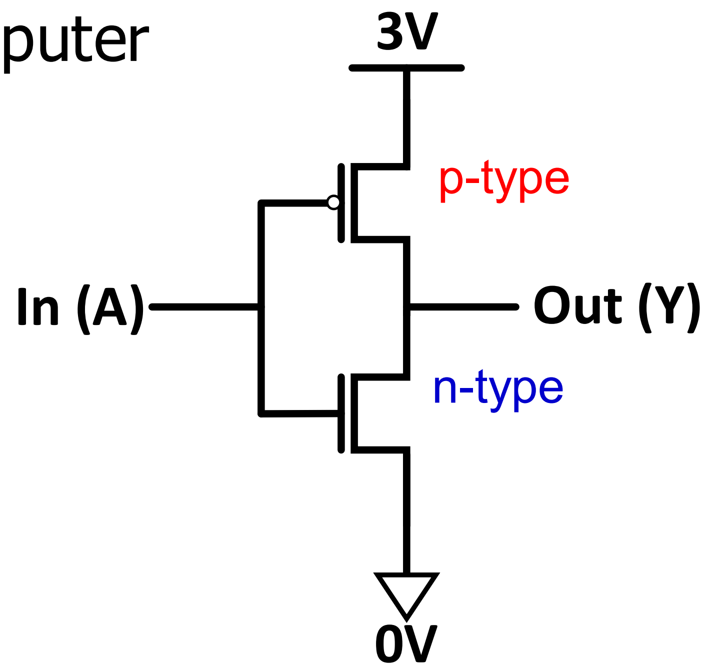
George Boole, "The Mathematical Analysis of Logic," 1847.

Making Logic Blocks Using CMOS Technology

- Modern computers use both n-type and p-type transistors, i.e. Complementary MOS (CMOS) technology

nMOS + pMOS = CMOS

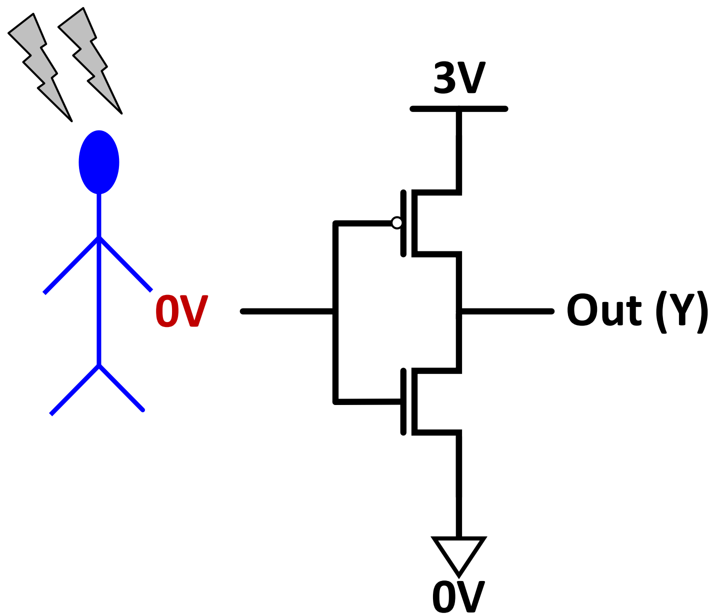
- The simplest logic structure that exists in a modern computer

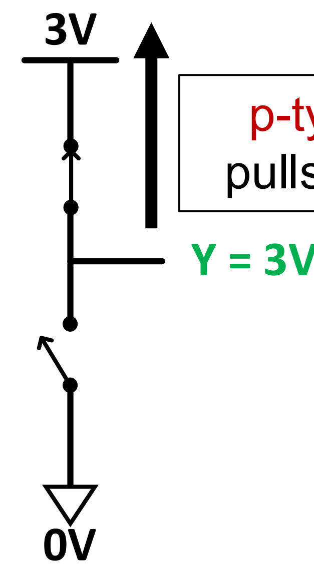


What does this circuit do?

Functionality of Our CMOS Circuit

What happens when the input is connected to 0V?



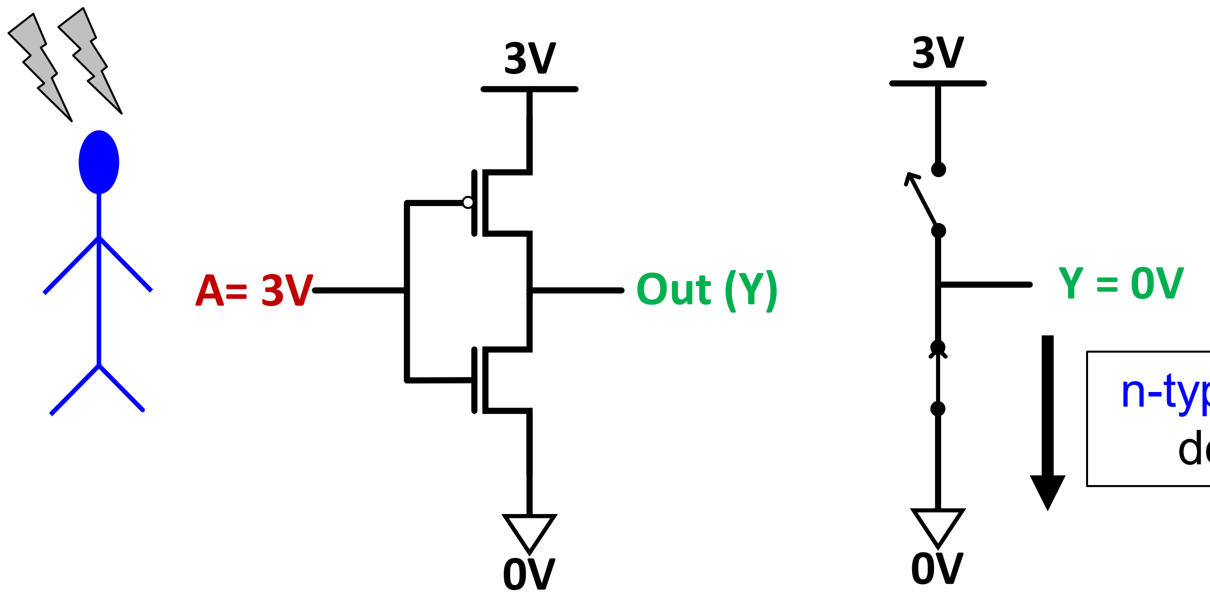


p-type transistor pulls up the output

p-type transistors are good at pulling u**p** the voltage

Functionality of Our CMOS Circuit

What happens when the input is connected to 3V?

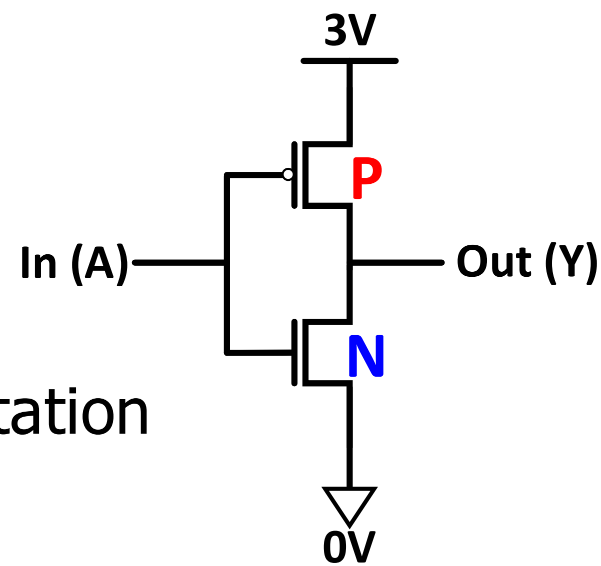


n-type transistor pulls down the output

n-type transistors are good at pulling dow**n** the voltage

CMOS NOT Gate (Inverter)

- This is actually the CMOS NOT Gate
- Why do we call it NOT?
  - If A = 0V then Y = 3V
  - If A = 3V then Y = 0V
- **Digital circuit:** one possible interpretation
  - Interpret 0V as logical (binary) 0 value
  - Interpret 3V as logical (binary) 1 value

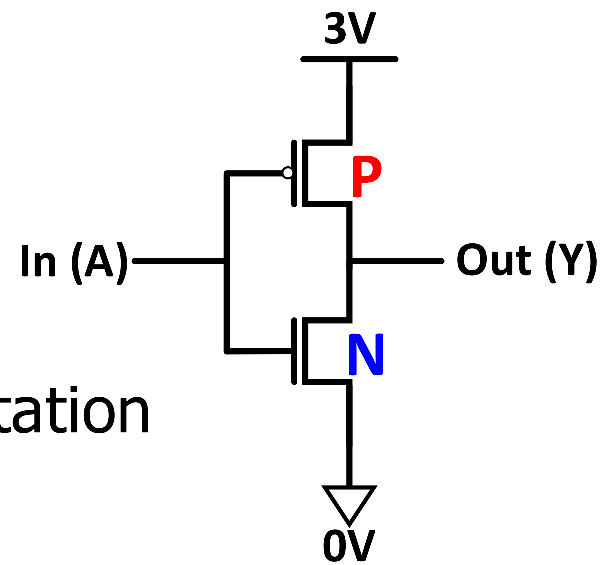


A	P	N	Y
0	**ON**	OFF	1
1	OFF	**ON**	0

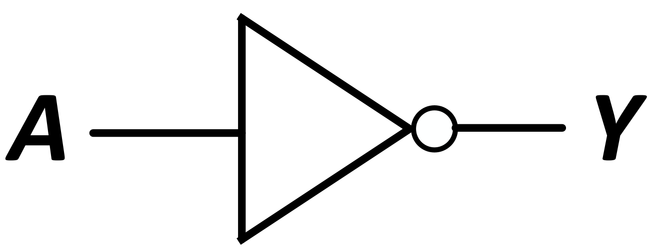
$$Y = \overline{A}$$

CMOS NOT Gate (Inverter)

- This is actually the CMOS NOT Gate
- Why do we call it NOT?
  - If A = 0V then Y = 3V
  - If A = 3V then Y = 0V
- **Digital circuit:** one possible interpretation
  - Interpret 0V as logical (binary) 0 value
  - Interpret 3V as logical (binary) 1 value



$$Y = \overline{A}$$



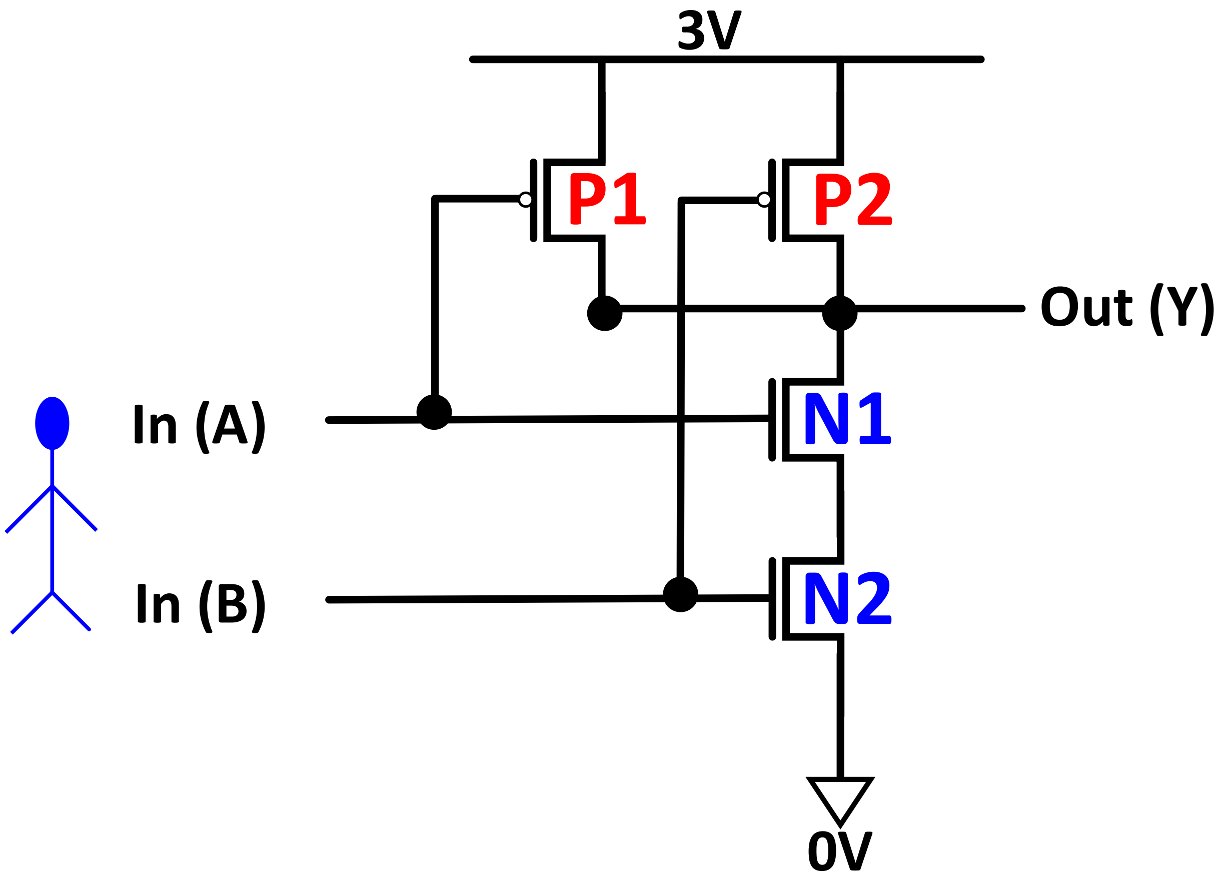
We call this a NOT gate
or an inverter
(bubble indicates inversion)

Truth table: shows the logical output of the circuit for each possible input

A	Y
0	1
1	0

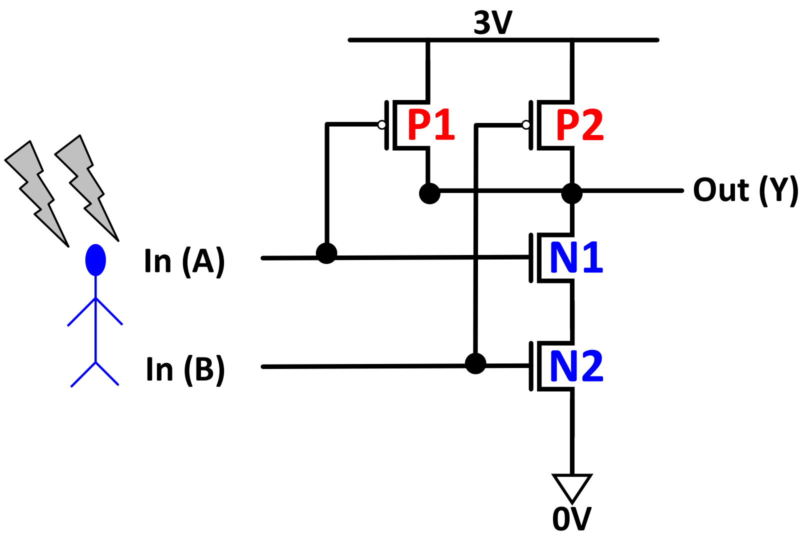
Another CMOS Gate: What Is This?

- Let's build more complex gates!



CMOS NAND Gate

- Let's build more complex gates!



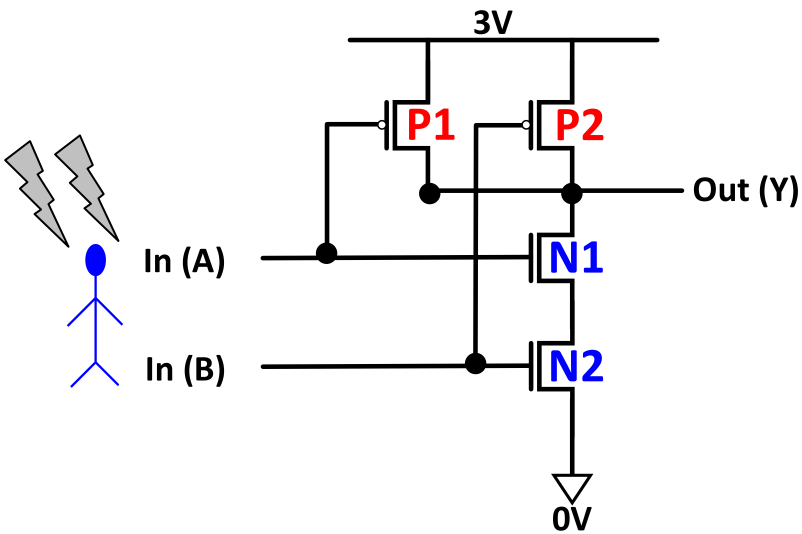
$$Y = \overline{A \cdot B} = \overline{AB}$$

A	B	P1	P2	N1	N2	Y
0	0	**ON**	**ON**	OFF	OFF	1
0	1	**ON**	OFF	OFF	ON	1
1	0	OFF	**ON**	ON	OFF	1
1	1	OFF	OFF	**ON**	**ON**	0

- P1 and P2 are in parallel; **only one must be ON to pull up the output to 3V**
- N1 and N2 are connected in series; **both must be ON to pull down the output to 0V**

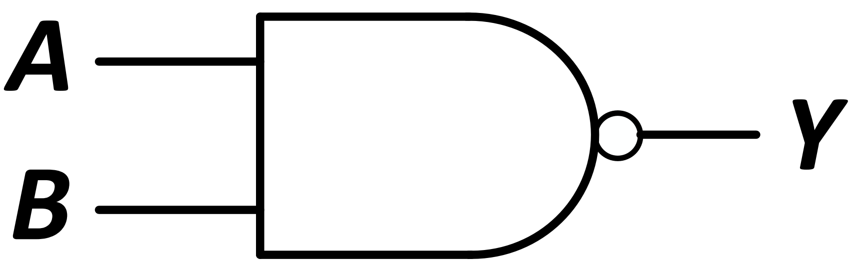
CMOS NAND Gate

- Let’s build more complex gates!



$$Y = \overline{A \cdot B} = \overline{AB}$$

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0



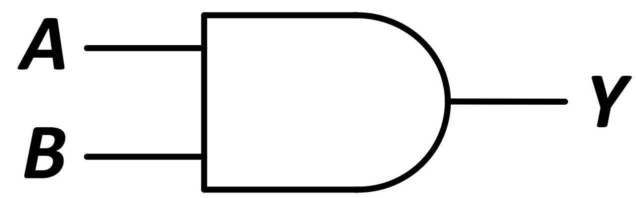
We call this a NAND gate
(bubble indicates inversion)

CMOS AND Gate

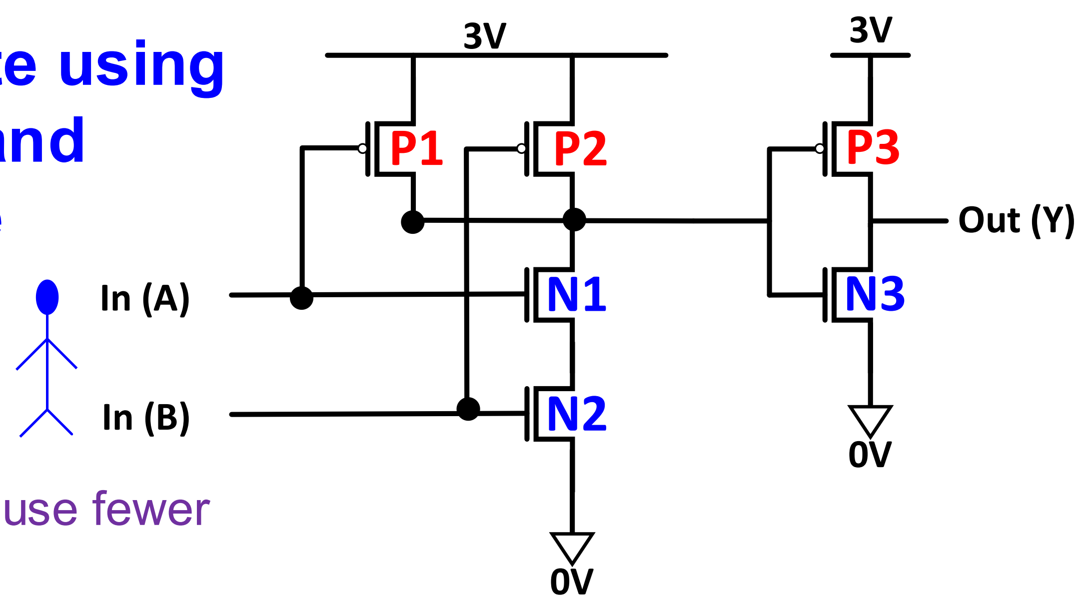
- How can we make an AND gate?

$$Y = A \cdot B = AB$$

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

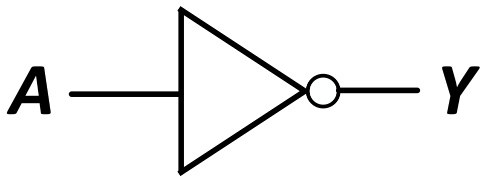


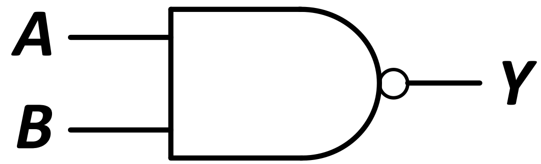
We make an AND gate using one NAND gate and one NOT gate

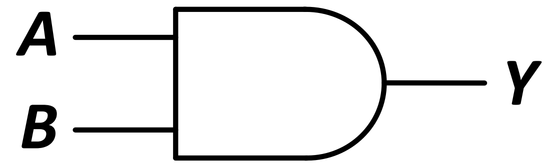


Food for thought: Can we not use fewer transistors for the AND gate?

CMOS NOT, NAND, AND Gates



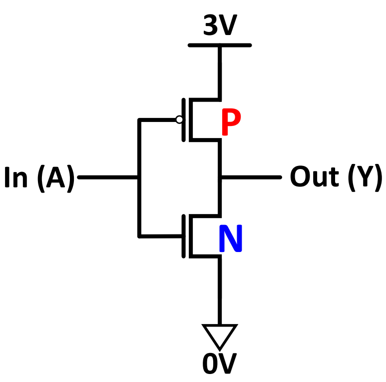


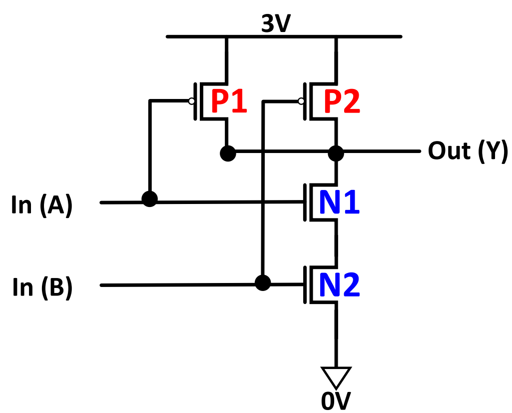


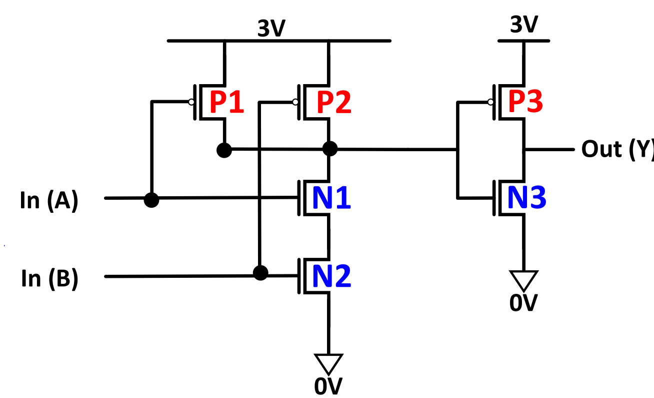
A	Y
0	1
1	0

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

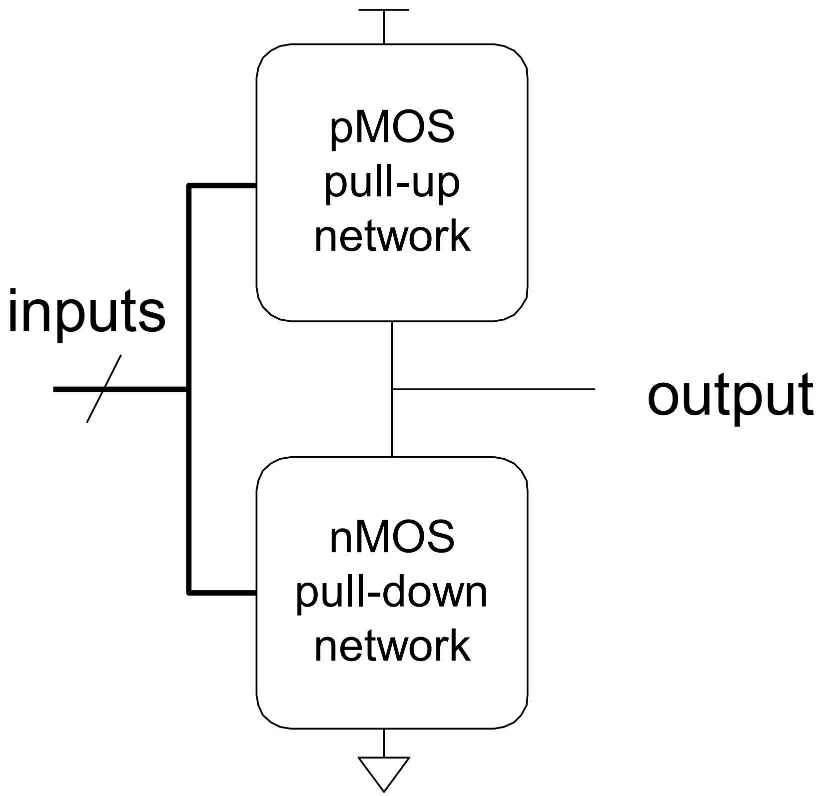






General CMOS Gate Structure

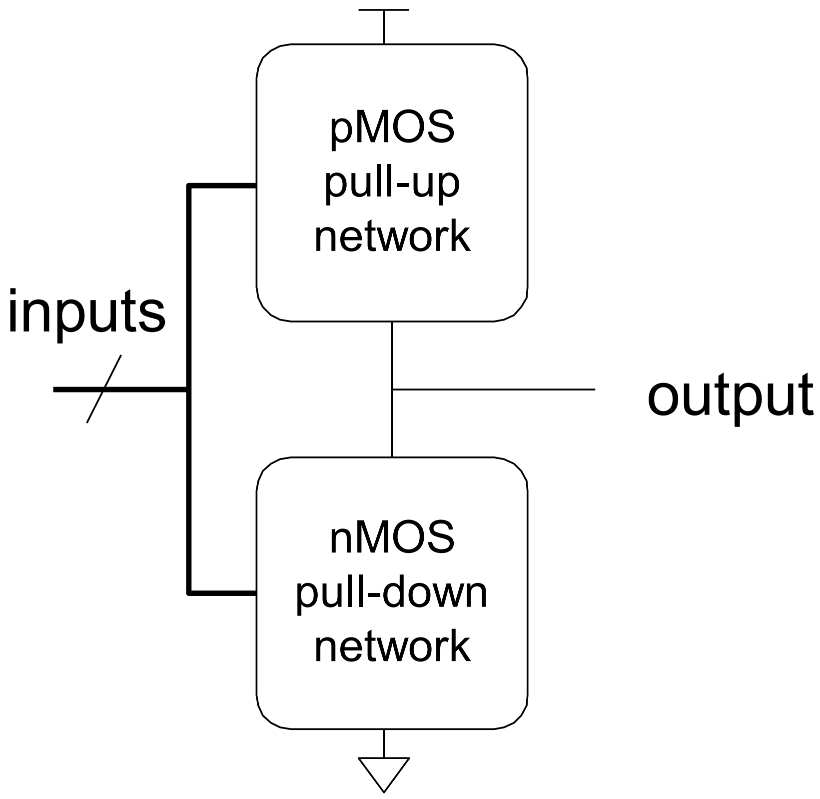
- The general form used to construct any inverting logic gate, such as: NOT, NAND, or NOR
  - The networks may consist of transistors in series or in parallel
  - When transistors are in parallel, the network is **ON** if one of the transistors is **ON**
  - When transistors are in series, the network is **ON** only if all transistors are **ON**



pMOS transistors are used for pull-u**p**
nMOS transistors are used for pull-dow**n**

General CMOS Gate Structure (II)

- Exactly one network should be ON, and the other network should be OFF at any given time
  - If both networks are ON at the same time, there is a short circuit → likely incorrect operation
  - If both networks are OFF at the same time, the output is floating → undefined

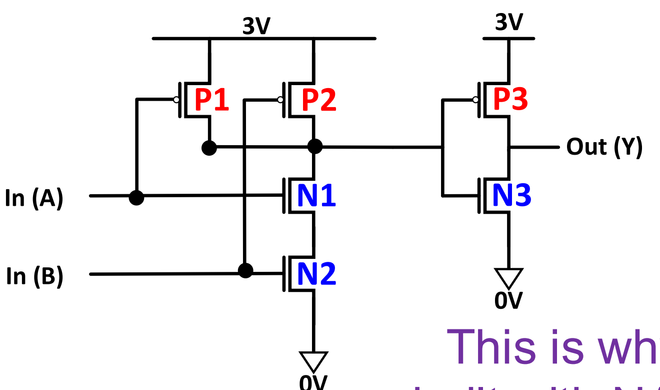


pMOS transistors are used for pull-u**p**
nMOS transistors are used for pull-dow**n**

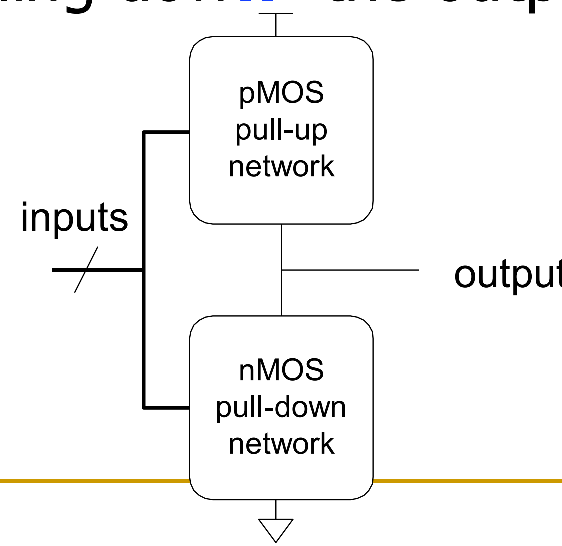
Digging Deeper: Why This Structure?

- MOS transistors are **imperfect** switches
- pMOS transistors pass 1's well but 0's poorly (holes carry charge)
- nMOS transistors pass 0's well but 1's poorly (electrons carry charge)

- **p**MOS transistors are good at "pulling u**p**" the output
- **n**MOS transistors are good at "pulling dow**n**" the output



This is why AND is built with NAND + NOT



See Section 1.7 in H&H

Digging Deeper: Latency

- Which one is slower?
  - Transistors in series
  - Transistors in parallel

- Series connections are slower than parallel connections
  - More resistance on the wire

- How do you alleviate this latency?
  - See H&H Section 1.7.8 for an example: pseudo-nMOS Logic

Used in the past when pMOS transistors could not be fabricated well

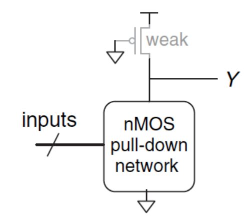


Figure 1.39 Generic pseudo-nMOS gate

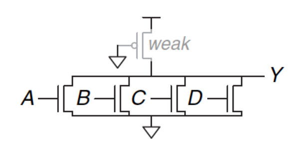


Figure 1.40 Pseudo-nMOS four-input NOR gate

Digging Deeper: Power Consumption

- **Dynamic Power Consumption**
  - Power used to charge capacitance as signals change (0 ←→ 1)
  - $C * V^2 * f$
    - C = capacitance of the circuit (wires and gates)
    - V = supply voltage
    - f = charging frequency of the capacitor

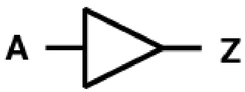
- **Static Power Consumption**
  - Power used when signals do not change
  - $V * I_{leakage}$
    - supply voltage * leakage current

- **Energy Consumption**
  - Power * Time

See more in H&H Chapter 1.8

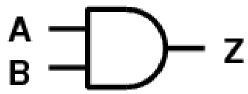
Common Logic Gates

Buffer



A	Z
0	0
1	1

AND



A	B	Z
0	0	0
0	1	0
1	0	0
1	1	1

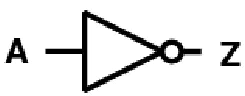
OR

A	B	Z
0	0	0
0	1	1
1	0	1
1	1	1

XOR

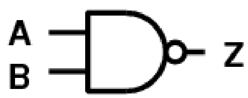
A	B	Z
0	0	0
0	1	1
1	0	1
1	1	0

Inverter



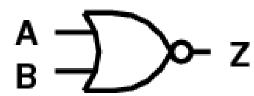
A	Z
0	1
1	0

NAND



A	B	Z
0	0	1
0	1	1
1	0	1
1	1	0

NOR



A	B	Z
0	0	1
0	1	0
1	0	0
1	1	0

XNOR

A	B	Z
0	0	1
0	1	0
1	0	0
1	1	1

Larger Gates

- We can extend the gates to more than 2 inputs
- Example: 3-input AND gate, 10-input NOR gate
- See your readings

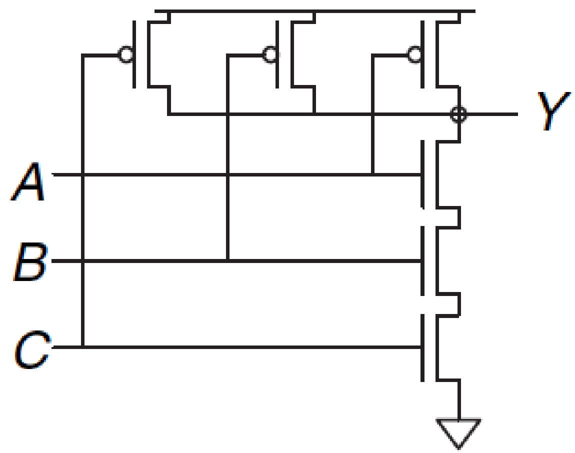


Figure 1.35 **Three-input NAND gate schematic**

Digital Design & Computer Arch.

Lecture 1: Introduction: Fundamentals, Transistors, Gates

Prof. Onur Mutlu

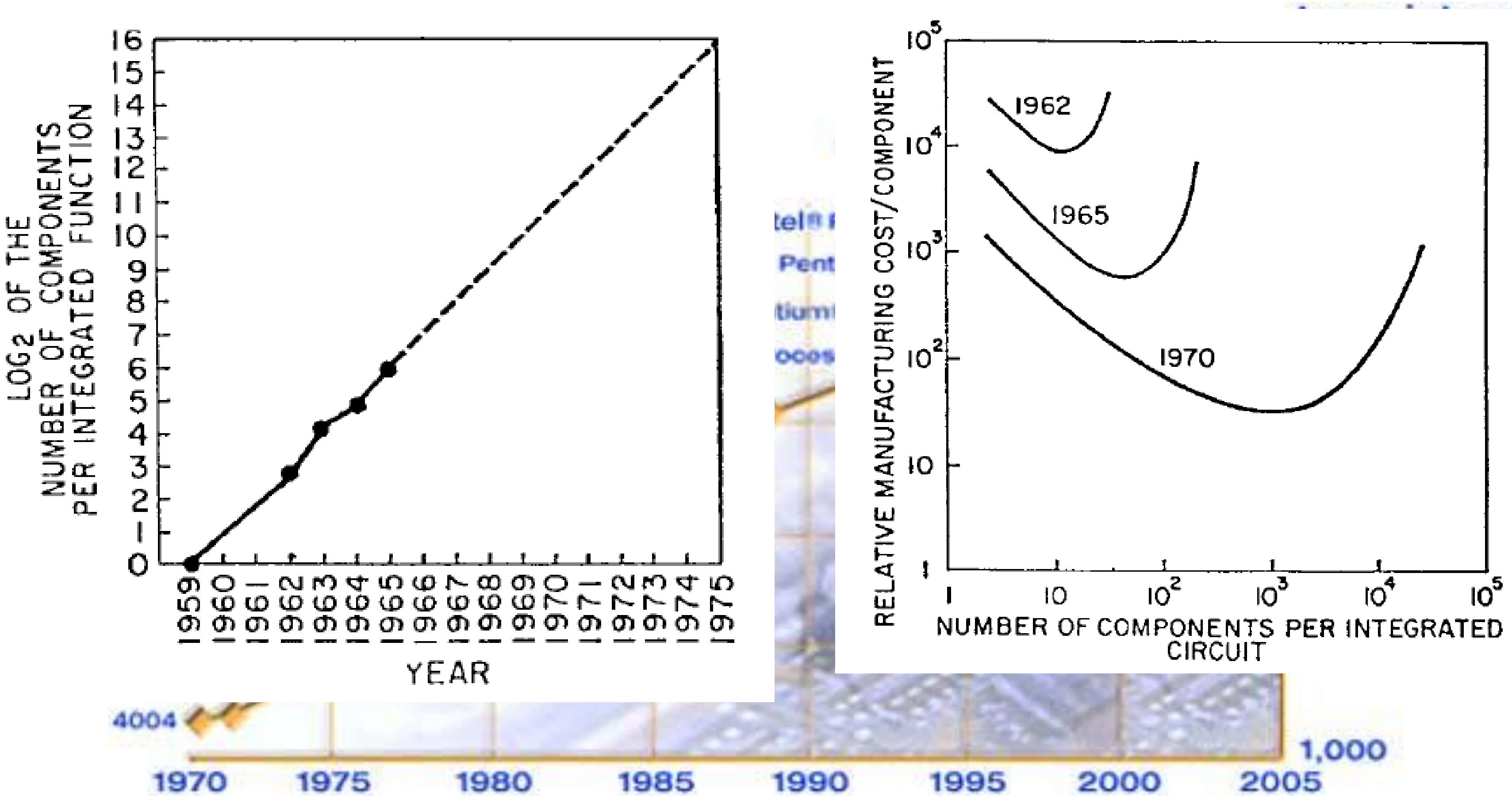
ETH Zürich

Spring 2026

19 February 2026

Aside: Moore's Law:
Enabler of Many Gates on a Chip

An Enabler: Moore's Law

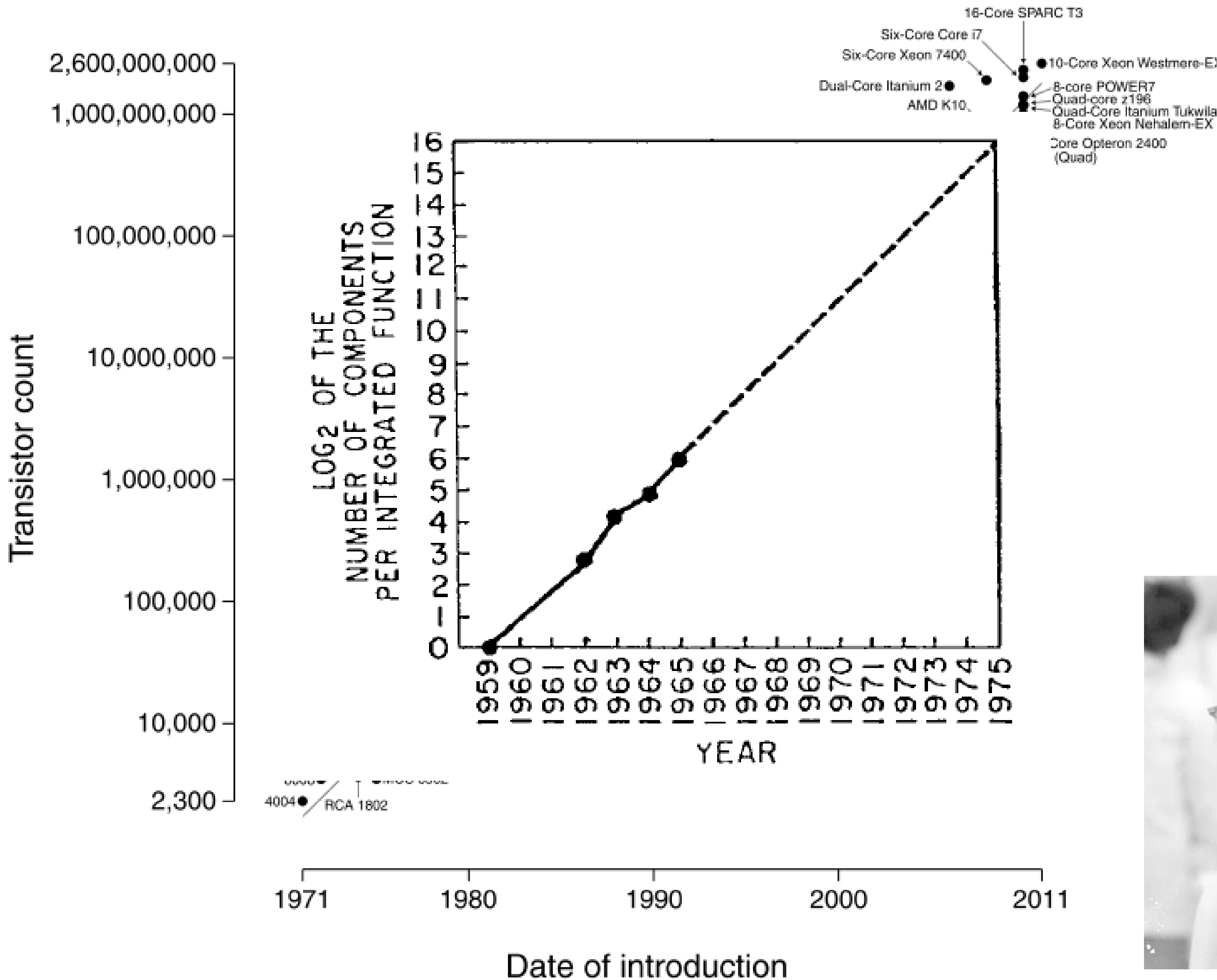


Moore, "Cramming more components onto integrated circuits," Electronics Magazine, 1965.

Component counts double every other year

Image source: Intel

Microprocessor Transistor Counts 1971-2011 & Moore's Law



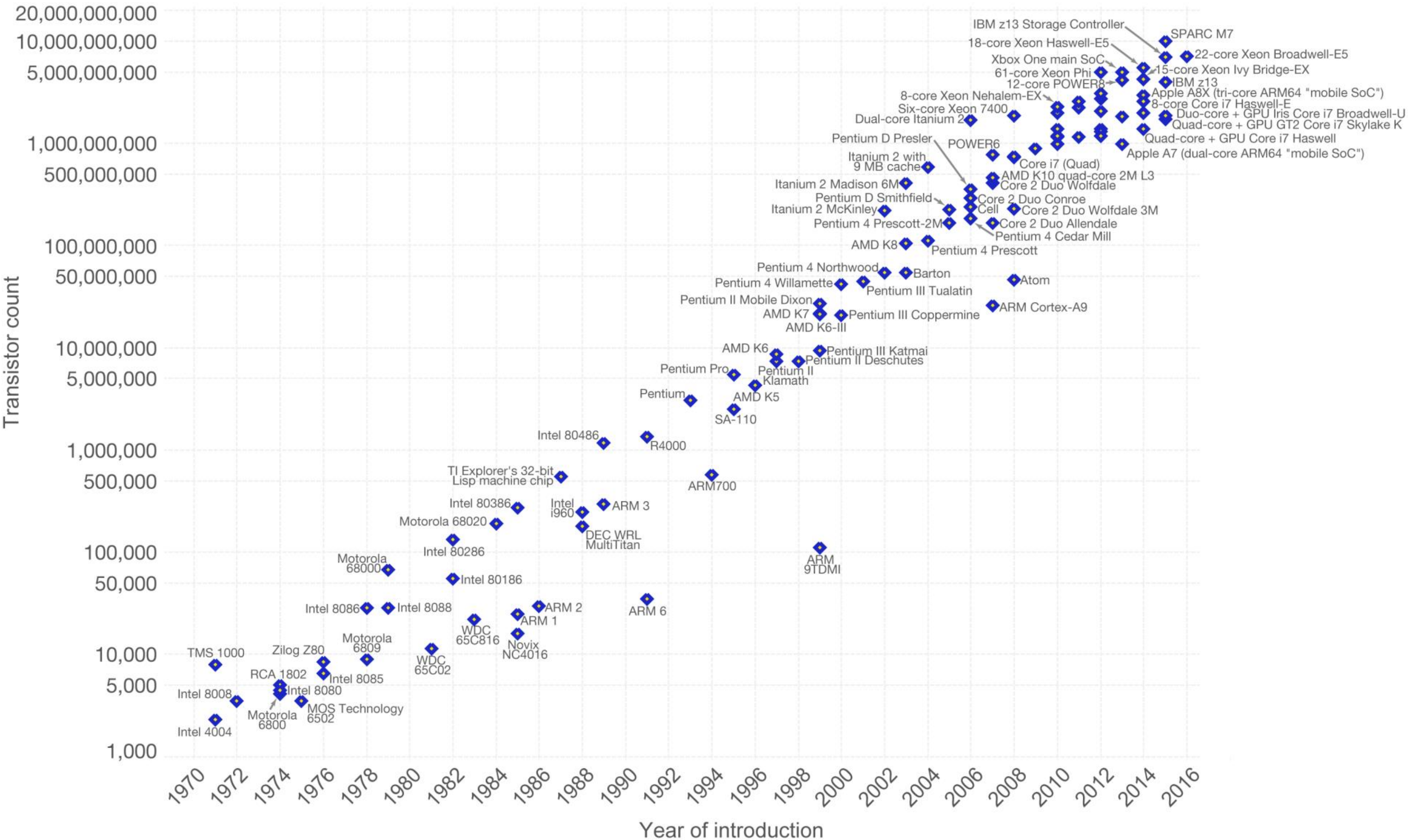
Number of transistors on an integrated circuit doubles ~ every two years

Image source: Wikipedia

Moore's Law – The number of transistors on integrated circuit chips (1971-2016)

Our World in Data

Moore's law describes the empirical regularity that the number of transistors on integrated circuits doubles approximately every two years. This advancement is important as other aspects of technological progress – such as processing speed or the price of electronic products – are strongly linked to Moore's law.



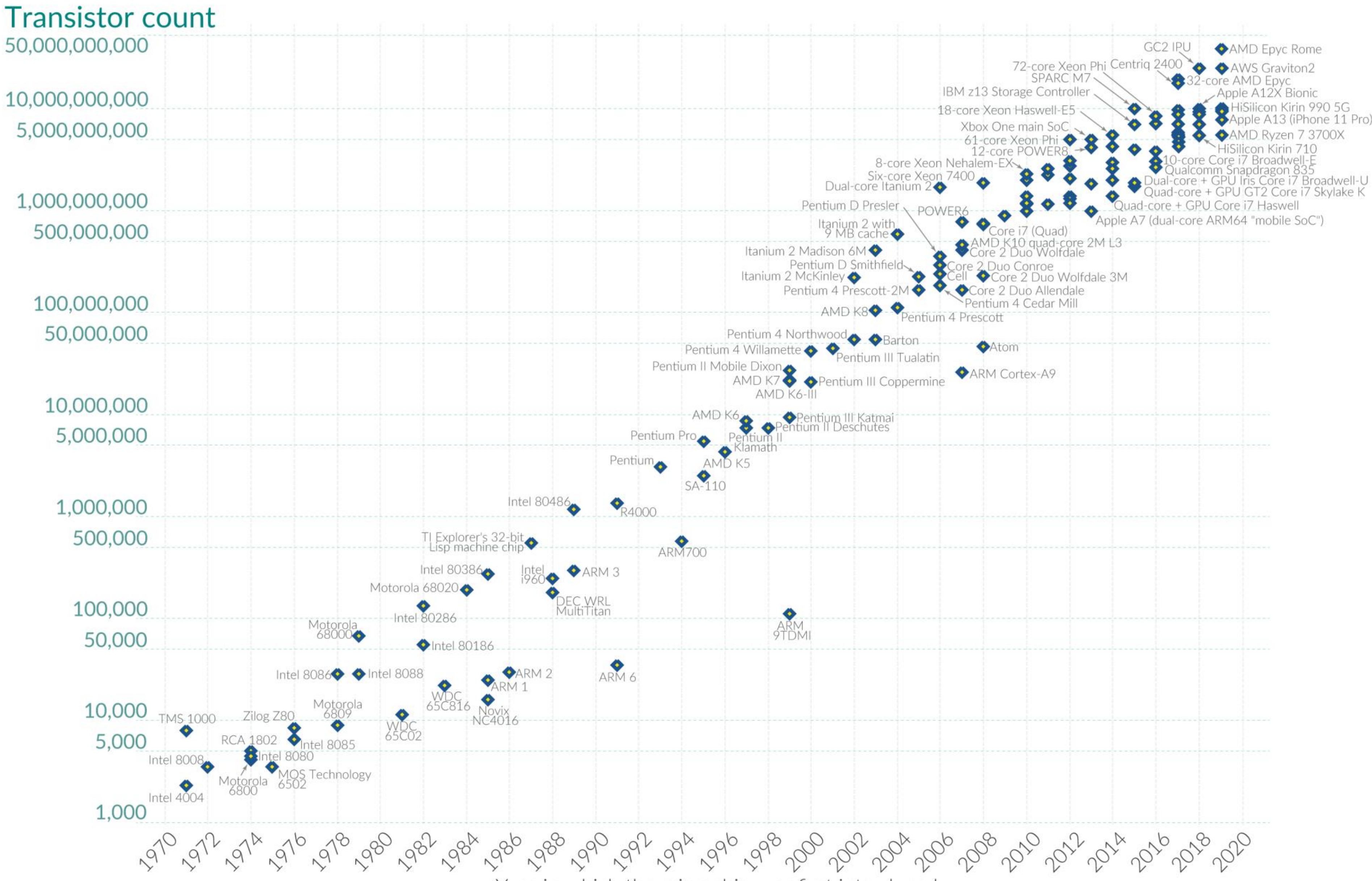
Data source: Wikipedia (https://en.wikipedia.org/wiki/Transistor_count)

The data visualization is available at OurWorldinData.org. There you find more visualizations and research on this topic.

Moore's Law: The number of transistors on microchips doubles every two years

Our World in Data

Moore's law describes the empirical regularity that the number of transistors on integrated circuits doubles approximately every two years. This advancement is important for other aspects of technological progress in computing – such as processing speed or the price of computers.



Data source: Wikipedia (wikipedia.org/wiki/Transistor_count)

OurWorldinData.org – Research and data to make progress against the world's largest problems.

Recommended Reading

- Moore, "Cramming more components onto integrated circuits," Electronics Magazine, 1965.

- Only 3 pages

- A quote:

  "*With unit cost falling as the number of components per circuit rises, by 1975 economics may dictate squeezing as many as 65 000 components on a single silicon chip.*"

- Another quote:

  "*Will it be possible to remove the heat generated by tens of thousands of components in a single silicon chip?*"

How Do We Keep Moore's Law: Innovation

- **Manufacturing smaller transistors/structures**
  - Some structures are already a few atoms in size
- **Using materials with better properties**
  - Copper instead of Aluminum (better conductor)
  - Hafnium Oxide, air for insulators
  - Making sure all materials are compatible is the challenge
- **Enabling precision manufacturing**
  - Extreme ultraviolet (EUV) light to pattern <10nm structures
- **Creating new device technologies**
  - FinFET, Gate All Around transistor, Single Electron Transistor…

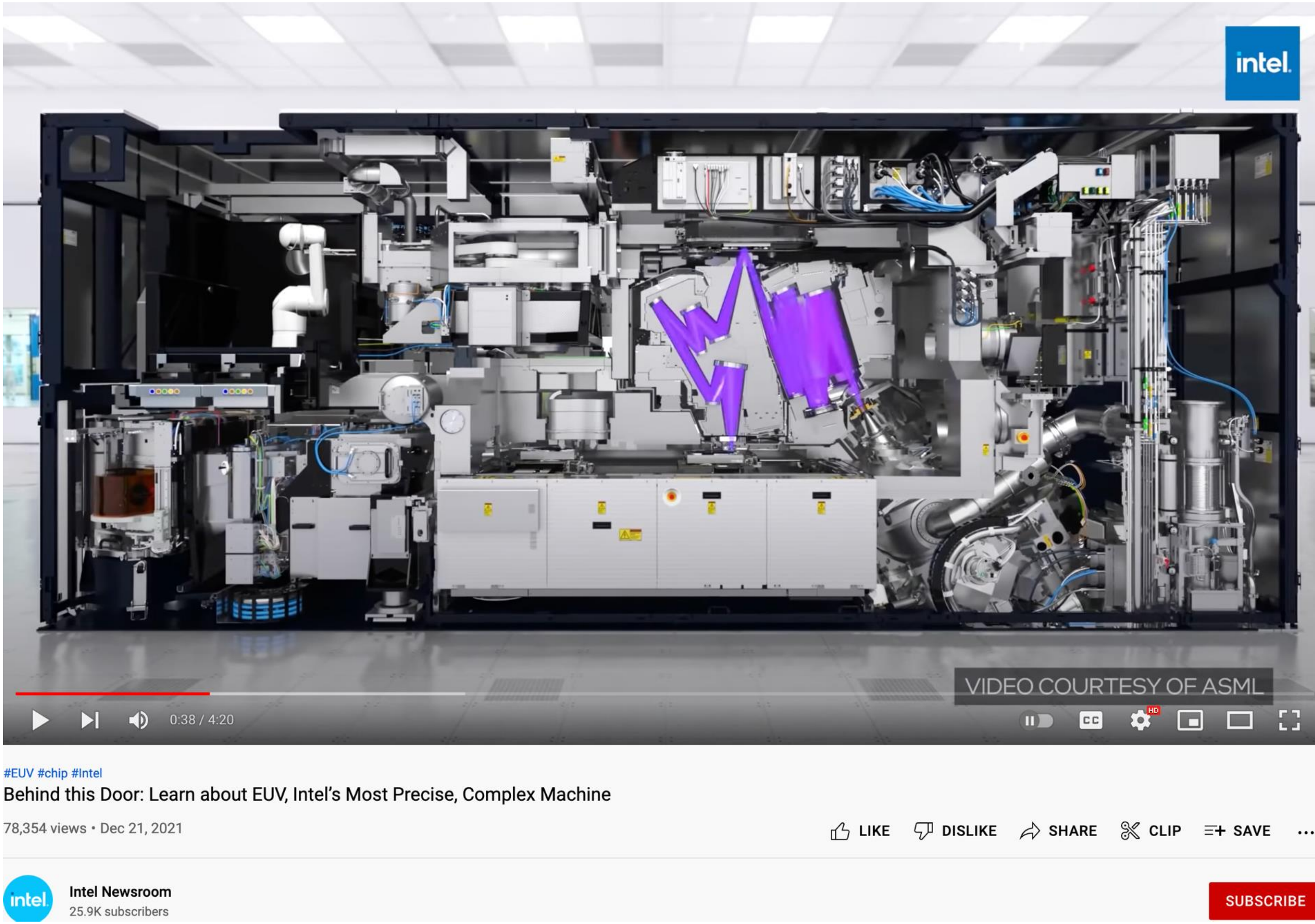
A 5-Minute Video on Transistor Innovation

https://www.youtube.com/watch?v=Z7M8etXUEUU

A 5-Minute Video on Transistor Innovation

https://www.youtube.com/watch?v=Z7M8etXUEUU

Enabling Manufacturing Tech: EUV



https://www.youtube.com/watch?v=Jv40Viz-KTc

Enabling Manufacturing Tech: EUV

Extreme ultraviolet lithography

11 languages

Article Talk Read Edit View history

From Wikipedia, the free encyclopedia

Extreme ultraviolet lithography (also known as **EUV** or **EUVL**) is an optical lithography technology used in semiconductor device fabrication to make integrated circuits (ICs). It uses extreme ultraviolet (EUV) wavelengths, roughly spanning a 2% FWHM bandwidth near *13.5 nm* (13.36nm - 13.65nm at 50% power), using a laser-pulsed *tin plasma*, to produce a pattern by using a reflective photomask to expose a substrate covered by photoresist. EUV (10-124nm) is the band between X-Rays (0.1-10nm) and overlapping slightly with UVC (100-280nm). It is currently applied only in the most advanced semiconductor device fabrication.

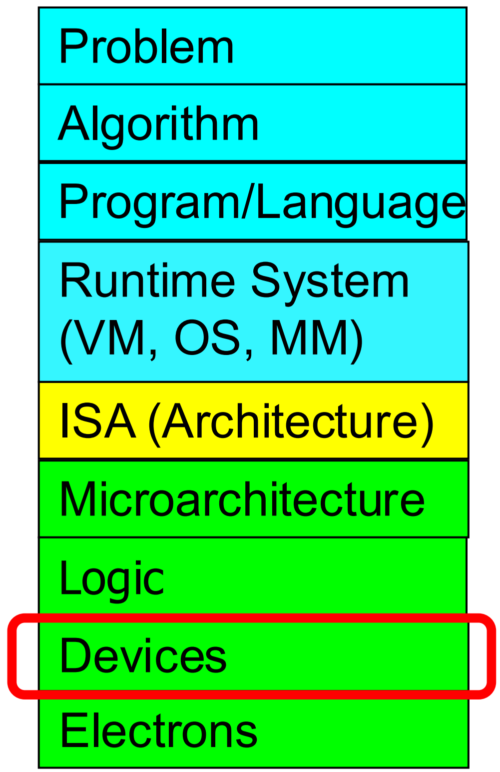
As of 2022, ASML Holding is the only company who produces and sells EUV systems for chip production, targeting 5 nm and 3 nm. At the 2019 International Electron Devices Meeting (IEDM), TSMC reported use of EUV for 5 nm in contact, via, metal line, and cut layers, where the cuts can be applied to fins, gates or metal lines.[1][2] At IEDM 2020, TSMC reported their 5 nm minimum metal pitch to be reduced 30% from that of 7 nm,[3] which was 40 nm.[4] Samsung's 5 nm is lithographically the same design rule as 7 nm, with a minimum metal pitch of 36 nm.[5]

History [edit]

In the 1960s, visible light was used for IC-production, with wavelengths as small as 435 nm (mercury "g line"). Later UV light was used, with wavelength of at first 365nm (mercury "i line"), then excimer wavelengths first of 248 nm (krypton fluoride laser) and then 193 nm (argon fluoride laser), which was called deep UV. The next step, going even smaller, was dubbed Extreme UV or EUV. The EUV technology was considered impossible by many. EUV is absorbed by glass and even air, so instead of using lenses, as before, to focus the beams of light, mirrors in a vacuum would be needed and a reliable production of EUV was also problematic. The then leading producers of steppers, Japanese companies Canon and Nikon gave up trying. And some even predicted the end of Moore's law.

https://en.wikipedia.org/wiki/Extreme_ultraviolet_lithography

Innovation At the Bottom Enables Computing



Historical: Opportunities at the Bottom

There's Plenty of Room at the Bottom

From Wikipedia, the free encyclopedia

"**There's Plenty of Room at the Bottom: An Invitation to Enter a New Field of Physics**" was a lecture given by physicist Richard Feynman at the annual American Physical Society meeting at Caltech on December 29, 1959.[1] Feynman considered the possibility of direct manipulation of individual atoms as a more powerful form of synthetic chemistry than those used at the time. Although versions of the talk were reprinted in a few popular magazines, it went largely unnoticed and did not inspire the conceptual beginnings of the field. Beginning in the 1980s, nanotechnology advocates cited it to establish the scientific credibility of their work.

https://en.wikipedia.org/wiki/There%27s_Plenty_of_Room_at_the_Bottom

Historical: Opportunities at the Bottom (II)

There's Plenty of Room at the Bottom

From Wikipedia, the free encyclopedia

Feynman considered some ramifications of a general ability to manipulate matter on an atomic scale. He was particularly interested in the possibilities of denser computer circuitry, and microscopes that could see things much smaller than is possible with scanning electron microscopes. These ideas were later realized by the use of the scanning tunneling microscope, the atomic force microscope and other examples of scanning probe microscopy and storage systems such as Millipede, created by researchers at IBM.

Feynman also suggested that it should be possible, in principle, to make nanoscale machines that "arrange the atoms the way we want", and do chemical synthesis by mechanical manipulation.

He also presented the possibility of "swallowing the doctor", an idea that he credited in the essay to his friend and graduate student Albert Hibbs. This concept involved building a tiny, swallowable surgical robot.

https://en.wikipedia.org/wiki/There%27s_Plenty_of_Room_at_the_Bottom

Extra Assignment: Moore's Law (I)

- **Paper review**
- G.E. Moore. "Cramming more components onto integrated circuits," Electronics magazine, 1965

- **Optional Assignment for Your Benefit**
  - **Write a 1-page individual review**
    - What are your key takeaways? What did you learn?
    - What surprised you about the content presented? What excited you?
    - How do you think Moore's law evolved since the paper was written?
    - What do you think (future) solutions should be like?